Chapter Eight Numerical Solution of Ordinary Differential Equations

8.0 Introduction

This chapter concerns numerical problems involving ordinary differential equations. The central problem is to solve a single first-order equation when one point on the solution curve is given. Later sections are devoted to systems of equations, higher-order equations, and two-point boundary-value problems.

8.1 The Existence and Uniqueness of Solutions

Our model is an initial-value problem written in the form

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{1}$$

Here x is an unknown function of t that we hope to construct from the information given in Equations (1), where $x' = dx(t)/dt$. The second of the two equations in (1) specifies one particular value of the function $x(t)$. The first equation gives the slope of the curve x at any point t. Of course, the function f must be specified. For a concrete example, we can take

$$\begin{cases} x' = x\tan(t+3) \\ x(-3) = 1 \end{cases} \tag{2}$$

We would like to determine $x(t)$ on an interval containing the initial point t_0. The *analytic* solution of this initial-value problem is $x(t) = \sec(t+3)$, as we can easily verify. Since $\sec t$ becomes infinite at $t = \pm\pi/2$, our solution is valid only for $-\pi/2 < t + 3 < \pi/2$. The example in (2) is exceptional because it has a simple analytic solution from which numerical values are readily calculated. Typically, for problems of the type in (1), analytic solutions are *not* available and numerical methods must be used.

Existence

Will every initial-value problem of the form in Equations (1) have a solution? No. Some assumptions must be made about f, and even then we can expect the solution to exist only in a neighborhood of $t = t_0$. As an example of what could happen, consider

$$\begin{cases} x' = 1 + x^2 \\ x(0) = 0 \end{cases} \tag{3}$$

The solution curve starts at $t = 0$ with slope 1; that is, $x'(0) = 1$. Because the slope is positive, $x(t)$ is *increasing* near $t = 0$. Therefore, the expression $1 + x^2$ is also increasing. Hence, x' is increasing. Because x and x' are both increasing and are related by the equation $x' = 1 + x^2$, we can expect that at some finite value of t there will be no solution; that is, $x(t) = +\infty$. As a matter of fact, this occurs at $t = \pi/2$ because the analytic solution of (3) is $x(t) = \tan t$.

THEOREM 1 First Existence Theorem, Initial-Value Problem

If f is continuous in a rectangle R centered at (t_0, x_0), say

$$R = \left\{(t,x) : |t - t_0| \le \alpha, \quad |x - x_0| \le \beta\right\} \tag{4}$$

then the initial-value problem (1) has a solution $x(t)$ for $|t - t_0| \le \min(\alpha, \beta/M)$, where M is the maximum of $|f(t,x)|$ in the rectangle R.

EXAMPLE 1 Prove that the initial-value problem

$$\begin{cases} x' = (t + \sin x)^2 \\ x(0) = 3 \end{cases}$$

has a solution on the interval $-1 \le t \le 1$.

Solution In this example, $f(t, x) = (t + \sin x)^2$ and $(t_0, x_0) = (0, 3)$. In the rectangle

$$R = \{(t, x) : |t| \leq \alpha, |x - 3| \leq \beta\}$$

the magnitude of f is bounded by $|f(t, x)| \leq (\alpha + 1)^2 \equiv M$. We want $\min(\alpha, \beta/M) \geq 1$, and so we can let $\alpha = 1$. Then $M = 4$, and our objective is met by letting $\beta \geq 4$. The existence theorem asserts that a solution of the initial-value problem exists on the interval $|t| \leq \min(\alpha, \beta/M) = 1$. ■

Uniqueness

It can happen, even if f is continuous, that the initial-value problem (1) does not have a unique solution. A simple example of this phenomenon is given by the problem

$$\begin{cases} x' = x^{2/3} \\ x(0) = 0 \end{cases}$$

It is obvious that the 0 function, $x(t) \equiv 0$, is a solution of this problem. Another solution is the function $x(t) = \frac{1}{27}t^3$.

To prove that the initial-value problem (1) has a *unique* solution in a neighborhood of $t = t_0$, it is necessary to assume somewhat more about f. Here is the usual theorem on this.

THEOREM 2 Uniqueness Theorem, Initial-Value Problem

If f and $\partial f/\partial x$ are continuous in the rectangle R defined by (4), then the initial-value problem (1) has a unique solution in the interval $|t - t_0| < \min(\alpha, \beta/M)$.

In both Theorems 1 and 2, the interval on the t-axis in which the solution is asserted to exist may be smaller than the base of the rectangle in which we have defined $f(t, x)$. The next theorem is of a different type that allows us to infer the existence and unicity of a solution on a prescribed interval $[a, b]$. (See Henrici [1962, p. 15].)

THEOREM 3 Second Existence Theorem, Initial-Value Problem

If f is continuous in the strip $a \leq t \leq b$, $-\infty < x < \infty$ and satisfies there an inequality

$$|f(t, x_1) - f(t, x_2)| \leq L|x_1 - x_2| \tag{5}$$

then the initial-value problem (1) has a unique solution in the interval $[a, b]$.

Inequality (5) is called a **Lipschitz condition** in the second variable. For a function of one variable, such a condition would assert simply

$$|g(x_1) - g(x_2)| \leq L|x_1 - x_2| \tag{6}$$

We see immediately that this condition is *stronger* than continuity because if x_2 approaches x_1, then the right-hand side in (6) approaches 0, and this forces $g(x_2)$ to approach $g(x_1)$. The condition (6) is weaker than having a bounded derivative. Indeed, if $g'(x)$ exists everywhere and does not exceed L in modulus, then by the Mean-Value Theorem,

$$|g(x_1) - g(x_2)| = |g'(\xi)|\,|x_1 - x_2| \le L|x_1 - x_2|$$

EXAMPLE 2 Show that the function $g(x) = \sum_{i=1}^{n} a_i|x - w_i|$ satisfies a Lipschitz condition with the constant $L = \sum_{i=1}^{n} |a_i|$.

Solution

$$\begin{aligned}
|g(x_1) - g(x_2)| &= \left| \sum_{i=1}^{n} a_i|x_1 - w_i| - \sum_{i=1}^{n} a_i|x_2 - w_i| \right| \\
&= \left| \sum_{i=1}^{n} a_i \Big\{ |x_1 - w_i| - |x_2 - w_i| \Big\} \right| \\
&\le \sum_{i=1}^{n} |a_i|\, \big| |x_1 - w_i| - |x_2 - w_i| \big| \\
&\le \sum_{i=1}^{n} |a_i||x_1 - x_2| = L|x_1 - x_2|
\end{aligned}$$

■

PROBLEMS 8.1

1. Find two solutions of the initial-value problem

$$\begin{cases} x' = x^{1/3} \\ x(0) = 0 \end{cases}$$

Hint: Try $x = ct^{\lambda}$, or observe that the equation is separable.

2. **a.** Use Theorem 1, on initial-value problem existence, to predict in what interval a solution of the initial-value problem (3) exists. Find the largest interval.

   b. Repeat Part **a** for the initial-value problem (2).

3. Show that $x = -t^2/4$ and $x = 1 - t$ are solutions of the initial-value problem

$$\begin{cases} 2x' = \sqrt{t^2 + 4x} - t \\ x(2) = -1 \end{cases}$$

Why does this not contradict Theorem 2, on initial-value problem uniqueness?

4. Solve the initial-value problem $x' = f(t, x)$, $x(0) = 0$ in these special cases:

   a. $f(t, x) = t^3$

   b. $f(t, x) = (1 - t^2)^{-1/2}$

c. $f(t,x) = (1+t^2)^{-1}$

d. $f(t,x) = (t+1)^{-1}$

5. Solve the initial-value problem $x' = f(t,x)$, $x(0) = 0$ in the following cases. Use the fact that $dt/dx = (dx/dt)^{-1}$ when $dx/dt \neq 0$.

a. $f(t,x) = x^{-2}$

b. $f(t,x) = 1 + x^2$

c. $f(t,x) = (\sin x + \cos x)^{-1}$

6. Use Theorem 1, on initial-value problem existence, to show that the initial-value problem

$$\begin{cases} x' = \sqrt{|x|} \\ x(0) = 0 \end{cases}$$

has a solution on the entire real line.

7. Show by using Theorem 1, on initial-value problem existence, that the initial-value problem

$$\begin{cases} x' = \tan x \\ x(0) = 0 \end{cases}$$

has a solution in the interval $|t| < \pi/4$.

8. Let f be a continuous function of one variable, defined on all of $\mathbb{R}$. Let $M(r)$ denote the maximum of $|f(x)|$ for $|x| \leq r$. If $M(r) = o(r)$ as $r \to \infty$, then the initial-value problem

$$\begin{cases} x' = f(x) \\ x(0) = 0 \end{cases}$$

has a solution on all of $\mathbb{R}$. Prove this assertion.

9. Prove that the initial-value problem

$$\begin{cases} x' = t^2 + e^x \\ x(0) = 0 \end{cases}$$

has a unique solution in the interval $|t| \leq 0.351$.

10. Prove that if $f(t,x)$ is continuous and bounded in the domain $a \leq t \leq b$, $-\infty < x < \infty$, then the initial-value problem

$$\begin{cases} x' = f(t,x) \\ x(a) = \alpha \end{cases}$$

has a solution in the interval $a \leq t \leq b$.

11. Let R denote the rectangle in the tx-plane defined by $|t - t_0| \leq \alpha$, $|x - x_0| \leq \beta$. Let f be a continuous function defined on this rectangle and satisfying $\beta \geq \alpha |f(t,x)|$. Prove that the initial-value problem $x' = f(t,x)$, $x(t_0) = x_0$ has a solution on the interval $|t - t_0| \leq \alpha$.

12. Establish that the initial-value problem

$$\begin{cases} x' = 1 + x + x^2 \cos t \\ x(0) = 0 \end{cases}$$

has a solution in the interval $-1/3 \leq t \leq 1/3$.

13. Show that the initial-value problem

$$\begin{cases} x' = \sqrt{|x|} \\ x(0) = 0 \end{cases}$$

has two solutions, and indicate why Theorem 2, on initial-value problem uniqueness, does not apply.

14. Prove that the problem

$$\begin{cases} x' = tx^{2/3} \\ x(0) = 1 \end{cases}$$

has a solution in the interval $-2 \le t \le 2$. Is there more than one solution?

15. Show that the initial-value problem

$$\begin{cases} x' = 2te^{-x} \\ x(0) = 0 \end{cases}$$

has a unique solution in the interval $-\infty < t < \infty$. Is there a choice of α and β in the existence theorem that allows this conclusion to be drawn?

16. Let $f(t, x)$ be continuous on the rectangle defined by $|t| \le 3$, $|x| \le 4$, and assume that $|f(t, x)| \le 7$ for all points in this rectangle. What is the largest interval in which the initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(0) = 0 \end{cases}$$

must have a solution?

17. Find an interval in which we can be sure that the initial-value problem

$$\begin{cases} x' = \sec x \\ x(0) = 0 \end{cases}$$

has a unique solution.

18. Use each of the three theorems in this section to predict where this problem has a solution, and then solve the problem explicitly, comparing the theory to the fact:

$$\begin{cases} x' = x^2 \\ x(0) = 1 \end{cases}$$

19. Prove that the initial-value problem

$$\begin{cases} x' = 1 + x^2 \\ x(0) = 0 \end{cases}$$

has a solution in the interval $[-1, 1]$. Prove that this example does not satisfy the hypotheses of Theorem 3. Explain why this example does not contradict Theorem 3.

COMPUTER PROBLEM 8.1

1. Using a symbolic manipulation program, find the solution of the differential equation $x' + x = (1 + e^t)^{-1}$.

8.2 Taylor-Series Method

In the numerical solution of differential equations, we rarely expect to obtain the solution directly as a *formula* giving $x(t)$ as a function of t. Instead, we usually construct a table of function values of the form

t_0	t_1	t_2	t_3	$\cdots$	t_m
x_0	x_1	x_2	x_3	$\cdots$	x_m

(1)

Here, x_i is the computed approximate value of $x(t_i)$, our notation for the *exact* solution at t_i. From a table such as (1), a spline function or other approximating function can be constructed. However, most numerical methods for solving ordinary differential equations produce such a table first.

We consider again the initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{2}$$

where f is a prescribed function of two variables, and (t_0, x_0) is a single given point through which the solution curve passes. A *solution* of (2) is a function $x \mapsto x(t)$ such that $dx(t)/dt = f(t, x(t))$ for all t in some neighborhood of t_0, and $x(t_0) = x_0$.

Illustration

For the Taylor-series method, it is necessary to assume that various partial derivatives of f exist. To illustrate the method, we take a concrete example:

$$\begin{cases} x' = \cos t - \sin x + t^2 \\ x(-1) = 3 \end{cases} \tag{3}$$

At the heart of the procedure is the Taylor series for x, which we write as

$$x(t+h) = x(t) + hx'(t) + \frac{h^2}{2!}x''(t) + \frac{h^3}{3!}x'''(t) + \frac{h^4}{4!}x^{(4)}(t) + \cdots \tag{4}$$

The derivatives appearing here can be obtained from the differential equation in (3). They are

$$\begin{aligned} x'' &= -\sin t - x'\cos x + 2t \\ x''' &= -\cos t - x''\cos x + (x')^2 \sin x + 2 \\ x^{(4)} &= \sin t - x'''\cos x + 3x'x''\sin x + (x')^3 \cos x \end{aligned}$$

At this point, our patience wears thin and we decide to use only terms up to and including h^4 in Formula (4). The terms that we have *not* included start with a term in h^5, and they constitute collectively the **truncation error** inherent in our procedure. The resulting numerical method is said to be of order 4. (The *order* of the

Taylor-series method is n if terms up to and including $h^n x^{(n)}(t)/n!$ are used.) Notice that in differentiating terms such as $\sin x$ with respect to t, we must think of it as $d\{\sin[x(t)]\}/dt$ and employ the *chain rule* of differentiation. This, of course, accounts for the complexity of the formulas for x'', x''', ... We could perform various substitutions to obtain formulas for x'', x''', ... containing no derivatives of x on the right-hand side. It is not necessary to do this if the formulas are used in the order listed. They are recursive in nature.

Here is an algorithm to solve the initial-value problem (3), starting at $t = -1$ and progressing in steps of $h = 0.01$. We desire a solution in the t-interval $[-1, 1]$, and thus we must take 200 steps.

input $M \leftarrow 200;\ h \leftarrow 0.01;\ t \leftarrow -1.0;\ x \leftarrow 3.0$
output $0, t, x$
for $k = 1$ **to** M **do**
  $x' \leftarrow \cos t - \sin x + t^2$
  $x'' \leftarrow -\sin t - x' \cos x + 2t$
  $x''' \leftarrow -\cos t - x'' \cos x + (x')^2 \sin x + 2$
  $x^{(4)} \leftarrow \sin t + ((x')^3 - x''') \cos x + 3x'x'' \sin x$
  $x \leftarrow x + h(x' + \frac{h}{2}(x'' + \frac{h}{3}(x''' + \frac{h}{4}(x^{(4)}))))$
  $t \leftarrow t + h$
  output k, t, x
end do

What can be said of the errors in our computed solution? At each step, the **local truncation error** is $\mathcal{O}(h^5)$ because we have *not* included terms involving $h^5, h^6, \ldots$ from the Taylor series. Thus, as $h \to 0$, the behavior of the local errors should be similar to Ch^5. Unfortunately, we do not know C. But h^5 is 10^{-10} since $h = 10^{-2}$. So with good luck, the error in each step should be roughly of the magnitude 10^{-10}. After several hundred steps, these small errors could accumulate and spoil the numerical solution. At each step (except the first), the estimate x_k of $x(t_k)$ already contains errors, and further computations continue to add to these errors. These remarks should make us quite cautious about accepting blindly all the decimal digits in a numerical solution of a differential equation!

When the preceding algorithm was programmed and run, the solution at $t = 1$ was $x_{200} = 6.42194$. Here is a sample of the output from that computer program:

k	t	x
0	−1.00000	3.00000
1	−0.99000	3.01400
2	−0.98000	3.02803
3	−0.97000	3.04209
4	−0.96000	3.05617
5	−0.95000	3.07028
6	−.94000	3.08443
7	−.93000	3.09861
⋮	⋮	⋮

k	t	x
196	0.96000	6.36566
197	0.97000	6.37977
198	0.98000	6.39386
199	0.99000	6.40791
200	1.00000	6.42194

In a subsequent computer run, the differential equation was integrated with this value of x_{200} as the *initial* condition and with $h = -0.01$. The result of this second computer run was $x_{200} \approx 3.00000$ at $t = -0.99999$. The close agreement with the original initial value leads us to think that the numerical solution is accurate to about six significant figures, the number of digits shown.

In the example just discussed, it is not difficult to estimate the local truncation error in each step of the numerical solution. To do this, we recall that the error term in the Taylor series (4) is of the form

$$E_n = \frac{1}{(n+1)!} h^{n+1} x^{(n+1)}(t + \theta h) \qquad (0 < \theta < 1)$$

This is the error that is present when the last power of h included in the sum is h^n. In the example, we have taken $n = 4$ and $h = 0.01$. One can estimate $x^{(5)}(t + \theta h)$ by a simple finite-difference approximation and arrive at

$$E_4 \approx \frac{1}{5!} h^5 \left[\frac{x^{(4)}(t+h) - x^{(4)}(t)}{h} \right] = \frac{h^4}{120} \left[x^{(4)}(t+h) - x^{(4)}(t) \right] \qquad \textbf{(5)}$$

With a little extra programming, this estimate of E_4 can be incorporated into the algorithm and computed. We do not give the details here because it is Computer Problem **8.2.19** (p. 539). When this feature is added to the program, the computed output shows that our estimate of E_4 never exceeds 3.42×10^{-11} in absolute value. The program was run in double precision on a computer similar to the `Marc-32`.

Pros and Cons

What are the advantages and disadvantages of the Taylor-series method? For disadvantages, the method depends on repeated differentiation of the given differential equation (unless we intend to use only the method of order 1). Hence, the function $f(t, x)$ must possess partial derivatives in the region where the solution curve passes in the tx-plane. Such an assumption is, of course, not necessary for the existence of a solution. Also, the necessity of carrying out the preliminary analytic work is a decided drawback. For example, an error made at this stage might be overlooked and remain undetected. Finally, the various derivatives must be separately programmed.

Advantages are the conceptual simplicity of the procedure and its potential for very high precision. Thus, if we could easily obtain 20 derivatives of $x(t)$, then there is nothing to prevent us from using the method with order 20 (i.e., terms up to and including the one involving h^{20}). With such a high order, the same accuracy can be obtained with a larger step size, say $h = 0.2$. Fewer steps are required to traverse the given interval, and this reduces the amount of computation. On the other hand, the calculations in each step are more burdensome. Some examples in which a high-order Taylor-series method can be used are among the problems.

In recent years, symbol-manipulating programs have become available for carrying out various routine mathematical calculations of a nonnumerical type. Differentiation and integration of rather complicated expressions can thus be turned over to the computer! Of course, these programs apply only to a restricted class of functions, but this class is broad enough to include all the functions that one encounters in the typical calculus textbook. With the use of such programs, the Taylor-series method of, say, order 20, could be used without difficulty.

Errors

In numerically solving a differential equation, several types of errors arise. These are conveniently classified as follows:

1. Local truncation error
2. Local roundoff error
3. Global truncation error
4. Global roundoff error
5. Total error

The **local truncation error** is the error made in one step when we replace an infinite process by a finite one. In the Taylor-series method, we replace the infinite Taylor series for $x(t+h)$ by a partial sum. The local truncation error is inherent in any algorithm that we might choose and is quite independent of the roundoff error. **Roundoff error** is, of course, caused by the limited precision of our computing machines, and its magnitude depends on the word length of the computer (or on the number of bits in the mantissa of the floating-point machine numbers).

In the Taylor-series method, if we retain terms up to and including h^n in the series, then the local truncation error is the sum of all the remaining terms that we do *not* include. By Taylor's Theorem, these terms can be compressed into a single term of the form

$$\frac{h^{n+1}}{(n+1)!}x^{(n+1)}(\xi)$$

for some nearby point ξ. We say that the local truncation error is $\mathcal{O}(h^{n+1})$. An error of this sort is present in each step of the numerical solution. The accumulation of all the many local truncation errors gives rise to the **global truncation** error. Again, this error will be present even if all calculations are performed using exact arithmetic. It is an error that is associated with the method and is independent of the computer on which the calculations are performed. If the local truncation errors are $\mathcal{O}(h^{n+1})$, then the global truncation error must be $\mathcal{O}(h^n)$ because the number of steps necessary to reach an arbitrary point T, having started at t_0, is $(T-t_0)/h$.

The **global roundoff error** is the accumulation of local roundoff errors in prior steps. The **total error** is the sum of the global truncation error and the global roundoff error. If the global truncation error is $\mathcal{O}(h^n)$, we say that the numerical procedure is of **order** n.

Euler's Method

The Taylor-series method with $n = 1$ is called **Euler's method**. It looks like this:

$$x(t + h) = x(t) + hf(t, x)$$

This formula has the obvious advantage of *not* requiring any differentiation of f. This advantage is offset by the necessity of taking small values for h to gain acceptable precision. Still, the method serves as a useful example and is of great importance theoretically because existence theorems can be based on it. See Henrici [1962, pp. 15–25] for such an existence theorem.

Delay Differential Equations

A special type of differential equation, called a **delay differential equation** or a **differential equation with retarded argument**, arises in some practical problems. Population models and mixing problems often have this special feature, which is that the value of $x'(t)$ is related to values of the function x at previous values of t. For example, we might have

$$x'(t) = f\big(x(t - 1)\big)$$

If we know the value of x at $t - 1$, the differential equation enables us to compute the value of $x'(t)$. To integrate the differential equation starting at $t = 0$, we shall need the history of $x(t)$ starting at $t = -1$. Thus, the values of $x(t)$ on the interval $[-1, 0]$ will have to be supplied to us as *initial values*.

An example of a specific and well-defined problem of this type might be

$$\begin{cases} x'(t) = x(t - 1) & (t \geq 0) \\ x(t) = t^2 & (-1 \leq t \leq 0) \end{cases} \tag{6}$$

The second of these equations gives the needed initial values for $x(t)$. If t is restricted to the interval $[0, 1]$, then $t - 1$ is in $[-1, 0]$, and hence

$$\begin{cases} x'(t) = x(t - 1) = (t - 1)^2 & (0 \leq t \leq 1) \\ x(0) = 0 \end{cases}$$

The solution of this is easily obtained by integrating and supplying an appropriate constant of integration:

$$x(t) = \tfrac{1}{3}(t - 1)^3 + \tfrac{1}{3} \qquad (0 \leq t \leq 1)$$

If the solution is to be continued into the next interval $[1, 2]$, another step like the first can be taken. Thus, for t in $[1, 2]$, we have

$$\begin{cases} x'(t) = x(t - 1) = \tfrac{1}{3}(t - 2)^3 + \tfrac{1}{3} & (1 \leq t \leq 2) \\ x(1) = \tfrac{1}{3} \end{cases}$$

The solution of this is

$$x(t) = \tfrac{1}{12}(t-2)^4 + \tfrac{1}{3}t - \tfrac{1}{12} \qquad (1 \le t \le 2)$$

The solution can be continued indefinitely to the right by similar calculations.

For more complicated equations, such as

$$x'(t) = \sin\left[x(t-1)^3\right] + \log\left[x(t) + t^5\right]$$

one must resort to numerical procedures. The Taylor-series method is available but is not without some drawbacks, such as determining the derivatives, which may be complicated and error-prone. Consider, for example,

$$\begin{cases} x'(t) = 2x(t-1) + x(t) & (t > 0) \\ x(t) = t^2 & (-1 \le t \le 0) \end{cases} \tag{7}$$

To compute the solution on the interval [0, 1], one can proceed in steps of length h by using a short Taylor expansion:

$$x(t+h) = x(t) + hx'(t) + \frac{h^2}{2}x''(t) + \frac{h^3}{6}x'''(t)$$

As usual, in the Taylor-series method, we must provide formulas for x', x'', and x''' by using the differential equation. In this example, if we consider only the interval [0, 1], we can use

$$\begin{aligned} x'(t) &= 2x(t-1) + x(t) = 2(t-1)^2 + x(t) \\ x''(t) &= 2x'(t-1) + x'(t) = 4(t-2)^2 + 2(t-1)^2 + x'(t) \\ x'''(t) &= 2x''(t-1) + x''(t) = 8(t-3)^2 + 8(t-2)^2 + 2(t-1)^2 + x''(t) \end{aligned}$$

The numerical solution produces values of $x(t)$ at discrete points in the interval [0, 1]. At the same time, we should store the values of $x'(t)$, $x''(t)$, and $x'''(t)$ at these same discrete points for use in the next interval. If we do not change the value of h, we can proceed in each interval in the same way using the appropriate saved values.

For the theory of delay differential equations, see the books by Driver [1977], Kuang [1993], and Diekmann, Van Gils, Verduyn Lunel, and Walther [1995]. Also, a code called DELSOL, used for the solution of systems of delay differential equations, is described in Willé and Baker [1992].

PROBLEMS 8.2

1. In Computer Problem **8.2.3** (p. 537), it is necessary to compute derivatives of the function e^{-t^2}. Show that the nth derivative of this function is of the form $e^{-t^2}P_n(t)$, where the polynomials P_n are determined recursively from the formulas

$$\begin{cases} P_0 = 1 \\ P_{n+1}(t) = P_n'(t) - 2tP_n(t) \end{cases}$$

Show that, for example,

$$P_4(t) = 12 - 48t^2 + 16t^4$$
$$P_5(t) = -120t + 160t^3 - 32t^5$$

2. Verify that the function $x(t) = t^2/4$ solves the initial-value problem

$$\begin{cases} x' = \sqrt{x} \\ x(0) = 0 \end{cases}$$

Apply the Taylor-series method of order 1, and explain why the numerical solution differs from the solution $t^2/4$.

3. Compute $x(0.1)$ by solving the differential equation

$$\begin{cases} x' = -tx^2 \\ x(0) = 2 \end{cases}$$

with one step of the Taylor-series method of order 2. (Use a calculator.)

4. Using the ordinary differential equation

$$\begin{cases} x' = x^2 + xe^t \\ x(0) = 1 \end{cases}$$

and one step of the Taylor-series method of order 3, calculate $x(0.01)$.

5. Consider the ordinary differential equation

$$\begin{cases} 5tx' + x^2 = 2 \\ x(4) = 1 \end{cases}$$

Calculate $x(4.1)$ using one step of the Taylor-series method of order 2.

6. An **integral equation** is an equation involving an unknown function within an integration. For example, here is a typical integral equation (of a type known by the name **Volterra**):

$$x(t) = \int_0^t \cos(s + x(s))\, ds + e^t$$

By differentiating this integral equation, obtain an equivalent initial-value problem for the unknown function.

7. If the Taylor-series method is used to solve an initial-value problem involving the differential equation

$$x' = \cos(tx)$$

what are the formulas for x'', x''', and $x^{(4)}$?

8. Let $x' = f(t, x)$. Determine x'', x''', and $x^{(4)}$ from this equation.

COMPUTER PROBLEMS 8.2

1. Write and test a computer program to solve the following differential equation with initial condition

$$\begin{cases} x' = x + e^t + tx \\ x(1) = 2 \end{cases}$$

on the interval $[1, 3]$. Use the Taylor series of order 5 and $h = 0.01$.

2. Write and test a computer program for solving the following initial-value problem:

$$\begin{cases} x' = 1 + x^2 - t^3 \\ x(0) = -1 \end{cases}$$

Use the Taylor-series method of order 4, with h a binary machine number near 0.01. Find the solution in the interval $[0, 2]$. Account for any peculiar phenomenon in the solution.

3. Methods for solving initial-value problems also can be used to compute definite or indefinite integrals. For example, we can compute

$$\int_0^2 e^{-s^2}\,ds$$

by solving the initial-value problem

$$\begin{cases} x' = e^{-t^2} \\ x(0) = 0 \end{cases}$$

on the t-interval $[0, 2]$. Do this, using the Taylor-series method of order 4. From a table of the **error function**

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-s^2}\,ds$$

we obtain $x(2) \approx 0.88208\,13907$. (See Abramowitz and Stegun [1964, p. 311].)

4. (Continuation) Use Problem **8.2.1** (p. 535) to write a computer program for the Taylor-series method of order 6 applicable to the integral in the preceding computer problem. Test the code with $h = 0.01$ to see whether the same value of $x(2)$ is obtained. Print a table of the function $\frac{1}{2}\sqrt{\pi}\operatorname{erf}(t)$ in steps of 0.01 from $t = 0$ to $t = 2$.

5. Solve the initial-value problem

$$\begin{cases} x' = 1 + x^2 \\ x(0) = 0 \end{cases}$$

on the interval $[0, 1.56]$ using the Taylor-series method of order 4 with $h = 0.01$. Then use the computed value of $x(1.56)$ as the initial value to integrate back to $t = 0$. Compare the results and explain what happened.

6. The equation $\arctan(x/t) = \ln\sqrt{x^2 + t^2}$ defines x implicitly as a function of t. Verify that this implicit function is a solution of the initial-value problem

$$\begin{cases} x' = (t + x)/(t - x) \\ x(1) = 0 \end{cases}$$

Prepare a table of the function $x(t)$ on the interval $[0, 2]$ with steps of ± 0.01. Use the Taylor-series method of order 4.

7. The function

$$\varphi(t) = \int_0^t \sin s^2\,ds$$

is known as a **Fresnel integral**. Prepare a table of this function on the interval [0, 10]. Use the Taylor-series method of order 5 with $h = 0.1$. If possible, obtain a computer plot of the function.

8. (Continuation) Show that the derivatives of the function $f(t) = \sin t^2$ are

$$f^{(n)}(t) = P_n(t)\sin t^2 + Q_n(t)\cos t^2$$

where P_n and Q_n are computed recursively as follows:

$$\begin{cases} P_0 = 1 \\ P_{n+1} = P_n' - 2tQ_n \end{cases} \qquad \begin{cases} Q_0 = 0 \\ Q_{n+1} = Q_n' + 2tP_n \end{cases}$$

Use this to generate a table of P_n and Q_n values for $n = 0$ to 6. Modify the code in the preceding computer problem to become order 7. Test, as in that computer problem, first with $h = 0.1$ and then with $h = 0.2$. Compare the value of $\varphi(10)$ obtained with $h = 0.2$ to the value obtained in the preceding computer problem.

9. The function

$$x(t) = \int_0^t (1 - k\sin^2\theta)^{1/2}\,d\theta$$

where k is a parameter in [0, 1], is an **elliptic integral of the second kind.** Make a table of this function, when $k = 1/2$, for the interval $0 \le t \le \pi/2$, using the Taylor-series method of order 3 with $h = 0.01$.

10. Solve the initial-value problem

$$\begin{cases} x' = -\dfrac{3t^2x + x^2}{2t^3 + 3tx} \\ x(1) = -2 \end{cases}$$

by the Taylor-series method of order 2 on the interval [0, 1] with steps $h = -0.01$. Use the implicit solution $t^3x^2 + tx^3 + 4 = 0$ as a check on the computed solution. Also verify the correctness of the implicit solution.

11. Prepare a table of a **dilogarithm** function

$$f(x) = -\int_0^x \frac{\ln(1-t)}{t}\,dt$$

on the interval $-2 \le x \le 0$ by solving an appropriate initial-value problem. (Compare with Problem **6.7.14**, p. 391.)

12. The integral $\int \sqrt{1 + x^3}\,dx$ is one that cannot be obtained by the methods of elementary calculus. (It is an **elliptic integral**.) Prepare a table of the function

$$f(x) = \int_0^x \sqrt{1 + t^3}\,dt$$

on the interval $0 \le x \le 5$ by solving a suitable initial-value problem. Use the Taylor-series method of order 3 with $h = 1/64$.

13. Consider the initial-value problem $x' = 1 - xt^{-1}$, $x(2) = 2$. Prove that

$$\begin{cases} x'' = (1 - 2x')t^{-1} \\ x^{(n)} = -nx^{(n-1)}t^{-1} \qquad (n \ge 3) \end{cases}$$

Write a computer program to solve this initial-value problem by the Taylor-series method of order 10. Test your program using $h = 1$ and the interval $[2, 20]$. What is the analytic solution of the problem? Compare it to your numerical solution. (See Conte and de Boor [1980].)

14. Write a computer program to solve the initial-value problem

$$\begin{cases} x' = t^2 + x^2 + 2tx \\ x(0) = 7 \end{cases}$$

using the Taylor-series method. A solution is wanted in the interval $-2 \le t \le 2$. Include terms up to and including h^3. Use step size 0.01.

15. Write a program to compute a table of values for the function

$$x(t) = \int_2^t \sin(u^3)\, du$$

We want to cover the interval $2 \le t \le 5$ in steps of 0.05. Do this by setting up an appropriate initial-value problem. Then carry out the numerical solution using the Taylor-series method of order 3. Estimate the accuracy of the resulting table.

16. Solve the integral equation in Problem **8.2.6** (p. 537) using the Taylor-series method of order 3 on the equivalent initial-value problem. Obtain the solution on $[0, 3]$ using steps of 0.01.

17. Using a symbol-manipulating program, solve the initial-value problem given by Equation (3) in the text, using the Taylor-series method of orders 4 and 20. Compare to the numerical results given in the text. Compare these two Taylor-series methods using various step sizes h. Determine how large h can be for the method of order 20 to obtain the same accuracy as the method of order 4 with $h = 0.01$.

18. (Continuation) Repeat for the initial-value problem in Computer Problem **8.2.6** (p. 537).

19. Reprogram the pseudocode (p. 531) by including E_4 given in Equation (5).

20. Solve numerically the initial-value problem

$$\begin{cases} (x')^2 - 2tx' - x\cos t = 0 \\ x(0) = 0 \end{cases}$$

on the interval $0 \le t \le 1$. (The trivial solution is not the one wanted here!) What would you have done if $(x')^2$ had been $(x')^3$?

21. Numerically solve Equation (6) and compare to the true solution.

22. Carry out the numerical solution of Equation (7) using the Taylor-series method of order 3 over the interval $[0, 1]$ with step size $h = 0.01$.

8.3 Runge-Kutta Methods

The Taylor-series method of the preceding section has the drawback of requiring some analysis prior to programming it. Thus, if we wish to use the fourth-order Taylor-series method on the general problem

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{1}$$

we have to determine formulas for x'', x''', and $x^{(4)}$ by successive differentiation in (1). Then these functions will have to be programmed.

The Runge-Kutta methods avoid this difficulty, although they do imitate the Taylor-series method by means of clever combinations of values of $f(t, x)$. We illustrate by deriving a second-order Runge-Kutta procedure.

Second-Order Runge-Kutta Method

Let us begin with the Taylor series for $x(t+h)$:

$$x(t+h) = x(t) + hx'(t) + \frac{h^2}{2!}x''(t) + \frac{h^3}{3!}x'''(t) + \cdots \tag{2}$$

From the differential equation, we have

$$\begin{aligned} x'(t) &= f \\ x''(t) &= f_t + f_x x' = f_t + f_x f \\ x'''(t) &= f_{tt} + f_{tx} f + (f_t + f_x f) f_x + f(f_{xt} + f_{xx} f) \\ &\vdots \end{aligned}$$

Here subscripts denote partial derivatives, and the chain rule of differentiation is used repeatedly. The first three terms in Equation (2) can be written now in the form

$$\begin{aligned} x(t+h) &= x + hf + \tfrac{1}{2}h^2(f_t + ff_x) + \mathcal{O}(h^3) \\ &= x + \tfrac{1}{2}hf + \tfrac{1}{2}h[f + hf_t + hff_x] + O(h^3) \end{aligned} \tag{3}$$

where x means $x(t)$, f means $f(t, x)$, and so on. We are able to eliminate the partial derivatives with the aid of the first few terms in the Taylor series in two variables (as given in Section 1.1, p. 11):

$$f(t+h, x+hf) = f + hf_t + hff_x + \mathcal{O}(h^2)$$

Equation (3) can be rewritten as

$$x(t+h) = x + \tfrac{1}{2}hf + \tfrac{1}{2}hf(t+h, x+hf) + \mathcal{O}(h^3)$$

Hence, the formula for advancing the solution is

$$x(t+h) = x(t) + \frac{h}{2}f(t, x) + \frac{h}{2}f\big(t+h, x+hf(t, x)\big)$$

or equivalently,

$$x(t+h) = x(t) + \tfrac{1}{2}(F_1 + F_2) \tag{4}$$

where

$$\begin{cases} F_1 = hf(t, x) \\ F_2 = hf(t+h, x+F_1) \end{cases}$$

This formula can be used repeatedly to advance the solution one step at a time. It is called a **second-order Runge-Kutta method**. It is also known as **Heun's method**.

In general, second-order Runge-Kutta formulas are of the form

$$x(t+h) = x + w_1 hf + w_2 hf(t + \alpha h, x + \beta hf) + \mathcal{O}(h^3) \tag{5}$$

where w_1, w_2, α, and β are parameters at our disposal. Equation (5) can be rewritten with the aid of the Taylor series in two variables as

$$x(t+h) = x + w_1 hf + w_2 h[f + \alpha h f_t + \beta h f f_x] + \mathcal{O}(h^3) \tag{6}$$

Comparing Equations (3) and (6), we see that we should impose these conditions:

$$\begin{cases} w_1 + w_2 = 1 \\ w_2\alpha = \frac{1}{2} \\ w_2\beta = \frac{1}{2} \end{cases} \tag{7}$$

One solution is $w_1 = w_2 = \frac{1}{2}$, $\alpha = \beta = 1$, which is the one corresponding to Heun's method in Equation (4). The system of Equations (7) has solutions other than this one, such as the one obtained by letting $w_1 = 0$, $w_2 = 1$, $\alpha = \beta = \frac{1}{2}$. The resulting formula from (5) is called the **modified Euler method**:

$$x(t+h) = x(t) + F_2$$

where

$$\begin{cases} F_1 = hf(t, x) \\ F_2 = hf(t + \frac{1}{2}h, x + \frac{1}{2}F_1) \end{cases}$$

Compare this to the standard Euler method, described in Section 8.2 (p. 534).

Fourth-Order Runge-Kutta Method

The higher-order Runge-Kutta formulas are very tedious to derive, and we shall not do so. The formulas are rather elegant, however, and are easily programmed once they have been derived. Here are the formulas for the *classical* **fourth-order Runge-Kutta method**:

$$x(t+h) = x(t) + \tfrac{1}{6}(F_1 + 2F_2 + 2F_3 + F_4) \tag{8}$$

where

$$\begin{cases} F_1 = hf(t, x) \\ F_2 = hf(t + \frac{1}{2}h, x + \frac{1}{2}F_1) \\ F_3 = hf(t + \frac{1}{2}h, x + \frac{1}{2}F_2) \\ F_4 = hf(t + h, x + F_3) \end{cases}$$

This is called a fourth-order method because it reproduces the terms in the Taylor series up to and including the one involving h^4. The error is therefore $\mathcal{O}(h^5)$. Exact expressions for the h^5 error term are available.

EXAMPLE 1 Give an algorithm incorporating the Runge-Kutta method of order 4 for solving the following initial-value problem:

$$\begin{cases} x' = t^{-2}(tx - x^2) \\ x(1) = 2 \end{cases} \tag{9}$$

on the interval $[1, 3]$, using steps of $h = 1/128$.

Solution Because this is a numerical experiment to gauge the effectiveness of the procedure, a problem has been chosen with a known analytic solution. The solution of (9) is given by $x(t) = (\frac{1}{2} + \ln t)^{-1}t$. The values of the error are printed by the computer program.

input $M \leftarrow 256;\ t \leftarrow 1.0;\ x \leftarrow 2.0;\ h \leftarrow 0.0078125$
define $f(t, x) = (tx - x^2)/t^2$
define $u(t) = t/(\frac{1}{2} + \ln t)$
$e \leftarrow |u(t) - x|$
output $0, t, x, e$
for $k = 1$ **to** M **do**
    $F_1 \leftarrow hf(t, x)$
    $F_2 \leftarrow hf(t + \frac{1}{2}h, x + \frac{1}{2}F_1)$
    $F_3 \leftarrow hf(t + \frac{1}{2}h, x + \frac{1}{2}F_2)$
    $F_4 \leftarrow hf(t + h, x + F_3)$
    $x \leftarrow x + (F_1 + 2F_2 + 2F_3 + F_4)/6$
    $t \leftarrow t + h$
    $e \leftarrow |u(t) - x|$
    output k, t, x, e
end do

Some of the output from a computer program based on this algorithm follows:

k	t	x	e
0	1.00000	2.00000	
1	1.00781	1.98473	1.19×10^{-07}
2	1.01563	1.97016	0.
3	1.02344	1.95623	0.
4	1.03125	1.94293	1.19×10^{-07}
5	1.03906	1.93020	1.19×10^{-07}
6	1.04688	1.91802	0.
7	1.05469	1.90637	1.19×10^{-07}
8	1.06250	1.89521	1.19×10^{-07}
9	1.07031	1.88452	0.
⋮	⋮	⋮	⋮

k	t	x	e
249	2.94531	1.86387	7.15×10^{-07}
250	2.95313	1.86569	5.96×10^{-07}
251	2.96094	1.86750	7.15×10^{-07}
252	2.96875	1.86932	5.96×10^{-07}
253	2.97656	1.87115	4.77×10^{-07}
254	2.98438	1.87297	5.96×10^{-07}
255	2.99219	1.87480	5.96×10^{-07}
256	3.00000	1.87663	5.96×10^{-07}

■

Errors

We turn now to a discussion of **truncation error** in the Runge-Kutta procedure. Roughly described, the local truncation error is the error that arises in each step simply because our methods cannot take into account all the terms of a Taylor series. This error is inevitable and will be present even if the calculations are executed with *infinite* precision. In the case of the fourth-order Runge-Kutta method, the formulas involve a local truncation error of $\mathcal{O}(h^5)$. This order of error corresponds exactly to the error in the Taylor-series method when terms up to h^4 are included in Taylor's formula.

At the very first step in the fourth-order Runge-Kutta procedure, a value of $x(t_0 + h)$ is computed by the algorithm. On the other hand, there is a correct solution, $x^*(t_0 + h)$, which we shall not know. The local truncation error in this step is, by definition,

$$x^*(t_0 + h) - x(t_0 + h)$$

The theory of the Runge-Kutta algorithm indicates that this truncation error behaves like Ch^5 for small values of h. Here C is a number independent of h but dependent on t_0 and on the function x^*. To estimate Ch^5, we assume that C does *not* change as t changes from t_0 to $t_0 + h$. Let v be the value of the approximate solution at $t_0 + h$, obtained by taking one step of length h from t_0. Let u be the approximate solution at $t_0 + h$, obtained by taking *two* steps of size $h/2$ from t_0. These are both *computable*. By the assumption made, we have

$$x^*(t_0 + h) = v + Ch^5$$
$$x^*(t_0 + h) = u + 2C(h/2)^5$$

By subtraction, we obtain from these two equations:

$$\text{Local truncation error } = Ch^5 = (u - v)/(1 - 2^{-4})$$

Thus, the local truncation error is approximately $u - v$.

In a computer realization of the Runge-Kutta method, the approximate truncation error can be occasionally monitored, by computing $|u - v|$, to ensure that it remains below a specified tolerance. If it does not, the step size can be decreased (usually halved) to improve the local truncation error. On the other hand, if the local truncation error is far below a permitted threshold, then the step size can be doubled.

As we saw in the derivation of the Runge-Kutta method of order 2, a number of parameters must be selected. A similar selection process occurs in establishing higher-order Runge-Kutta methods. Consequently, there is not just *one* Runge-Kutta method of each order, but a family of methods. As shown in the following table, the number of required function evaluations increases more rapidly than the order of the Runge-Kutta methods:

Number of function evaluations	1	2	3	4	5	6	7	8
Maximum order of Runge-Kutta method	1	2	3	4	4	5	6	6

Unfortunately, this makes the higher-order Runge-Kutta methods less attractive than the classical fourth-order method, because they are more expensive to use.

Adaptive Runge-Kutta-Fehlberg Method

In an effort to devise a procedure for automatically adjusting the step size in the Runge-Kutta method, Fehlberg [1969] turned to a fourth-order method with five function evaluations and a fifth-order method with six function evaluations. At first glance, this does not seem to be promising because his fourth-order method required more function evaluations than the classical method. Moreover, these two methods together would require a total of eleven function evaluations. However, Fehlberg was able to select the parameters in these methods to obtain two formulas of different orders with the same points for the function evaluations. Hence, only *six* function evaluations are required, and as a bonus we obtain an estimate for the local truncation error on which to base the control of the step size. The resulting **Runge-Kutta-Fehlberg method** is of order 5 and utilizes two formulas, having orders 4 and 5. These formulas give different approximate values of the solution, and we denote them by $x(t+h)$ and $\overline{x}(t+h)$:

$$x(t+h) = x(t) + \sum_{i=1}^{6} a_i F_i \tag{10}$$

$$\overline{x}(t+h) = x(t) + \sum_{i=1}^{6} b_i F_i \tag{11}$$

The quantities F_i are computed from formulas of the type

$$F_i = hf\left(t + c_i h, x + \sum_{j=1}^{i-1} d_{ij} F_j\right) \qquad (1 \le i \le 6) \tag{12}$$

Formula (10) is of the fifth order, and Formula (11) is of the fourth order. The difference, $e = x(t+h) - \overline{x}(t+h)$, can be interpreted as an estimate of the local truncation error associated with the less accurate Formula (11). Thus, e can be used to monitor the step size. Because Formula (10) is more accurate, it is used to provide the output in the algorithm. Note that

$$e = x(t+h) - \overline{x}(t+h) = \sum_{i=1}^{6} (a_i - b_i) F_i \tag{13}$$

The values of the coefficients are given in the following table:

i	a_i	$a_i - b_i$	c_i	d_{ij}
1	$\frac{16}{135}$	$\frac{1}{360}$	0	0
2	0	0	$\frac{1}{4}$	$\frac{1}{4}$
3	$\frac{6656}{12825}$	$-\frac{128}{4275}$	$\frac{3}{8}$	$\frac{3}{32}, \frac{9}{32}$
4	$\frac{28561}{56430}$	$-\frac{2197}{75240}$	$\frac{12}{13}$	$\frac{1932}{2197}, -\frac{7200}{2197}, \frac{7296}{2197}$
5	$-\frac{9}{50}$	$\frac{1}{50}$	1	$\frac{439}{216}, -8, \frac{3680}{513}, -\frac{845}{4104}$
6	$\frac{2}{55}$	$\frac{2}{55}$	$\frac{1}{2}$	$-\frac{8}{27}, 2, -\frac{3544}{2565}, \frac{1859}{4104}, -\frac{11}{40}$

An adaptive routine using the preceding formulas attempts to keep the magnitude of the local truncation error, e, below a certain prescribed tolerance, δ. The solution is sought in a certain interval $[a, b]$, and an initial value $x(a) = \alpha$ is also given. An upper bound, M, on the number of steps also would be prescribed. Notice that the procedure is *conservative* because the local truncation error in a fourth-order method is used to control the step size while the numerical solution is actually being computed with a fifth-order formula. An algorithm to carry out the procedure is given here. Since the local truncation error, as estimated by $x(t) - \overline{x}(t)$, is Ch^5, it seems sensible to double the step size when the doubling would lead to $C(2h)^5 < \delta/4$. Thus, we double h when the apparent local truncation error is less than $\delta/128$.

Another formula in common use to control the error per step is:

$$h \leftarrow 0.9h\big[\delta/|e|\big]^{1/(1+p)}$$

where p is the order of the first formula in the pair of Runge-Kutta formulas. (See, for example, Hull, Enright, Fellen, and Sedgwick [1972] and Shampine, Watts, and Davenport [1976].) The same formula is used for both increasing and decreasing the step size h. In one case, a rejected step size is recomputed, whereas in the other a new step size is established.

Here is an **adaptive Runge-Kutta-Fehlberg algorithm**:

input $a, \alpha, b, h, \delta, M$
$t \leftarrow a;\ x \leftarrow \alpha$
`iflag` $\leftarrow 1$
while $k < M$ **do**
  $d \leftarrow b - t$
  if $|d| \leq |h|$ **then**
    `iflag` $\leftarrow 0$
    $h \leftarrow d$
  end if
  $s \leftarrow t$
  $y \leftarrow x$
  compute $F_1, F_2, \dots, F_6$ [Equation (12)]
  compute x [Equation (10)]

```
compute e        [Equation (13)]
    t ← t + h
    output k, t, x, e
if iflag = 0 then stop
if |e| ≥ δ then
        t ← s
        h ← h/2
        x ← y
        k ← k − 1
else
    if |e| < δ/128 then h ← 2h
end if
end do
```

Embedded Runge-Kutta procedures are composed of a pair of formulas of orders p and q (usually $q = p + 1$) that share the same function evaluation points. Generally they provide an efficient technique for solving nonstiff initial-value problems. Many higher-order embedded Runge-Kutta formulas have been derived in recent years. Examples of some of these are given in the problems. Although higher-order Runge-Kutta methods with complicated coefficients have been derived, the classic fourth-order formula remains the most popular except for adaptive schemes. Additional information on Runge-Kutta methods can be found in numerous sources—for example, Butcher [1987], Fehlberg [1969], Gear [1971], Jackson, Enright, and Hull [1978], Prince and Dormand [1981], Shampine and Gordon [1975], Thomas [1986], and Verner [1978].

PROBLEMS 8.3

1. Write out the second-order Runge-Kutta formulas when $\alpha = 2/3$.
2. If an initial-value problem involving the differential equation $x + 2tx' + xx' = 3$ is to be solved using a Runge-Kutta method, what function must be programmed?
3. Prove that the Runge-Kutta formula (8) is of order 4 in the special case that $f(t, x)$ is independent of x. Show that in this case the Runge-Kutta formula is equivalent to Simpson's rule. (See Equation (6) in Section 7.2, p. 483.)
4. Derive the **modified Euler's method**,
$$x(t+h) = x(t) + hf\big(t + \tfrac{1}{2}h, x(t) + \tfrac{1}{2}hf\big(t, x(t)\big)\big)$$
by performing Richardson's extrapolation on Euler's method using step sizes h and $h/2$. *Hint:* Assume the error term is Ch^2.
5. Derive the **third-order Runge-Kutta formula**
$$x(t+h) = x(t) + \tfrac{1}{9}(2F_1 + 3F_2 + 4F_3)$$
where
$$\begin{cases} F_1 = hf(t, x) \\ F_2 = hf(t + \tfrac{1}{2}h, x + \tfrac{1}{2}F_1) \\ F_3 = hf(t + \tfrac{3}{4}h, x + \tfrac{3}{4}F_2) \end{cases}$$

Show that it agrees with the Taylor-series method of order 3 for the differential equation $x' = x + t$.

6. Prove that when the fourth-order Runge-Kutta method is applied to the problem $x' = \lambda x$, the formula for advancing this solution will be

$$x(t+h) = \left[1 + h\lambda + \tfrac{1}{2}h^2\lambda^2 + \tfrac{1}{6}h^3\lambda^3 + \tfrac{1}{24}h^4\lambda^4\right]x(t)$$

7. (Continuation) Prove that the local truncation error in the preceding problem is $\mathcal{O}(h^5)$.

COMPUTER PROBLEMS 8.3

1. Write a computer program to solve an initial-value problem $x' = f(t, x)$ with $x(t_0) = x_0$ on an interval $t_0 \le t \le t_m$ or $t_m \le t \le t_0$. Use the fourth-order Runge-Kutta method. Test it on this example:

$$\begin{cases} (e^t + 1)x' + xe^t - x = 0 \\ x(0) = 3 \end{cases}$$

Determine the *analytic* solution and compare it to the computed solution on the interval $-2 \le t \le 0$. Use $h = -0.01$.

2. Using various values of λ, such as 5, -5, or -10, numerically solve the following initial-value problem using the fourth-order Runge$-$Kutta method:

$$\begin{cases} x' = \lambda x + \cos t - \lambda \sin t \\ x(0) = 0 \end{cases}$$

Compare the numerical solution to the analytic solution on the interval $[0, 5]$. Use step size $h = 0.01$. What effect does λ have on the numerical accuracy?

3. Program and test the Runge-Kutta-Fehlberg algorithm described in the text. For test cases, use the examples in Computer Problems **8.2.7**, **11**, **15** (pp. 537–539).

4. Write and execute a program to solve this initial-value problem:

$$\begin{cases} x' = e^{xt} + \cos(x - t) \\ x(1) = 3 \end{cases}$$

Use the fourth-order Runge-Kutta formulas with $h = 0.01$. Stop the computation just *before* the solution overflows.

5. Solve $x' = x^2$, $x(0) = 1$ on the interval $[0, 2]$ using the adaptive Runge-Kutta-Fehlberg procedure. Show that the true solution is $x(t) = 1/(1-t)$. What happens in the algorithm near the discontinuity at $t = 1$?

6. Numerically compare the following **fourth-order Runge-Kutta-Gill method** with the classical Runge-Kutta method:

$$x(t+h) = x(t) + \tfrac{1}{6}\left[F_1 - \left(2 + \sqrt{2}\right)F_2 + \left(2 + \sqrt{2}\right)F_3 + F_4\right]$$

where

$$\begin{cases} F_1 = hf(t, x) \\ F_2 = hf\left(t + \tfrac{1}{2}h, x + \tfrac{1}{2}F_1\right) \\ F_3 = hf\left(t + \tfrac{1}{2}h, x + \tfrac{1}{2}\left(\sqrt{2} - 1\right)F_1 + \tfrac{1}{2}\left(2 - \sqrt{2}\right)F_2\right) \\ F_4 = hf\left(t + h, x - \tfrac{1}{2}\sqrt{2}F_2 + \tfrac{1}{2}\left(2 + \sqrt{2}\right)F_3\right) \end{cases}$$

7. Numerically compare the following **fifth-order Runge-Kutta method** with the classical Runge-Kutta method on a problem with a known solution:

$$x(t+h) = 7x(t) + \tfrac{1}{24}F_1 + \tfrac{5}{48}F_4 + \tfrac{27}{56}F_5 + \tfrac{125}{336}F_6$$

where

$$\begin{cases} F_1 = hf(t,x) \\ F_2 = hf\left(t + \tfrac{1}{2}h,\ x + \tfrac{1}{2}F_1\right) \\ F_3 = hf\left(t + \tfrac{1}{2}h,\ x + \tfrac{1}{4}F_1 + \tfrac{1}{4}F_2\right) \\ F_4 = hf\left(t + h,\ x - F_2 + 2F_3\right) \\ F_5 = hf\left(t + \tfrac{2}{3}h,\ x + \tfrac{7}{27}F_1 + \tfrac{10}{27}F_2 + \tfrac{1}{27}F_4\right) \\ F_6 = hf\left(t + \tfrac{1}{5}h,\ x + \tfrac{28}{625}F_1 - \tfrac{1}{5}F_2 + \tfrac{546}{625}F_3 + \tfrac{54}{625}F_4 - \tfrac{378}{625}F_5\right) \end{cases}$$

8. The **Runge-Kutta-Merson method** of order 5 is given by the coefficients in the following table. Write and test an adaptive routine based on this method.

i	a_i	$a_i - b_i$	c_i	d_{ij}
1	$\frac{1}{6}$	$\frac{1}{15}$	0	0
2	0	0	$\frac{1}{3}$	$\frac{1}{3}$
3	0	$-\frac{3}{10}$	$\frac{1}{3}$	$\frac{1}{6}, \frac{1}{6}$
4	$\frac{2}{3}$	$\frac{4}{15}$	$\frac{1}{2}$	$\frac{1}{8}, 0, \frac{3}{8}$
5	$\frac{1}{6}$	$-\frac{1}{30}$	1	$\frac{1}{2}, 0, -\frac{3}{2}, 2$

9. Write and test an adaptive routine based on the **fifth-order Runge-Kutta-Verner method** given by the coefficients in the following table (see Verner [1978]):

i	a_i	$a_i - b_i$	c_i	d_{ij}
1	$\frac{3}{80}$	$\frac{33}{640}$	0	0
2	0	0	$\frac{1}{18}$	$\frac{1}{18}$
3	$\frac{4}{25}$	$-\frac{132}{325}$	$\frac{1}{6}$	$-\frac{1}{12}, \frac{1}{4}$
4	$\frac{243}{1120}$	$\frac{891}{2240}$	$\frac{2}{9}$	$-\frac{2}{81}, \frac{4}{27}, \frac{8}{81}$
5	$\frac{77}{160}$	$-\frac{33}{320}$	$\frac{2}{3}$	$\frac{40}{33}, -\frac{4}{11}, -\frac{56}{11}, \frac{54}{11}$
6	$\frac{73}{700}$	$-\frac{73}{700}$	1	$-\frac{369}{73}, \frac{72}{73}, \frac{5380}{219}, -\frac{12285}{584}, \frac{2695}{1752}$
7	0	$\frac{891}{8329}$	$\frac{8}{9}$	$-\frac{8716}{891}, \frac{656}{297}, \frac{39520}{891}, -\frac{416}{11}, \frac{52}{27}$
8	0	$\frac{2}{35}$	1	$\frac{3015}{256}, -\frac{9}{4}, -\frac{4219}{78}, \frac{5985}{128}, -\frac{539}{394}, \frac{693}{3328}$

10. A pair of **embedded Runge-Kutta formulas** of orders 6 and 5 are given by the coefficients in the following table (see Prince and Dormand [1981]). Write and test an adaptive routine using them.

i	a_i	b_i	c_i	d_{ij}
1	$\frac{821}{10800}$	$\frac{61}{864}$	0	0
2	0	0	$\frac{1}{10}$	$\frac{1}{10}$
3	$\frac{19683}{71825}$	$\frac{98415}{321776}$	$\frac{2}{9}$	$-\frac{2}{81}, \frac{20}{81}$
4	$\frac{175273}{912600}$	$\frac{16807}{146016}$	$\frac{3}{7}$	$\frac{615}{1372}, -\frac{270}{343}, \frac{1053}{1372}$
5	$\frac{395}{3672}$	$\frac{1375}{7344}$	$\frac{3}{5}$	$\frac{3243}{5500}, -\frac{54}{55}, \frac{50949}{71500}, \frac{4998}{17875}$
6	$\frac{785}{2704}$	$\frac{1375}{5408}$	$\frac{4}{5}$	$-\frac{26492}{37125}, \frac{72}{55}, \frac{2808}{23375}, -\frac{24206}{37125}, \frac{338}{459}$

7	$\frac{3}{50}$	$-\frac{37}{1120}$	1	$\frac{5561}{2376}, -\frac{35}{11}, -\frac{24117}{31603}, \frac{899983}{200772}, -\frac{5225}{1836}, \frac{3925}{4056}$
8	0	$\frac{1}{10}$	1	$\frac{465467}{266112}, -\frac{2945}{1232}, -\frac{5610201}{14158144}, \frac{10513573}{3212352}, -\frac{424325}{205632}, \frac{376225}{454272}, 0$

11. Consider the initial-value problem $x' = -kx$, $x(0) = 1$ on the interval [0, 1] with various values for the **decay constant** k. Show that the analytical solution is $x(t) = e^{-kt}$. For $k = 5$, compare the behavior of Euler's method and the Runge-Kutta-Gill method (given in Computer Problem **8.3.6**, p. 547) using various values for the step size h, such as 0.2, 0.4, 0.6, 0.8, and 1.0. Repeat using $k = 25$. For Euler's method the condition $0 \le hk \le 2$ must be satisfied in order to solve this problem correctly, whereas $0 \le hk < 2.8$ is needed for the Runge-Kutta-Gill method. These are the **regions of stability** for this problem using these methods. For a particular problem and method, we can either adjust the step size h to try to stay within the region of stability or alternatively use a higher-order method with a larger region of stability. As discussed by Thomas [1986], both of these methods become unworkable for large values of k.

8.4 Multistep Methods

The Taylor-series method and the Runge-Kutta method for solving the initial-value problem are **single-step methods** because they do not use any knowledge of prior values of $x(t)$ when the solution is advanced from t to $t + h$. If $t_0, t_1, t_2, \ldots, t_i$ are steps along the t-axis, then x_{i+1} (the approximate value of $x(t_{i+1})$) depends only on x_i, and knowledge of the approximate values $x_{i-1}, x_{i-2}, \ldots, x_0$ is *not* used.

More efficient procedures can be devised if some prior values of the solution are utilized at each step. The principle involved here is as follows: We are numerically solving the initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{1}$$

We have prescribed steps $t_0, t_1, t_2, \ldots, t_n$ on the t-axis. (They will not always be equally spaced.) If the true solution is denoted by $x(t)$, then integrating Equation (1) gives us

$$\int_{t_n}^{t_{n+1}} x'(t)\,dt = x(t_{n+1}) - x(t_n)$$

and then

$$x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(t, x(t))\,dt \tag{2}$$

The integral on the right can be approximated by a numerical quadrature scheme, and the result will be a formula for generating the approximate solution step-by-step.

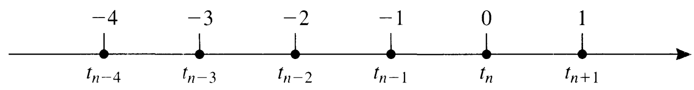

FIGURE 8.1
Points involved in order 5 Adams-Bashforth formula (4).

Adams-Bashforth Formula

Suppose the resulting formula is of the following type:

$$x_{n+1} = x_n + af_n + bf_{n-1} + cf_{n-2} + \cdots \tag{3}$$

where f_i denotes $f(t_i, x_i)$. An equation of this type is called an **Adams-Bashforth formula**. Here is the **Adams-Bashforth formula of order 5**, based on equally spaced points $t_i = t_0 + ih$ for $0 \le i \le n$:

$$x_{n+1} = x_n + \frac{h}{720}\big[1901 f_n - 2774 f_{n-1} + 2616 f_{n-2} - 1274 f_{n-3} + 251 f_{n-4}\big] \tag{4}$$

How were these coefficients determined? We start with the intention of approximating the integral in Equation (2) as

$$\int_{t_n}^{t_{n+1}} f\big(t, x(t)\big)\, dt \approx h\big[Af_n + Bf_{n-1} + Cf_{n-2} + Df_{n-3} + Ef_{n-4}\big] \tag{5}$$

The coefficients A, B, C, D, and E are determined by requiring that Equation (5) be exact whenever the integrand is a polynomial of degree ≤ 4. By Problem **8.3.6** (p. 547), there is no loss of generality in assuming $t_n = 0$ and $h = 1$, as sketched in Figure 8.1.

We take as a basis for Π_4 the following five polynomials:

$$\begin{aligned}
p_0(t) &= 1\\
p_1(t) &= t\\
p_2(t) &= t(t+1)\\
p_3(t) &= t(t+1)(t+2)\\
p_4(t) &= t(t+1)(t+2)(t+3)
\end{aligned}$$

When these are substituted in the equation

$$\int_0^1 p_n(t)\, dt = Ap_n(0) + Bp_n(-1) + Cp_n(-2) + Dp_n(-3) + Ep_n(-4)$$

we obtain five equations for the determination of the coefficients A, B, C, D, and E. This system of equations is

$$\begin{cases} A + B + C + D + E = 1 \\ -B - 2C - 3D - 4E = 1/2 \\ 2C + 6D + 12E = 5/6 \\ -6D - 24E = 9/4 \\ 24E = 251/30 \end{cases} \tag{6}$$

The coefficients of Formula (4) are obtained by back substitution.

The procedure just illustrated is called the **method of undetermined coefficients**. In principle, it can be used to obtain similar formulas of higher order and in a variety of other situations. (See Section 7.2, p. 482.)

Alternatively, we could obtain the values of these unknown coefficients directly from Equation (4) by letting $x = p_n$ and using $f = p_n'$ because $x' = f$. A system of equations similar to (6) is obtained.

Adams-Moulton Formula

In numerical practice, the Adams-Bashforth formulas are rarely used by themselves. They are used in conjunction with other formulas to enhance the precision. To see how this might be possible, we return to Equation (2) and suppose that we use a numerical quadrature formula that involves f_{n+1}. Then Equation (3) will have the form

$$x_{n+1} = x_n + af_{n+1} + bf_n + cf_{n-1} + \cdots \tag{7}$$

Here is a formula of this type, known as the **Adams-Moulton formula of order 5**:

$$x_{n+1} = x_n + \frac{h}{720}\left[251f_{n+1} + 646f_n - 264f_{n-1} + 106f_{n-2} - 19f_{n-3}\right] \tag{8}$$

This formula also can be derived using the method of undetermined coefficients. Notice that it cannot be used directly to advance the solution because x_{n+1} occurs on both sides of the equation! Remember that f_i stands for $f(t_i, x_i)$, and so the term involving f_{n+1} can be computed only after x_{n+1} is known. However, a very satisfactory algorithm, called a **predictor-corrector method**, uses the Adams-Bashforth formula (4) to **predict** a tentative value for x_{n+1}, say x^*_{n+1}, and then the Adams-Moulton formula (8) to compute a *corrected* value of x_{n+1}. So in Formula (8), we evaluate f_{n+1} as $f(t_{n+1}, x^*_{n+1})$, using the predicted value x^*_{n+1} obtained from Formula (4).

In using this predictor-corrector method, a special procedure must be employed to start the method because initially only x_0 is known. Of course, a Runge-Kutta method is ideal for obtaining x_1, x_2, x_3, x_4. Usually formulas of the same order are used together. Therefore, fifth-order Runge-Kutta methods, such as those in Computer Problems **8.3.8–10** (pp. 548–549), could be used in combination with the Adams-Bashforth formula (4) and the Adams-Moulton formula (8).

Still another method can be used to obtain the value of x_{n+1} in Equation (8). After all, that equation states that x_{n+1} is a fixed point of a certain mapping, namely the one defined by

$$\phi(z) = \tfrac{251}{720}h\, f(t_{n+1}, z) + C$$

where C is composed of all the other terms in Equation (8). Therefore, the method of functional iteration suggests itself as a means to compute x_{n+1}. The theory of functional iteration, as developed in Section 3.4 (p. 100), tells us that the sequence determined by the equation

$$z_{k+1} = \phi(z_k) \qquad (k \geq 0)$$

will converge to a fixed point of ϕ under appropriate hypotheses. In particular, Problem **3.4.31** (p. 108) can be used here. If ξ is the fixed point of ϕ (and thus the value of x_{n+1} that we seek), then we should start our iteration with a point z_0 in an interval centered at ξ in which $|\phi'(z)| < 1$. It is necessary to assume that ϕ' is continuous. In the case at hand,

$$\phi'(z) = \frac{251}{720} h \frac{\partial f(t_{n+1}, z)}{\partial z}$$

This can be made as small as we please by decreasing the step size, h. In practice, only one or two steps are required in this iteration to find the value of x_{n+1}.

Analysis of Linear Multistep Methods

In the remainder of this section, we take up the theory of linear multistep methods in general. The format of any such method is

$$a_k x_n + a_{k-1} x_{n-1} + \cdots + a_0 x_{n-k} = h\big[b_k f_n + b_{k-1} f_{n-1} + \cdots + b_0 f_{n-k}\big] \qquad \textbf{(9)}$$

This is called a **k-step method**. The coefficients a_i and b_i are given. As before, x_i denotes an approximation to the solution at t_i, with $t_i = t_0 + ih$. The letter f_i denotes $f(t_i, x_i)$. Formula (9) is used to compute x_n, assuming that $x_0, x_1, \ldots, x_{n-1}$ are already known. Thus, we may assume that $a_k \neq 0$. The coefficient b_k can be 0. If $b_k = 0$, the method is said to be **explicit**. In this case, x_n can be computed directly from Equation (9) in an elementary fashion. If, however, $b_k \neq 0$, then on the right the term f_n involves the unknown x_n, and Equation (9) implicitly determines x_n. The method is then said to be **implicit**.

The accuracy of a numerical solution of a differential equation is largely determined by the **order** of the algorithm used. The order indicates how many terms in a Taylor-series solution are being simulated by the method. For example, the Adams-Bashforth method in Equation (4) is said to be of order 5 because it produces approximately the same accuracy as the Taylor-series method with terms h, h^2, h^3, h^4, and h^5. The error can then be expected to be $\mathcal{O}(h^6)$ in each step of a solution using Equation (4). The intuitive idea of order will be made more precise in the following paragraphs.

In correspondence with the multistep method in Equation (9), we define a linear functional L by writing

$$Lx = \sum_{i=0}^{k} \big[a_i x(ih) - hb_i x'(ih)\big]a \qquad \textbf{(10)}$$

Here we let $k = n$ to simplify the notation and assume that the first value in Equation (9) begins at $t = 0$ rather than at $t = n - k$. The operation Lx can be applied to any function x that is differentiable. However, in the following analysis, we assume that x is represented by its Taylor series at $t = 0$. By using the Taylor series for x, one can express L in this form:

$$Lx = d_0 x(0) + d_1 h x'(0) + d_2 h^2 x''(0) + \cdots \tag{11}$$

To compute the coefficients, d_i, in Equation (11), we write the Taylor series for x and x':

$$x(ih) = \sum_{j=0}^{\infty} \frac{(ih)^j}{j!} x^{(j)}(0)$$

$$x'(ih) = \sum_{j=0}^{\infty} \frac{(ih)^j}{j!} x^{(j+1)}(0)$$

These series are then substituted into Equation (10). Rearranging the result in powers of h, we obtain an equation having the form of (11), with these values of d_i:

$$\begin{cases} d_0 = \sum_{i=0}^{k} a_i \\ d_1 = \sum_{i=0}^{k} (ia_i - b_i) \\ d_2 = \sum_{i=0}^{k} \left(\frac{1}{2} i^2 a_i - i b_i\right) \\ \quad \vdots \\ d_j = \sum_{i=0}^{k} \left(\frac{i^j}{j!} a_i - \frac{i^{j-1}}{(j-1)!} b_i\right) \qquad (j \geq 1) \end{cases} \tag{12}$$

Here we use $0! = 1$ and $i^0 = 1$, of course.

THEOREM 1 Theorem on Multistep Method Properties

These three properties of the multistep method (9) *are equivalent:*

1. $d_0 = d_1 = \cdots = d_m = 0$
2. $Lp = 0$ *for each polynomial* p *of degree* $\leq m$.
3. Lx *is* $\mathcal{O}(h^{m+1})$ *for all* $x \in C^{m+1}$.

Proof If Property **1** is true, then Equation (11) has the form

$$Lx = d_{m+1} h^{m+1} x^{(m+1)}(0) + \cdots \tag{13}$$

If x is a polynomial of degree $\leq m$, then $x^{(j)}(t) = 0$ for all $j > m$, and thus $Lx = 0$ from Equation (13). So Property **1** implies Property **2**.

Assume that Property **2** is true. If $x \in C^{m+1}$, then by Taylor's Theorem we can write $x = p + r$, where p is a polynomial of degree $\leq m$ and r is a function whose first m derivatives vanish at 0. Since $Lp = 0$, Equation (11) gives us

$$Lx = Lr = d_{m+1}h^{m+1}r^{(m+1)}(0) = \mathcal{O}(h^{m+1})$$

and Property **2** implies Property **3**. Finally, assume that Property **3** is true. Then in Equation (11), we must have the condition $d_0 = d_1 = \cdots = d_m = 0$. Hence, Property **3** implies Property **1**. ■

The **order** of the multistep method in Equation (9) is the unique natural number m such that

$$d_0 = d_1 = \cdots = d_m = 0 \neq d_{m+1}$$

EXAMPLE 1 What is the order of the method described by the following equation?

$$x_n - x_{n-2} = \tfrac{1}{3}h(f_n + 4f_{n-1} + f_{n-2})$$

Solution The vector (a_0, a_1, a_2) is $(-1, 0, 1)$, and the vector (b_0, b_1, b_2) is $(\frac{1}{3}, \frac{4}{3}, \frac{1}{3})$. Thus, the d_i are

$$\begin{aligned}
d_0 &= a_0 + a_1 + a_2 = 0 \\
d_1 &= -b_0 + (a_1 - b_1) + (2a_2 - b_2) = 0 \\
d_2 &= \left(\tfrac{1}{2}a_1 - b_1\right) + (2a_2 - 2b_2) = 0 \\
d_3 &= \left(\tfrac{1}{6}a_1 - \tfrac{1}{2}b_1\right) + \left(\tfrac{4}{3}a_2 - 2b_2\right) = 0 \\
d_4 &= \left(\tfrac{1}{24}a_1 - \tfrac{1}{6}b_1\right) + \left(\tfrac{2}{3}a_2 - \tfrac{4}{3}b_2\right) = 0 \\
d_5 &= \left(\tfrac{1}{120}a_1 - \tfrac{1}{24}b_1\right) + \left(\tfrac{4}{15}a_2 - \tfrac{2}{3}b_2\right) = -\tfrac{1}{90}
\end{aligned}$$

The order of the method is 4. ■

Other characteristics being equal, we might prefer a method of high order to one of low order. If we desire to produce a k-step method of the form (9) of order $2k$, we simply write down the $2k + 1$ equations

$$d_0 = d_1 = \cdots = d_{2k} = 0$$

By Equation (12), this is a system of $2k + 1$ linear homogeneous equations in the $2k + 2$ unknowns a_i and b_i $(0 \leq i \leq k)$. By elementary linear algebra, this system has a nontrivial solution; Dahlquist [1956] proved that there is a solution with $a_k \neq 0$ (which is necessary in the method). We shall see in Section 8.5 (p. 558), however, that some characteristics of a multistep method other than its order *must* be taken into account. Chief among these is *stability*, defined in that section. Dahlquist also proved that a stable k-step method, as in Equation (9), cannot have an order greater than $k + 2$.

PROBLEMS 8.4

1. Derive the first-order (one-step) Adams-Moulton formula and verify that it is equivalent to the trapezoid rule. (See Section 7.2, p. 481.)

2. Verify the correctness of the system of equations in (6).

3. Use the method of undetermined coefficients to derive Equation (8).

4. Use the method of undetermined coefficients to derive the **fourth-order Adams-Bashforth formula**

$$x_{n+1} = x_n + \frac{h}{24}[55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}]$$

5. Derive the **fourth-order Adams-Moulton formula**

$$x_{n+1} = x_n + \frac{h}{24}[9f_{n+1} + 19f_n - 5f_{n-1} + f_{n-2}]$$

6. Prove that if every element of Π_m is correctly integrated by the formula

$$\int_0^1 f(x)\,dx \approx \sum_{i=-n}^{n} A_i f(i)$$

then the same is true of the formula

$$\int_{t_0}^{t_0+h} f(x)\,dx \approx h \sum_{i=-n}^{n} A_i f(t_0 + ih)$$

7. Derive the **second-order Adams-Bashforth formula**

$$x_{n+1} = x_n + h\left[\tfrac{3}{2}f_n - \tfrac{1}{2}f_{n-1}\right]$$

8. **a.** Using the method of undetermined coefficients, determine A and B in the following Adams-Bashforth formula. (Do not change to a corresponding numerical integration formula.)

$$x_{n+1} = x_n + h[Af_n + Bf_{n-1}]$$

   b. Repeat Part **a** by changing it to a corresponding numerical integration formula.

   c. Repeat the problem by integrating the Newton interpolation polynomial based on the nodes x_n and x_{n-1}.

9. **a.** Use the method of undetermined coefficients to find A and B in the implicit Adams-Moulton formula of second order

$$x_{n+1} = x_n + h[Af_{n+1} + Bf_n]$$

   b. Use a numerical integration formula to derive the formula.

   c. Use an interpolation formula to derive the formula.

10. **a.** Use the method of undetermined coefficients to derive a multistep formula of the form

$$x_{n+1} = x_n + h[Af_{n+1} + Bf_n + Cf_{n-1}]$$

   b. Repeat Part **a** for

$$x_{n+1} = x_n + h[Af_n + Bf_{n-1} + Cf_{n-2}]$$

11. Derive an implicit multistep formula based on Simpson's rule (involving uniformly spaced points x_{n-1}, x_n, x_{n+1}) for numerically solving the ordinary differential equation $x' = f$.

12. The formula

$$x_{n+1} = (1 - A)x_n + Ax_{n-1} + \frac{h}{12}\left[(5 - A)x'_{n+1} + 8(1 + A)x'_n + (5A - 1)x'_{n-1}\right]$$

is known to be exact for all polynomials of degree m or less for all A. Determine A so that it will be exact for all polynomials of degree $m + 1$. Find A and m.

13. Compute the coefficients in a multistep method of the form

$$x_{n+1} = x_n + h[Af_n + Bf_{n-2} + Cf_{n-4}]$$

The formula should correctly integrate an equation $x' = f(t, x)$ when the right-hand side is of the form $f(t, x) = a + bt + ct^2$.

14. Find a method of order 4 in the form of Equation (9), taking $k = 2$.

15. Prove that the multistep method of Equation (9) is of order m if and only if $Lp_i = 0$ for $0 \le i \le m$ and $Lp_{m+1} \ne 0$, where $p_i(t) = t^i$ and L is as given in Equation (10).

16. (Continuation) Determine the order of this method by the procedure in the preceding problem:

$$x_n = x_{n-2} + 2hf_{n-1}$$

17. Determine the order of this method by computing $d_0, d_1, \ldots$:

$$x_n = x_{n-3} + \tfrac{3}{8}h[f_n + 3f_{n-1} + 3f_{n-2} + f_{n-3}]$$

18. Let $a_0, a_1, \ldots, a_m$ be given, and assume that $\sum_{i=0}^{k} a_i = 0$. Do there necessarily exist $b_0, b_1, \ldots, b_m$ so that the multistep method having these coefficients will be of order at least m? (A theorem or an example is required.)

19. Prove that for Euler's method $d_0 = d_1 = 0$ and $d_j = 1/j!$ for $j \ge 2$.

COMPUTER PROBLEMS 8.4

1. Write and test a subprogram or procedure for the fifth-order Adams-Bashforth-Moulton method coupled with a fifth-order Runge-Kutta method (see Computer Problems **8.3.7–9**, p. 548). Keep only the five most recent values of (t_i, x_i). Print statements should be included in the routine.

2. Write a computer program that calls the routine in the preceding computer problem and solves the initial-value problem

$$\begin{cases} x' = (t - e^{-t})/(x + e^x) \\ x(0) = 0 \end{cases}$$

on the interval $[-1, 1]$. Use $h = 1/238$. Verify (analytically) that the true solution is given implicitly by the equation $x^2 - t^2 + 2e^x - 2e^{-t} = 0$. Use this equation in the program to provide a check on the computed solution. Use the Runge-Kutta method to get started.

3. Compute the solution of

$$\begin{cases} y' = -2xy^2 \\ y(0) = 1 \end{cases}$$

at $x = 1.0$ using $h = 0.25$ and the fourth-order Adams-Bashforth-Moulton method (Problems **8.4.4–5**, p. 555) together with the fourth-order Runge-Kutta method. Give the computed solution to five significant digits at 0.25, 0.5, 0.75, and 1.0. Compare your results to the exact solution $y = 1/(1 + x^2)$.

4. Find a program in your computer center program library that is suitable for solving the initial-value problem for a system of first-order ordinary differential equations. Use the program to solve this problem:

$$\begin{cases} dx/dt = \sin x + \cos(yt) & x(-1) = 2.37 \\ dy/dt = e^{-xt} + [\sin(yt)]/t & y(-1) = -3.48 \end{cases}$$

The solution is desired with eight-decimal-place accuracy in the interval $[-1, 4]$. The solution is to be printed at intervals of 0.1 on the t-axis.

a. Obtain plots of the functions $x(t)$ and $y(t)$ for $-1 \leq t \leq 4$. Both curves should appear on the same plot. The maximum and minimum ordinates should be set as $+4$ and -4.

b. Obtain a plot of the curve defined parametrically as

$$\{(x(t), y(t)) : -1 \leq t \leq 4\}$$

(Values of t will not appear on this plot.) The curve obtained here is called an **orbit** of the original system of differential equations. (The word **trajectory** is also used.)

8.5 Local and Global Errors: Stability

The Adams-Bashforth and Adams-Moulton methods of Section 8.4 (pp. 550–551) are only two examples of multistep methods for solving an initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{1}$$

The term **multistep** refers to the fact that in the computation of x_n, some previously computed terms, $x_{n-1}, x_{n-2}, \ldots$ are required. Let us review a few matters discussed in the preceding section.

Implicit/Explicit and Convergent Methods

Any multistep method can be described by an equation of the form

$$a_k x_n + a_{k-1} x_{n-1} + \cdots + a_0 x_{n-k} = h[b_k f_n + b_{k-1} f_{n-1} + \cdots + b_0 f_{n-k}] \tag{2}$$

For example, the fifth-order Adams-Moulton formula would read as follows:

$$x_n - x_{n-1} = h\left[\tfrac{251}{720} f_n + \tfrac{646}{720} f_{n-1} - \tfrac{264}{720} f_{n-2} + \tfrac{106}{720} f_{n-3} - \tfrac{19}{720} f_{n-4}\right] \tag{3}$$

In these equations, f_i stands for $f(t_i, x_i)$. In solving problem (1) using a formula such as (2), we assume that starting values $x_0, x_1, \ldots, x_{k-1}$ have been obtained by some other method. Then Equation (2) is used with $n = k, k+1, \ldots,$ in succession. Observe that if $b_k \neq 0$, then Equation (2) contains the unknown ordinate x_n on *both* sides of the equation. In this case, the method is said to be **implicit**. If $b_k = 0$, the method is **explicit**. In our analysis, we assume that x_n satisfies Equation (2). In practice, x_n can be obtained by iteration, starting with a tentative value given by a *predictor* formula.

Two polynomials are associated with Equation (2):

$$\begin{cases} p(z) = a_k z^k + a_{k-1} z^{k-1} + \cdots + a_0 \\ q(z) = b_k z^k + b_{k-1} z^{k-1} + \cdots + b_0 \end{cases} \tag{4}$$

The analysis will show that certain desirable properties of the multistep method depend on the location of the roots of the polynomials p and q.

The multistep method defined by Equation (2) is said to be **convergent** under the following circumstance: imagine that numerical solutions of problem (1) are computed by using a variety of step sizes. We denote by $x(h, t)$ the approximate solution obtained by using step size h. As usual, the exact solution is written $x(t)$. Convergence means that

$$\lim_{h \to 0} x(h, t) = x(t) \qquad (t \text{ fixed}) \tag{5}$$

for all t in some interval $[t_0, t_m]$, provided only that the starting values obey the same equation, that is,

$$\lim_{h \to 0} x(h, t_0 + nh) = x_0 \qquad (0 \le n < k) \tag{6}$$

and that the function f satisfies the hypotheses of the basic existence theorem (Theorem 3 in Section 8.1, p. 526).

Stability and Consistency

Two other terms that are used are *stable* and *consistent*. The method is **stable** if all roots of p lie in the disk $|z| \le 1$ and if each root of modulus 1 is simple. The method is **consistent** if $p(1) = 0$ and $p'(1) = q(1)$. The main theorem in this subject can now be stated.

THEOREM 1 Theorem on Stability and Consistency, Multistep Method

For the multistep method of Equation (2) *to be convergent, it is necessary and sufficient that it be stable and consistent.*

Proof (*Stability is necessary.*) Suppose that the method is not stable. Then either p has a root λ satisfying $|\lambda| > 1$ or p has a root λ satisfying $|\lambda| = 1$ and $p'(\lambda) = 0$. In either case we consider a simple initial-value problem whose solution is $x(t) = 0$:

$$\begin{cases} x' = 0 \\ x(0) = 0 \end{cases} \tag{7}$$

The multistep method is governed by the equation

$$a_k x_n + a_{k-1} x_{n-1} + \cdots + a_0 x_{n-k} = 0 \tag{8}$$

This is a linear difference equation, one of whose solutions is $x_n = h\lambda^n$, where λ is one of the roots of p. If $|\lambda| > 1$, then for $0 \le n < k$ we have

$$|x(h, nh)| = h|\lambda^n| < h|\lambda|^k \to 0 \qquad (\text{as } h \to 0)$$

This establishes condition (6). But Equation (5) is violated because if $t = nh$, then $h = tn^{-1}$ and

$$|x(h, t)| = |x(h, nh)| = tn^{-1}|\lambda|^n \to \infty$$

On the other hand, if $|\lambda| = 1$ and $p'(\lambda) = 0$, then a solution of Equation (8) is $x_n = hn\lambda^n$. Condition (6) is fulfilled because if $0 \le n < k$, then

$$|x(h, nh)| = hn|\lambda|^n = hn < hk \to 0 \qquad (\text{as } h \to 0)$$

Condition (5) is violated because

$$|x(h, t)| = (tn^{-1})n|\lambda|^n = t \ne 0$$

■

Proof (*Consistency is necessary.*) Suppose that the method defined by Equation (2) is convergent. Consider the problem

$$\begin{cases} x' = 0 \\ x(0) = 1 \end{cases} \tag{9}$$

The exact solution is $x = 1$. Equation (2) again has the form (8). One solution is obtained by setting $x_0 = x_1 = \cdots = x_{k-1} = 1$ and then using (8) to generate the remaining values, $x_k, x_{k+1}, \ldots$. Since the method is convergent, $\lim_{n\to\infty} x_n = 1$. Putting this into Equation (8) yields the result $a_k + a_{k-1} + \cdots + a_0 = 0$, or in other words, $p(1) = 0$.

Now consider the initial-value problem

$$\begin{cases} x' = 1 \\ x(0) = 0 \end{cases} \tag{10}$$

whose exact solution is $x = t$. Equation (8) becomes

$$a_k x_n + a_{k-1}x_{n-1} + \cdots + a_0 x_{n-k} = h\big[b_k + b_{k-1} + \cdots + b_0\big] \tag{11}$$

Since the method is convergent, it is stable by the preceding proof. Hence $p(1) = 0$ and $p'(1) \ne 0$. One solution of Equation (11) is given by $x_n = (n + k)h\gamma$, with $\gamma = q(1)/p'(1)$. Indeed, substitution of this into the left side of Equation (11) yields

$$\begin{aligned} &h\gamma\big[a_k(n + k) + a_{k-1}(n + k - 1) + \cdots + a_0 n\big] \\ &\quad = nh\gamma(a_k + a_{k-1} + \cdots + a_0) + h\gamma\big[ka_k + (k - 1)a_{k-1} + \cdots + a_1\big] \\ &\quad = nh\gamma p(1) + h\gamma p'(1) \\ &\quad = h\gamma p'(1) = hq(1) = h\big[b_k + b_{k-1} + \cdots + b_0\big] \end{aligned}$$

Notice that the starting values in this numerical solution are consistent with the initial value $x(0) = 0$ because $\lim_{h\to 0}(n+k)h\gamma = 0$ for $n = 0, 1, \ldots, k-1$. Now the convergence condition demands that $\lim_{n\to\infty} x_n = t$ if $nh = t$. Hence, we have $\lim_{n\to\infty}(n+k)h\gamma = t$. Therefore, we conclude that $\gamma = 1$, or $p'(1) = q(1)$ because $\lim_{n\to\infty} kh = 0$. ■

The proof that stability and consistency together imply convergence is much more involved. The interested reader should consult the text by Henrici [1962, Section 5.3].

Milne Method

To illustrate Theorem 1, on multistep method stability and consistency, we analyze the **Milne method**, defined by

$$x_n - x_{n-2} = h\left[\tfrac{1}{3}f_n + \tfrac{4}{3}f_{n-1} + \tfrac{1}{3}f_{n-2}\right] \tag{12}$$

This is an implicit method and is characterized by the two polynomials

$$p(z) = z^2 - 1$$
$$q(z) = \tfrac{1}{3}z^2 + \tfrac{4}{3}z + \tfrac{1}{3}$$

Note that the roots of p are $+1$ and -1. They are simple roots. Furthermore, $p'(z) = 2z$, $p'(1) = 2$, and $q(1) = 2$. Thus, the conditions of consistency and stability are fulfilled. By the theorem, the Milne method is convergent.

Local Truncation Error

Our next task is to define and analyze the local truncation error that arises in using the multistep method (2). Suppose that Equation (2) is used to compute x_n, under the supposition that all previous values $x_{n-1}, x_{n-2}, \ldots$ are *correct*; that is, $x_i = x(t_i)$ for $i < n$. Here $x(t)$ denotes the solution of the differential equation. The **local truncation error** is then defined to be $x(t_n) - x_n$. This error is due solely to our modeling of the differential equation by a difference equation. In this definition, roundoff error is not included; we assume that x_n has been computed with complete precision from the difference Equation (2). We want to prove that if the method has order m (as defined in Section 8.4, p. 553), then the local truncation error will be $\mathcal{O}(h^{m+1})$. This is not true without qualification, because our analysis presupposes certain smoothness in f and in the exact solution function $x(t)$.

THEOREM 2 Theorem on Local Truncation Error, Multistep Method

If the multistep method (2) *is of order m, if $x \in C^{m+2}$, and if $\partial f/\partial x$ is continuous, then under the hypotheses of the preceding paragraph,*

$$x(t_n) - x_n = \left(\frac{d_{m+1}}{a_k}\right)h^{m+1}x^{(m+1)}(t_{n-k}) + \mathcal{O}(h^{m+2}) \tag{13}$$

(The coefficients d_k are defined in Section 8.4, p. 553.)

Proof It suffices to prove the equation when $n = k$, because x_n can be interpreted as the value of a numerical solution that began at the point t_{n-k}. Using the linear functional L, Equation (10) in Section 8.4 (p. 552), we have

$$Lx = \sum_{i=0}^{k} \left[a_i x(t_i) - hb_i x'(t_i)\right] = \sum_{i=0}^{k} \left[a_i x(t_i) - hb_i f\big(t_i, x(t_i)\big)\right] \qquad \textbf{(14)}$$

On the other hand, the numerical solution satisfies the equation

$$0 = \sum_{i=0}^{k} \left[a_i x_i - hb_i f(t_i, x_i)\right] \qquad \textbf{(15)}$$

Because we have assumed that $x_i = x(t_i)$ for $i < k$, the result of subtracting Equation (15) from Equation (14) will be

$$Lx = a_k\left[x(t_k) - x_k\right] - hb_k\left[f\big(t_k, x(t_k)\big) - f(t_k, x_k)\right] \qquad \textbf{(16)}$$

To the last term of Equation (16) we apply the Mean-Value Theorem, obtaining

$$\begin{aligned} Lx &= a_k\left[x(t_k) - x_k\right] - hb_k \frac{\partial f}{\partial x}(t_k, \xi)\left[x(t_k) - x_k\right] \\ &= [a_k - hb_k F]\left[x(t_k) - x_k\right] \end{aligned} \qquad \textbf{(17)}$$

where $F = \partial f(t_k, \xi)/\partial x$ for some ξ between $x(t_k)$ and x_k. Now recall from Equation (13) in Section 8.4 (p. 553) that if the method being used is of order m, then Lx will have the form

$$Lx = d_{m+1} h^{m+1} x^{(m+1)}(t_0) + \mathcal{O}(h^{m+2}) \qquad \textbf{(18)}$$

Combining Equations (17) and (18), we obtain (13). Here we can ignore $hb_k F$ in the denominator. (See Problem **8.5.8**, p. 564.) ■

Global Truncation Error

We now wish to establish a bound on the global truncation error in solving a differential equation numerically. At any given step in the solution process, say at t_n, the solution computed is denoted by x_n. Let us assume that all computations have been carried out with complete precision; that is, no roundoff error is involved. Of course, the true solution at t_n, $x(t_n)$ differs from the computed solution x_n because the latter has been obtained by formulas that approximate a Taylor series. The difference $x(t_n) - x_n$ is the **global truncation error**. It is *not* simply the sum of all local truncation errors that entered at previous points. To see that this is so, we must realize that each step in the numerical solution must use as its initial value the approximate ordinate computed at the preceding step. Since that ordinate is in error, the numerical process is in effect attempting to track the wrong solution curve. We therefore must begin the analysis by seeing how two solution curves differ if they are started with different initial conditions. In other words, we need to understand the effect of

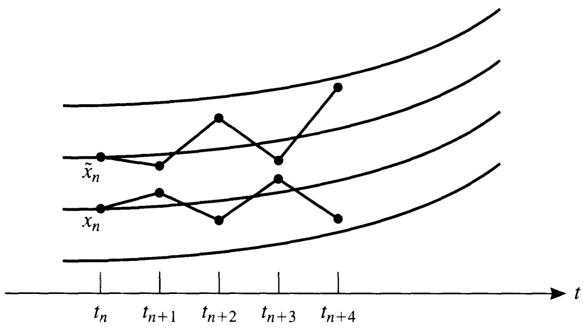

FIGURE 8.2
Effect of changing initial conditions

changing an initial value on the later ordinates of a solution curve. Figure 8.2 shows what we are trying to measure.

Consider the initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(0) = s \end{cases} \tag{19}$$

We assume that $\partial f/\partial x$, denoted here by f_x, is continuous and satisfies the condition $f_x(t, x) \le \lambda$ in the region defined by $0 \le t \le T$ and $-\infty < x < \infty$. The solution of (19) is a function of t, but to show its dependence on the initial value s, we write it as $x(t; s)$. Define $u(t) = \partial x(t; s)/\partial s$. We can obtain a differential equation—the **variational equation**—for u by differentiating with respect to s in the initial-value problem (19). The result is

$$\begin{cases} u'(t) = f_x(t, x)u \\ u(0) = 1 \end{cases} \tag{20}$$

EXAMPLE 1 Determine u explicitly in this initial-value problem:

$$\begin{cases} x' = x^2 \\ x(0) = s \end{cases} \tag{21}$$

Solution Here, $f(t, x) = x^2$ and $f_x = 2x$. Hence, the variational equation is

$$\begin{cases} u' = 2xu \\ u(0) = 1 \end{cases} \tag{22}$$

The solution of the initial-value problem (21) is $x(t) = s(1-st)^{-1}$. Hence, Equations (22) become

$$\begin{cases} u'(t) = 2s(1 - st)^{-1}u(t) \\ u(0) = 1 \end{cases}$$

and the solution of this is

$$u(t) = (1 - st)^{-2}$$ ■

THEOREM 3 Theorem on Variational Equation

If $f_x \leq \lambda$, then the solution of the variational equation (20) satisfies the inequality

$$|u(t)| \leq e^{\lambda t} \qquad (t \geq 0)$$

Proof From Equations (20), we have

$$u'/u = f_x = \lambda - \alpha(t) \tag{23}$$

in which $\alpha(t) \geq 0$. On integrating Equation (23), we get

$$\log|u| = \lambda t - \int_0^t \alpha(\tau)\,d\tau = \lambda t - A(t)$$

in which $A(t)$ denotes the indicated integral. Since $t \geq 0$, $A(t) \geq 0$, and consequently $\log|u| \leq \lambda t$. Thus, $|u| \leq e^{\lambda t}$ because the exponential function is increasing. ■

THEOREM 4 Theorem on Solution Curves, Initial-Value Problem

If the initial-value problem (19) is solved with initial values s and $s + \delta$, the solution curves at t differ by at most $|\delta|e^{\lambda t}$.

Proof By the Mean-Value Theorem, by the definition of u, and by Theorem 3, on the variational equation, we have

$$\begin{aligned} |x(t;s) - x(t;s+\delta)| &= \left|\frac{\partial}{\partial s}x(t, s+\theta\delta)\right| |\delta| \\ &= |u(t)|\,|\delta| \leq |\delta|e^{\lambda t} \end{aligned}$$ ■

THEOREM 5 Theorem on Global Truncation Error Bound

If the local truncation errors at $t_1, t_2, \ldots, t_n$ do not exceed δ in magnitude, then the global truncation error at t_n does not exceed $\delta(e^{n\lambda h} - 1)(e^{\lambda h} - 1)^{-1}$.

Proof Let truncation errors of $\delta_1, \delta_2, \ldots$ be associated with the numerical solution at points $t_1, t_2, \ldots$. In computing x_2 there was an error of δ_1 in the initial condition, and by Theorem 4, on initial-value problem solution curves, the effect of this at t_2 is at most $|\delta_1|e^{\lambda h}$. To this is added the truncation error at t_2. Thus the global truncation error at t_2 is at most

$$|\delta_1|e^{\lambda h} + |\delta_2|$$

The effect of this error at t_3 is (by Theorem 4) no greater than

$$\big(|\delta_1|e^{\lambda h} + |\delta_2|\big)e^{\lambda h}$$

To this is added the truncation error at t_3. Continuing in this way, we find that the global truncation error at t_n is no greater than

$$\sum_{k=1}^{n} |\delta_k| e^{(n-k)\lambda h} \leq \delta \sum_{k=0}^{n-1} e^{k\lambda h} = \delta(e^{n\lambda h} - 1)(e^{\lambda h} - 1)^{-1}$$ ■

THEOREM 6 Theorem on Global Truncation Error Approximation

If the local truncation errors in the numerical solution are $\mathcal{O}(h^{m+1})$, then the global truncation error is $\mathcal{O}(h^m)$.

Proof In Theorem 5, on global truncation error bound, let δ be $\mathcal{O}(h^{m+1})$. Since $e^z - 1$ is $\mathcal{O}(z)$ and $nh = t$, we find a decrease of 1 in the order from using the formula in Theorem 5. ■

PROBLEMS 8.5

1. Discuss these multistep methods in light of Theorem 1 (p. 558), on multistep method stability and consistency:

   a. $x_n - x_{n-2} = 2hf_{n-1}$

   b. $x_n - x_{n-2} = h\left[\frac{7}{3}f_{n-1} - \frac{2}{3}f_{n-2} + \frac{1}{3}f_{n-3}\right]$

   c. $x_n - x_{n-1} = h\left[\frac{3}{8}f_n + \frac{19}{24}f_{n-1} - \frac{5}{24}f_{n-2} + \frac{1}{24}f_{n-3}\right]$

2. A method is said to be **weakly unstable** if p has a zero λ such that $\lambda \neq 1$, $|\lambda| = 1$, and $q(\lambda) < \lambda p'(\lambda)$. Show that the Milne method given by Equation (12) is weakly unstable.

3. Show that every multistep method in which $p(z) = z^k - z^{k-1}$ and $\sum_{i=0}^{k} b_i = 1$ is stable, consistent, convergent, and weakly stable.

4. Determine the numerical characteristics of the multistep method whose equation is

$$x_n + 4x_{n-1} - 5x_{n-2} = h\,[4f_{n-1} + 2f_{n-2}]$$

5. Is there any reason for distrusting this numerical scheme for solving $x' = f(t, x)$?

$$x_n - 3x_{n-1} + 2x_{n-2} = h\,[f_n + 2f_{n-1} + f_{n-2} - 2f_{n-3}]$$

   Explain.

6. Which of these multistep methods is *convergent*?

   a. $x_n - x_{n-2} = h(f_n - 3f_{n-1} + 4f_{n-2})$

   b. $x_n - 2x_{n-1} + x_{n-2} = h(f_n - f_{n-1})$

   c. $x_n - x_{n-1} - x_{n-2} = h(f_n - f_{n-1})$

   d. $x_n - 3x_{n-1} + 2x_{n-2} = h(f_n + f_{n-1})$

   e. $x_n - x_{n-2} = h(f_n - 3f_{n-1} + 2f_{n-2})$

7. A multistep method is said to be **strongly stable** if $p(1) = 0$, $p'(1) \neq 0$, and all other roots of p satisfy the inequality $|z| < 1$. Prove that a strongly stable method is convergent, using Theorem 1, on multistep stability and consistency. Prove also that for any value of λ, a strongly stable method will solve the problem $x' = \lambda x$, $x(0) = 1$ without introducing any extraneous errors of exponential growth.

8. Prove that

$$\frac{Ah^{m+1} + \mathcal{O}(h^{m+2})}{B - Ch} = \frac{A}{B}h^{m+1} + \mathcal{O}(h^{m+2})$$

8.6 Systems and Higher-Order Ordinary Differential Equations

The standard form for a *system* of first-order differential equations is:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \dots, x_n) \\ x_2' = f_2(t, x_1, x_2, \dots, x_n) \\ \quad \vdots \\ x_n' = f_n(t, x_1, x_2, \dots, x_n) \end{cases} \tag{1}$$

In this system, n unknown functions, $x_1, x_2, \dots, x_n$, are to be determined. They are functions of the single independent variable, t, and the notation x_i' denotes the derivative dx_i/dt. For a concrete example, consider this system, in which x and y have been written in place of x_1 and x_2:

$$\begin{cases} x' = x + 4y - e^t \\ y' = x + y + 2e^t \end{cases} \tag{2}$$

The *general* solution of System (2) is

$$\begin{cases} x = 2ae^{3t} - 2be^{-t} - 2e^t \\ y = ae^{3t} + be^{-t} + \frac{1}{4}e^t \end{cases} \tag{3}$$

where a and b are arbitrary constants. The purported solution can be verified, of course, by differentiation and substitution in System (2). Notice that the example is a *linear* system in the unknown functions x and y.

In a well-defined physical problem having presumably a *unique* solution, the system of differential equations would be accompanied by auxiliary conditions that serve to determine the arbitrary constants in the *general* solution. Thus, if System (2) were accompanied by initial conditions

$$x(0) = 4 \qquad y(0) = \tfrac{5}{4}$$

then the solution would be

$$\begin{cases} x = 4e^{3t} + 2e^{-t} - 2e^t \\ y = 2e^{3t} - e^{-t} + \frac{1}{4}e^t \end{cases} \tag{4}$$

The initial-value problem for the general system in (1) consists of the n differential equations together with a prescribed *initial value* for t, say $t = t_0$, and a specification of the value of each function x_i at t_0.

Vector Notation

A convenient vector notation can be used to rewrite System (1). We let X denote the column vector whose components are $x_1, x_2, \dots, x_n$. These components are functions of t. Hence, X is a mapping of $\mathbb{R}$ (or an interval in $\mathbb{R}$) to $\mathbb{R}^n$. Similarly, let F

denote the column vector with components $f_1, f_2, \ldots, f_n$. Each of these is a function on $\mathbb{R}^{n+1}$ (or a subset thereof), and so F is a mapping of $\mathbb{R}^{n+1}$ to $\mathbb{R}^n$. System (1) can now be written as

$$X' = F(t, X) \tag{5}$$

An initial-value problem for System (5) would also include numerical values for the vector $X(t_0)$, where t_0 is the initial value of t.

A differential equation of high order can be converted to a system of first-order equations. Suppose that a single differential equation is given in the form

$$y^{(n)} = f(t, y, y', y'', \ldots, y^{(n-1)})$$

Here, of course, all derivatives are with respect to t: $y^{(i)} = d^i y/dt^i$. Next, we introduce new variables $x_1, x_2, \ldots, x_n$ according to these definitions:

$$x_1 = y \qquad x_2 = y' \qquad x_3 = y'' \qquad \cdots \qquad x_n = y^{(n-1)}$$

The new variables satisfy the following system of first-order differential equations:

$$\begin{cases} x_1' = x_2 \\ x_2' = x_3 \\ x_3' = x_4 \\ \quad\vdots \\ x_n' = f(t, x_1, x_2, x_3, \ldots, x_n) \end{cases}$$

This is a system of the form presented in Equation (5).

To solve differential equations using widely available software, it is almost always necessary to convert the problem into a system such as (5). We illustrate this process with two examples.

EXAMPLE 1 Convert the initial-value problem

$$\begin{cases} (\sin t)y''' + \cos(ty) + \sin(t^2 + y'') + (y')^3 = \log t \\ y(2) = 7 \\ y'(2) = 3 \\ y''(2) = -4 \end{cases} \tag{6}$$

into a system of first-order differential equations with initial values.

Solution We introduce new variables x_1, x_2, and x_3 as follows: $x_1 = y$, $x_2 = y'$, and $x_3 = y''$. The system of equations governing $X = (x_1, x_2, x_3)^T$ is

$$\begin{cases} x_1' = x_2 \\ x_2' = x_3 \\ x_3' = \left[\log t - x_2^3 - \sin(t^2 + x_3) - \cos(tx_1)\right]/\sin t \end{cases} \tag{7}$$

The initial conditions at $t = 2$ are $X = (7, 3, -4)^T$. ■

Systems of higher-order equations can be treated in a similar manner, as illustrated in the next example.

EXAMPLE 2 Convert the system shown into a system of first-order equations:

$$\begin{cases} (x'')^2 + te^y + y' = x' - x \\ y'y'' - \cos(xy) + \sin(tx'y) = x \end{cases} \tag{8}$$

The initial conditions are omitted in this example.

Solution After introducing the new variables $x_1 = x$, $x_2 = x'$, $x_3 = y$, and $x_4 = y'$, the problem can be written as

$$\begin{cases} x_1' = x_2 \\ x_2' = \left(x_2 - x_1 - x_4 - te^{x_3}\right)^{1/2} \\ x_3' = x_4 \\ x_4' = \left[x_1 - \sin(tx_2x_3) + \cos(x_1x_3)\right]/x_4 \end{cases} \tag{9}$$

■

Taylor-Series Method for Systems

The Taylor-series method, which was discussed in Section 8.2 (p. 530), can be applied to systems of first-order equations. We write a truncated Taylor series for each variable like this:

$$x_i(t+h) = x_i(t) + hx_i'(t) + \frac{h^2}{2!}x_i''(t) + \frac{h^3}{3!}x_i'''(t) + \cdots + \frac{h^n}{n!}x_i^{(n)}(t)$$

or in vector notation:

$$X(t+h) = X(t) + hX'(t) + \frac{h^2}{2!}X''(t) + \frac{h^3}{3!}X'''(t) + \cdots + \frac{h^n}{n!}X^{(n)}(t) \tag{10}$$

The derivatives appearing here can be obtained from the differential equation. Usually these derivatives must be computed in a particular order when used in a computer program. We must be sure that quantities needed at one step are available as the results from prior steps.

EXAMPLE 3 Write the Taylor-series algorithm of order 3 for the following initial-value problem. Use $|h| = 0.1$ and compute the solution on the interval $-2 \le t \le 1$.

$$\begin{cases} x' = x + y^2 - t^3 & x(1) = 3 \\ y' = y + x^3 + \cos t & y(1) = 1 \end{cases} \tag{11}$$

Solution The higher derivatives required are

$$\begin{aligned} x'' &= x' + 2yy' - 3t^2 \\ y'' &= y' + 3x^2x' - \sin t \\ x''' &= x'' + 2yy'' + 2(y')^2 - 6t \\ y''' &= y'' + 6x(x')^2 + 3x^2x'' - \cos t \end{aligned}$$

A suitable algorithm to carry out the computation follows:

input $t \leftarrow 1;\ x \leftarrow 3;\ y \leftarrow 1;\ h \leftarrow -0.1;\ M \leftarrow 30$
output $0, t, x, y$
for $k = 1$ **to** M **do**
  $x' \leftarrow x + y^2 - t^3$
  $y' \leftarrow y + x^3 + \cos t$
  $x'' \leftarrow x' + 2yy' - 3t^2$
  $y'' \leftarrow y' + 3x^2x' - \sin t$
  $x''' \leftarrow x'' + 2yy'' + 2(y')^2 - 6t$
  $y''' \leftarrow y'' + 6x(x')^2 + 3x^2x'' - \cos t$
  $x \leftarrow x + h(x' + \frac{1}{2}h(x'' + \frac{1}{3}h(x''')))$
  $y \leftarrow y + h(y' + \frac{1}{2}h(y'' + \frac{1}{3}h(y''')))$
  $t \leftarrow t + h$
  output k, t, x, y
end do

The numerical output from the computer implementation of this algorithm is not given here, but a computer plot of the two functions $x(t)$ and $y(t)$ is shown in Figure 8.3. Because plotting routines are not at all standardized, no purpose would be served in showing the computer program by which the graphs were obtained. It suffices to remark that usually we must give the automatic plotter two arrays, say $\{t_i : 0 \le i \le m\}$ and $\{x_i : 0 \le i \le m\}$, from which it would then plot the points (t_i, x_i) on a Cartesian grid. Usually the plotter can be instructed to connect these points with straight lines. Hence, it is imperative that the points be ordered with this in mind. Usually, of course, $t_0 < t_1 < \cdots < t_m$. If the plotted points are close to each other, the resulting curve will appear to be a smooth curve and not a sequence of line segments. ■

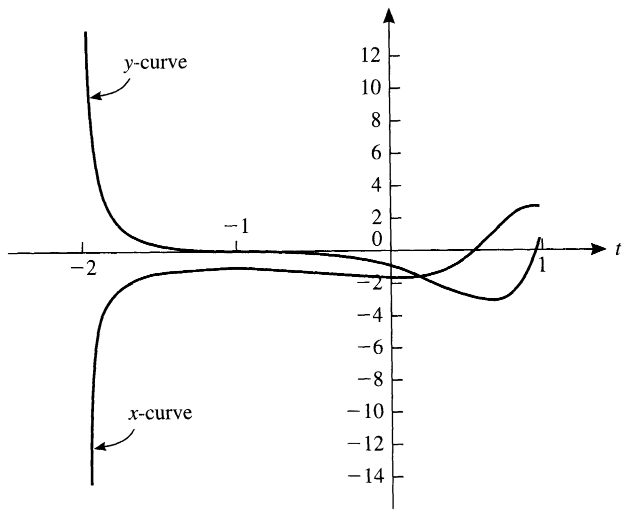

FIGURE 8.3
Solution curves for Example 3

From the theoretical standpoint, there is no loss of generality in assuming that the equations in System (1) do not contain t explicitly. We can write the system in the form

$$x_i' = f_i(x_0, x_1, \dots, x_n)$$

by introducing a variable $x_0 = t$. The differential equation for the new variable is simply $x_0' = 1$. By use of this device, we can write the system of equations (5) as

$$X' = F(X) \tag{12}$$

without t appearing explicitly. Here $X = (x_0, x_1, \dots, x_n)^T$. A system in the form of (12) is said to be **autonomous**.

EXAMPLE 4 Convert the initial-value problem of Example 1 into a system in which t does not appear explicitly.

Solution Let x_1, x_2, and x_3 be as in Example 1. Put $x_0 = t$. The new system is

$$\begin{cases} x_0' = 1 \\ x_1' = x_2 \\ x_2' = x_3 \\ x_3' = \left[\log x_0 - x_2^3 - \sin(x_0^2 + x_3) - \cos(x_0 x_1)\right] / \sin x_0 \end{cases} \tag{13}$$

and the initial condition is $X = (2, 7, 3, -4)^T$. ■

Other Methods for Systems

The Runge-Kutta procedure for systems of first-order equations is most easily written down in the case when our system is autonomous—that is, when it has the form of Equation (12). The classical **fourth-order Runge-Kutta formulas**, in vector form, are

$$X(t+h) = X(t) + \tfrac{1}{6}(F_1 + 2F_2 + 2F_3 + F_4) \tag{14}$$

where

$$\begin{cases} F_1 = hF(X) \\ F_2 = hF(X + \frac{1}{2}F_1) \\ F_3 = hF(X + \frac{1}{2}F_2) \\ F_4 = hF(X + F_3) \end{cases}$$

The programming of a routine to carry out this procedure is left as Computer Problem **8.6.4** (p. 572). Similar formulas can be given for the Runge-Kutta-Fehlberg method, discussed in Section 8.3 (p. 544).

Multistep methods also can be extended to apply to systems of equations. As an example, we give the vector form of the **Adams-Bashforth-Moulton predictor-**

corrector method (4) and (8) from Section 8.4 (pp. 550–551):

$$X^*(t+h) = X(t) + \frac{h}{720}\big[1901F\big(X(t)\big) - 2774F\big(X(t-h)\big) + 2616F\big(X(t-2h)\big) - 1274F\big(X(t-3h)\big) + 251F\big(X(t-4h)\big)\big]$$

$$X(t+h) = X(t) + \frac{h}{720}\big[251F\big(X^*(t+h)\big) + 646F\big(X(t)\big) - 264F\big(X(t-h)\big) + 106F\big(X(t-2h)\big) - 19F\big(X(t-3h)\big)\big]$$

As in the case of a single equation, a single-step procedure, such as a fifth-order Runge-Kutta method (Computer Problems **8.3.7–9**, p. 548), could be used to provide starting values:

$$X(t_0+h) \qquad X(t_0+2h) \qquad X(t_0+3h) \qquad X(t_0+4h)$$

PROBLEMS 8.6

1. Find the general solution of the system

$$\begin{cases} x' = 3x - 4y + e^t \\ y' = x - y - e^t \end{cases}$$

Hint: Try functions of the form e^t, te^t, and t^2e^t.

2. Write an autonomous system of first-order equations equivalent to

$$\begin{cases} x''' - \sin(x'') + e^t x' + 2t\cos x = 25 \\ x(0) = 5 \qquad x'(0) = 3 \qquad x''(0) = 7 \end{cases}$$

3. Write the third-order ordinary differential equation

$$\begin{cases} x''' + 2x'' - x' - 2x = e^t \\ x(8) = 3 \qquad x'(8) = 2 \qquad x''(8) = 1 \end{cases}$$

as an autonomous system of first-order equations.

4. Convert the system of second-order ordinary differential equations

$$\begin{cases} x'' - x'y = 3y'x\log t \\ y'' - 2xy' = 5x'y\sin t \end{cases}$$

into a system of first-order equations in which t does not appear explicitly.

5. Write

$$\begin{cases} y'' + yz = 0 \qquad y(0) = 1 \qquad y'(0) = 0 \\ z' + 2yz = 4 \qquad z(0) = 3 \end{cases}$$

as a system of first-order equations with initial conditions. Use vector notation.

6. Write an autonomous system of first-order equations equivalent to

$$\begin{cases} x''' - \left[\sin x'' + e^t x'\right]^2 + \cos x = 0 \\ x(0) = 3 \qquad x'(0) = 4 \qquad x''(0) = 5 \end{cases}$$

7. Explain how to solve this problem numerically, assuming the availability of a computer program for solving the initial-value problem for a system of first-order equations:

$$\begin{cases} x'' = x\cos t + e^t x' + 3t^2 + 7 \\ x(1) = 5 \qquad x'(1) = 9 \end{cases}$$

8. Write this system of higher-order ordinary differential equations as a system of first-order equations:

$$\begin{cases} y_1^{(n)} = f_1(t, y_1, y_1', \dots, y_1^{(n-1)}) \\ y_2^{(n)} = f_2(t, y_2, y_2', \dots, y_2^{(n-1)}) \\ \quad\vdots \\ y_m^{(n)} = f_m(t, y_m, y_m', \dots, y_m^{(n-1)}) \end{cases}$$

9. Convert the following system of higher-order differential equations into a system of first-order equations in which t does not appear explicitly:

$$\begin{cases} x''' - 5tx''y'' + \ln(x')z = 0 \\ y'' - \sin(ty) + 7tx'' = 0 \\ z' + 16ty' - e^t zx' = 0 \end{cases}$$

10. Write Euler's method, Heun's method, and the modified Euler's method for a system of equations in the form of Equation (12).

COMPUTER PROBLEMS 8.6

1. Write a computer program to solve this initial-value problem using the Taylor-series method. Include terms in h, h^2, and h^3 and continue the solution to $t = 1$. Let $h = 0.01$.

$$\begin{cases} x_1' = t + x_1^2 + x_2 & x_1(-1) = 0.43 \\ x_2' = t^2 - x_1 + x_2^2 & x_2(-1) = -0.69 \end{cases}$$

2. (A challenging problem) Follow the instructions in the preceding computer problem for this initial-value problem:

$$\begin{cases} x_1' = \sin x_1 + \cos(tx_2) & x_1(-1) = 2.37 \\ x_2' = t^{-1}\sin(tx_1) & x_2(-1) = -3.48 \end{cases}$$

The programming of x_2', x_2'', and x_2''' must be very carefully carried out because of the singularity at $t = 0$.

3. Solve numerically the system of differential equations

$$\begin{cases} x_1' = (-1 - 9c^2 + 12sc)x_1 + (12c^2 + 9sc)x_2 & x_1(0) = -12 \\ x_2' = (-12s^2 + 9s)x_1 + (-1 - 9s^2 - 12sc)x_2 & x_2(0) = 6 \end{cases}$$

in which $c = \cos 6t$ and $s = \sin 6t$. Integrate over the interval $[0, 10]$ with step size $h = 0.01$. Verify that the correct solution is

$$\begin{cases} x_1 = e^{-13t}(s - 2c) \\ x_2 = e^{-13t}(2s + c) \end{cases}$$

Does your computed solution compare favorably with the true solution? This example has been given by Lambert [1973].

4. Write a subprogram or procedure that takes one step of length h in the fourth-order Runge-Kutta procedure. Make it capable of handling a system of n differential equations with $n \leq 20$. Test your program by solving the following system on the interval $1 \leq t \leq 2$. Use $h = -0.01$.
$$\begin{cases} x' = x^{-2} + \log y + t^2 \\ y' = e^y - \cos x + (\sin t)x - (xy)^{-3} \\ x(2) = -2 \qquad y(2) = 1 \end{cases}$$
5. Solve and plot the resulting curve over the interval $[0, 5]$ for the ordinary differential equation $x'' + 192x = 0$ with $x(0) = \frac{1}{6}$, $x'(0) = 0$. This corresponds to an **undamped spring-mass system**.
6. Write and test a subroutine or procedure for the adaptive Runge-Kutta-Fehlberg algorithm that can handle systems of first-order differential equations.
7. Write and test a subroutine or procedure that is capable of handing systems of first-order differential equations for the fifth-order Adams-Bashforth-Moulton method coupled with a fifth-order Runge-Kutta method (see Computer Problems **8.3.7–9**, p. 548).

8.7 Boundary-Value Problems

In the preceding sections of this chapter, various methods have been considered for solving the initial-value problem. Thus, for example, the initial-value problem

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x'(a) = \beta \end{cases} \tag{1}$$

can be put into the form of a system of first-order equations by letting $x_1 = x$ and $x_2 = x'$:

$$\begin{cases} x_1' = x_2 & x_1(a) = \alpha \\ x_2' = f(t, x_1, x_2) & x_2(a) = \beta \end{cases} \tag{2}$$

System (2) can then be solved by one of the step-by-step methods described earlier.

The situation would be quite different, however, if the problem in (1) were changed to read

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x(b) = \beta \end{cases} \tag{3}$$

The difference lies in the specification of conditions at two different points, $t = a$ and $t = b$. The step-by-step procedures for the initial-value problem are not adapted to the solution of (3) because the numerical solution cannot begin without a *full complement* of initial values. In (3), we have a typical example of a **two-point boundary-value problem**. Such problems usually present difficulties of a greater magnitude than those of the initial-value problem.

Here is an example of a two-point boundary-value problem that can be solved without numerical work:

$$\begin{cases} x'' = -x \\ x(0) = 3 \qquad x(\frac{\pi}{2}) = 7 \end{cases} \tag{4}$$

First we can find the general solution of the differential equation, which is

$$x(t) = A \sin t + B \cos t \tag{5}$$

Then the constants A and B can be determined so that the boundary conditions are satisfied. Thus,

$$\begin{cases} 3 = x(0) = A \sin 0 + B \cos 0 = B \\ 7 = x(\frac{\pi}{2}) = A \sin \frac{\pi}{2} + B \cos \frac{\pi}{2} = A \end{cases}$$

The solution of (4) is then

$$x(t) = 7 \sin t + 3 \cos t$$

Further examples of this type are given in the problems.

The technique just illustrated is not effective if the general solution of the differential equation in (3) is unknown. Our interest here is in numerical methods by which any two-point boundary-value problem can be attacked.

Existence

Before considering numerical methods, a few comments about the existence question are appropriate. In general, we cannot assure the existence of a solution to (3) by simply assuming that f is a *nice* function. A simple example in support of this assertion is

$$\begin{cases} x'' = -x \\ x(0) = 3 \qquad x(\pi) = 7 \end{cases} \tag{6}$$

This is only slightly different from the problem in (4), but when we try to impose the boundary values on the general solution (5), we get the contradictory equations $3 = B$ and $7 = -B$. Thus, the problem in (6) has no solution.

Existence theorems for solutions of the two-point boundary-value problem (3) tend to be rather complicated, and we refer the reader to books by Stoer and Bulirsch [1980] and Keller [1968]. One elegant result in Keller [1968, p. 108] is the following.

THEOREM 1 Existence Theorem, Boundary-Value Problem

The boundary-value problem

$$\begin{cases} x'' = f(t, x) \\ x(0) = 0 \qquad x(1) = 0 \end{cases}$$

has a unique solution if $\partial f/\partial x$ is continuous, nonnegative, and bounded in the infinite strip defined by the inequalities $0 \le t \le 1$, $-\infty < x < \infty$.

EXAMPLE 1 Show that this two-point boundary-value problem has a unique solution:

$$\begin{cases} x'' = (5x + \sin 3x)e^t \\ x(0) = x(1) = 0 \end{cases}$$

Solution Let us try to use Theorem 1. Then

$$\frac{\partial f}{\partial x} = (5 + 3\cos 3x)e^t$$

This is continuous in the strip $0 \le t \le 1$, $-\infty < x < \infty$. Furthermore, it is bounded above by $8e$ and is nonnegative, since $3\cos 3x \ge -3$. The hypotheses of Theorem 1 are therefore fulfilled. ■

Changes of Variables

Although Theorem 1, on existence of boundary-value problems, applies to a special case, simple changes of variables can reduce more general problems to the special case. To do this, we begin by changing the t-interval. Suppose that the original problem is

$$\begin{cases} x'' = f(t, x) \\ x(a) = \alpha \qquad x(b) = \beta \end{cases} \tag{7}$$

where $x = x(t)$. A change of variable $t = a + (b - a)s$ is called for here. Notice that $s = 0$ corresponds to $t = a$, and $s = 1$ corresponds to $t = b$. Therefore, we define $y(s) = x(a + \lambda s)$, with $\lambda = b - a$. Then $y'(s) = \lambda x'(a + \lambda s)$ and $y''(s) = \lambda^2 x''(a + \lambda s)$. Likewise $y(0) = x(a) = \alpha$ and $y(1) = x(b) = \beta$. Thus, if x is a solution of (7), then y is a solution of the boundary-value problem

$$\begin{cases} y''(s) = \lambda^2 f\big(a + \lambda s, y(s)\big) \\ y(0) = \alpha \qquad y(1) = \beta \end{cases} \tag{8}$$

Conversely, if y is a solution of (8), then the function $x(t) = y((t - a)/(b - a))$ is a solution of (7).

THEOREM 2 First Theorem on Two-Point Boundary-Value Problems

Consider these two-point boundary-value problems:

1. $\begin{cases} x'' = f(t, x) \\ x(a) = \alpha \qquad x(b) = \beta \end{cases}$

2. $\begin{cases} y'' = g(t, y) \\ y(0) = \alpha \qquad y(1) = \beta \end{cases}$

in which

$$g(p, q) = (b - a)^2 f\big(a + (b - a)p, q\big)$$

*If y is a solution of Problem **2**, then the function x defined by*

$$x(t) = y((t-a)/(b-a))$$

*is a solution of Problem **1**. Moreover, if x is a solution of Problem **1**, then*

$$y(t) = x(a+(b-a)t)$$

*is a solution of Problem **2**.*

Proof It is a simple verification, as follows:

$$\begin{aligned}
x(a) &= y\left(\frac{a-a}{b-a}\right) = y(0) = \alpha \\
x(b) &= y\left(\frac{b-a}{b-a}\right) = y(1) = \beta \\
x'(t) &= y'\left(\frac{t-a}{b-a}\right)\frac{1}{b-a} \\
x''(t) &= y''\left(\frac{t-a}{b-a}\right)\frac{1}{(b-a)^2} \\
&= g\left(\frac{t-a}{b-a}, y\left(\frac{t-a}{b-a}\right)\right)\frac{1}{(b-a)^2} \\
&= (b-a)^2 f\left(a+(b-a)\frac{t-a}{b-a}, y\left(\frac{t-a}{b-a}\right)\right)\frac{1}{(b-a)^2} = f\big(t, x(t)\big) \quad \blacksquare
\end{aligned}$$

EXAMPLE 2 Show that these two-point boundary-value problems are equivalent:

$$\begin{cases} x'' = \sin(tx) + x^2 \\ x(1) = 3 \qquad x(4) = 7 \end{cases}$$

$$\begin{cases} y'' = 16\big\{\sin\big[(4s+1)y\big] + y^2\big\} \\ y(0) = 3 \qquad y(1) = 7 \end{cases}$$

Solution Use Theorem 2. ■

To reduce a two-point boundary-value problem

$$\begin{cases} y'' = g(t,y) \\ y(0) = \alpha \qquad y(1) = \beta \end{cases}$$

to one having homogeneous boundary values, we simply subtract from y a linear function that takes the values α and β at 0 and 1. Here is the theorem.

THEOREM 3 Second Theorem on Two-Point Boundary-Value Problems

Consider these two-point boundary-value problems:

1. $\begin{cases} y'' = g(t, y) \\ y(0) = \alpha \qquad y(1) = \beta \end{cases}$

2. $\begin{cases} z'' = h(t, z) \\ z(0) = 0 \qquad z(1) = 0 \end{cases}$

where

$$h(p, q) = g\bigl(p, q + \alpha + (\beta - \alpha)p\bigr)$$

If z solves Problem 2, then the function

$$y(t) = z(t) + \alpha + (\beta - \alpha)t$$

solves Problem 1. Moreover, if y solves Problem 1, then

$$z(t) = y(t) - [\alpha + (\beta - \alpha)t]$$

solves Problem 2.

Proof Again there is a straightforward verification:

$$\begin{aligned} y(0) &= z(0) + \alpha + (\beta - \alpha)0 = \alpha \\ y(1) &= z(1) + \alpha + (\beta - \alpha)1 = \beta \\ y''(t) &= z''(t) = h\bigl(t, z(t)\bigr) = g\bigl(t, z(t) + \alpha + (\beta - \alpha)t\bigr) \\ &= g\bigl(t, y(t)\bigr) \end{aligned}$$

■

EXAMPLE 3 Show that this problem has a unique solution:

$$\begin{cases} x'' = \bigl[5x - 10t + 35 + \sin(3x - 6t + 21)\bigr]e^t \\ x(0) = -7 \qquad x(1) = -5 \end{cases}$$

Solution Because the boundary values are not homogeneous in this problem, Theorem 1 does not apply immediately. To obtain an equivalent problem with 0 boundary values, we change the dependent variable as in Theorem 3. Let

$$z(t) = x(t) - \ell(t) \quad \text{and} \quad \ell(t) = -7 + 2t$$

Then

$$\begin{aligned} z'' = x'' &= \bigl[5x - 10t + 35 + \sin(3x - 6t + 21)\bigr]e^t \\ &= \bigl\{5(z + \ell) - 10t + 35 + \sin\bigl[3(z + \ell) - 6t + 21\bigr]\bigr\}e^t \\ &= \{5z + \sin 3z\}e^t \end{aligned}$$

The boundary values for the new variable z are

$$\begin{aligned} z(0) &= x(0) - \ell(0) = -7 + 7 = 0 \\ z(1) &= x(1) - \ell(1) = -5 - (-5) = 0 \end{aligned}$$

The boundary problem for z has a unique solution, as shown in Example 1. Hence the problem for x has a unique solution. ■

EXAMPLE 4 Convert this two-point boundary-value problem to an equivalent one with 0 boundary values on the interval [0, 1]:

$$\begin{cases} x'' = x^2 + 3 - t^2 + xt \\ x(3) = 7 \qquad x(5) = 9 \end{cases} \tag{9}$$

Solution By Theorem 2, an equivalent problem is

$$\begin{cases} y'' = g(t, y) \\ y(0) = 7 \qquad y(1) = 9 \end{cases} \tag{10}$$

with

$$g(t, y) = 4f(3 + 2t, y) = 4\big[y^2 + 3 - (3 + 2t)^2 + y(3 + 2t)\big]$$

By Theorem 3, another equivalent problem is

$$\begin{cases} z'' = h(t, z) \\ z(0) = 0 \qquad z(1) = 0 \end{cases} \tag{11}$$

with $h(t, z) = g(t, z + 7 + 2t)$. Expressing h in detail, we have

$$h(t, z) = 4\big[(z + 7 + 2t)^2 + 3 - (3 + 2t)^2 + (z + 7 + 2t)(3 + 2t)\big]$$

We can solve boundary-value problem (11) for z and obtain the solution of boundary-value problem (10) from the equation

$$y(t) = z(t) + 7 + 2t$$

The solution of boundary-value problem (9) is then obtained from

$$x(t) = y\left(\frac{t - 3}{2}\right)$$ ■

EXAMPLE 5 Suppose that we compute the solution $z(0.5) = 8.2$ for the boundary-value problem (11). What is the corresponding solution to boundary-value problem (9)?

Solution Clearly, $y(0.5) = z(0.5) + 7 + 2(0.5) = 16.2$ and $0.5 = (t - 3)/2$ implies $t = 4$. The answer is

$$x(4) = y(0.5) = z(0.5) + 8 = 16.2$$ ■

THEOREM 4 **Theorem on Unique Solution, Boundary-Value Problem**

Let f be a continuous function of (t, s), *where* $0 \le t \le 1$ *and* $-\infty < s < \infty$. *Assume that on this domain*

$$|f(t, s_1) - f(t, s_2)| \le k|s_1 - s_1| \qquad (k < 8)$$

Then the two-point boundary-value problem

$$\begin{cases} x'' = f(t, x) \\ x(0) = x(1) = 0 \end{cases} \tag{12}$$

has a unique solution in $C[0, 1]$.

Proof (*In outline*) By appealing to Problem **8.7.10** (p. 579), we know that the boundary-value problem (12) is equivalent to the integral equation

$$x(t) = \int_0^1 G(t, s) f\big(s, x(s)\big)\, ds \tag{13}$$

in which G is the Green's function of Problem **8.7.6** (p. 579). The integral equation (13) has the form $x = F(x)$, where F is the operation defined by the integral. By using Banach's Contraction Mapping Theorem in the space $C[0, 1]$, we conclude that F has a unique fixed point and that Equations (12) and (13) have unique solutions. ∎

EXAMPLE 6 Show that this problem has a unique solution:

$$\begin{cases} x'' = 2\exp(t \cos x) \\ x(0) = x(1) = 0 \end{cases} \tag{14}$$

Solution Here $f(t, s) = 2\exp(t \cos s)$. By the Mean-Value Theorem,

$$|f(t, s_1) - f(t, s_2)| = \left| \frac{\partial f}{\partial s}(t, s_3) \right| |s_1 - s_2|$$

The derivative needed here satisfies the relation

$$\left| \frac{\partial f}{\partial s} \right| = |2\exp(t \cos s)(-t \sin s)| \le 2e < 5.437 < 8$$

By Theorem 4, the two-point boundary-value problem (14) has a unique solution. ∎

PROBLEMS 8.7

1. Solve the two-point boundary-value problem $x'' = x$, $x(0) = 1$, $x(1) = 1$.
2. Determine all pairs (α, β) for which the problem $x'' = x$, $x(0) = \alpha$, $x(1) = \beta$ has a solution.
3. (Continuation) Repeat the preceding problem for $x'' = -x$, $x(0) = \alpha$, $x(\pi) = \beta$.
4. (Continuation) Give an example of a two-point boundary-value problem (p. 572) of type (3), for which there is more than one solution. *Hint:* Consider the preceding problem.
5. Solve the problem $x'' - 2x' + x = 0$, $x(0) = \alpha$, $x(1) = \beta$. Are there any pairs (α, β) for which no solution exists?

6. Show that the two-point boundary-value problem $x'' = f(t)$, $x(0) = x(1) = 0$ is solved by the formula

$$x(t) = \int_0^1 G(t, s) f(s)\, ds$$

in which

$$G(t, s) = \begin{cases} s(t-1) & 0 \le s \le t \le 1 \\ t(s-1) & 0 \le t \le s \le 1 \end{cases}$$

The function G is known as the **Green's function** for this problem.

7. Consider these two boundary-value problems:

i. $\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x(b) = \beta \end{cases}$

ii. $\begin{cases} x'' = h^2 f(a + th, x, h^{-1}x') \\ x(0) = \alpha \qquad x(1) = \beta \end{cases}$

Show that if x is a solution of boundary-value problem **ii**, then the function $y(t) = x((t-a)/h)$ solves boundary-value problem **i**, where $h = b - a$.

8. Illustrate Theorem 2 with the boundary-value problem

$$\begin{cases} x'' = t + x^2 - 3x' \\ x(3) = \alpha \qquad x(7) = \beta \end{cases}$$

This is equivalent to

$$\begin{cases} x'' = 48 + 64t + 16x^2 - 12x' \\ x(0) = \alpha \qquad x(1) = \beta \end{cases}$$

9. Use Theorem 4 to establish that this two-point problem has a unique solution:

$$\begin{cases} x'' = \cos(tx) \\ x(0) = 1 \qquad x(1) = 4 \end{cases}$$

10. (Continuation) Show that the boundary-value problem

$$\begin{cases} x'' = f(t, x) \\ x(0) = 0 \qquad x(1) = 0 \end{cases}$$

is equivalent to the integral equation

$$x(t) = \int_0^1 G(t, s) f\big(s, x(s)\big)\, ds$$

in which G is the Green's function of Problem **8.7.6** (above).

11. Prove that the following two-point boundary-value problem has a unique solution:

$$\begin{cases} x'' = (t^3 + 5)x + \sin t \\ x(0) = x(1) = 0 \end{cases}$$

12. Prove that this problem has a unique solution:

$$\begin{cases} x'' = \tan^{-1} x + 2x + \cos t \\ x(0) = x(1) = 0 \end{cases}$$

13. Solve the boundary-value problem

$$\begin{cases} x'' = x^2 \\ x(0) = 2/3 \qquad x(1) = 3/8 \end{cases}$$

14. (Continuation) Solve the preceding problem when the boundary conditions are simplified to $x(0) = 0$, $x(1) = 1$. Refer to Davis [1962].

15. Solve the boundary-value problem

$$\begin{cases} x'' = x^3 \\ x(0) = 1 \qquad x(1) = 2 - \sqrt{2} \end{cases}$$

16. Prove that if x solves the differential equation $x'' = f(x)$, then so does y, where $y(t) = x(t + c)$.

17. Prove that this problem has a unique solution:

$$\begin{cases} x'' = \frac{1}{2} \exp\left\{\frac{1}{2}(t+1)\cos(x + 7 - 3t)\right\} \qquad (-1 \le t \le 1) \\ x(-1) = -10 \qquad x(1) = -4 \end{cases}$$

18. Find a two-point boundary-value problem of the form

$$\begin{cases} z'' = h(s, z) \\ z(0) = z(1) = 0 \end{cases}$$

which is equivalent to this problem:

$$\begin{cases} x'' = \cos(x^2 t^2) \\ x(3) = 5 \qquad x(7) = 12 \end{cases}$$

19. Suppose that v and w are known functions of t having these properties:

$$\begin{cases} v'' = \cos t + v e^t + v' t^2 \qquad v(1) = 5 \qquad v(3) = 42 \\ w'' = \cos t + w e^t + w' t^2 \qquad w(1) = 8 \qquad w(3) = 9 \end{cases}$$

What function solves this two-point boundary-value problem?

$$\begin{cases} x'' = \cos t + x e^t + x' t^2 \\ x(1) = 7 \quad x(3) = 20 \end{cases}$$

20. (Multiple-choice question)

a. The two-point boundary-value problem

$$\begin{cases} x'' = -4x \\ x(0) = 1 \qquad x(\pi/2) = -1 \end{cases}$$

is an example having

i. no solution
ii. exactly two solutions
iii. exactly one solution
iv. no solution in elementary functions
v. more than one solution.

b. Repeat Part **a** for the two-point boundary-value problem

$$\begin{cases} x'' = -4x \\ x(0) = 1 \qquad x(\pi/2) = 2 \end{cases}$$

21. Consider the two-point boundary-value problem

$$\begin{cases} x'' = f(t, x) \\ x(0) = x(1) = 0 \end{cases}$$

For which function(s) f can we be sure that a unique solution exists?

i. $f(t, x) = t^2(1 + x^2)^{-1}$

ii. $f(t, x) = t \sin x$

iii. $f(t, x) = (\tan t)(\tan x)$

iv. $f(t, x) = t/x$

v. $f(t, x) = x^{1/3}$

22. Consider the two-point boundary-value problem

$$\begin{cases} x'' = f(t, x) \\ x(0) = x(1) = 0 \end{cases}$$

For which function(s) f can we be sure that a unique solution exists?

i. $f(t, x) = t^2 x^3$

ii. $f(t, x) = t(\tan^{-1} x)$

iii. $f(t, x) = t x^{4/3}$

iv. $f(t, x) = \log(1 + x^2)$

v. $f(t, x) = |tx|$

23. Prove that if $x(t)$ is a solution of

$$\begin{cases} x''(t) = f\big(t, x(t)\big) \\ x(a) = \alpha \qquad x(b) = \beta \end{cases}$$

then $y(s) = x(a + (b - a)s)$ is a solution of

$$\begin{cases} y''(s) = (b - a)^2 f\big(a + (b - a)s, y(s)\big) \\ y(0) = \alpha \qquad y(1) = \beta \end{cases}$$

and consequently $z(s) = y(s) - [\alpha + (\beta - \alpha)s]$ is a solution of

$$\begin{cases} z''(s) = (b - a)^2 f\big(a + (b - a)s, z(s) + \alpha + (\beta - \alpha)s\big) \\ z(0) = 0 \qquad z(1) = 0 \end{cases}$$

8.8 Boundary-Value Problems: Shooting Methods

The two-point boundary-value problem being considered is

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x(b) = \beta \end{cases} \tag{1}$$

One natural way to attack this problem is to solve the related initial-value problem, with a *guess* as to the appropriate initial value $x'(a)$. Then we can integrate the equation to obtain an approximate solution, hoping that $x(b) = \beta$. If $x(b) \neq \beta$, then the guessed value of $x'(a)$ can be altered and we can try again. This process is called **shooting**, and there are ways of doing it systematically.

Let us denote the guessed value of $x'(a)$ by z, so that the corresponding initial-value problem is

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x'(a) = z \end{cases} \tag{2}$$

The solution of this initial-value problem will be denoted by x_z. The objective is to select z so that $x_z(b) = \beta$. We put

$$\phi(z) \equiv x_z(b) - \beta$$

so that our objective is simply to solve the equation $\phi(z) = 0$ for z. The methods considered in Chapter 3 for solving a single nonlinear equation are applicable here. For example, the bisection method, the secant method, and Newton's method are all available, in addition to functional iteration. The function ϕ is an *expensive* one to compute because each value of $\phi(z)$ is obtained by numerically solving an initial-value problem.

Secant Method

Let us recall how the secant method is applied to solve an equation $\phi(z) = 0$. With two values of $\phi(z)$ at hand, say $\phi(z_1)$ and $\phi(z_2)$, we pretend that ϕ is *linear*. Using the equation of the straight line through the points $(z_2, \phi(z_2))$ and $(z_1, \phi(z_1))$, we have

$$\phi(z) - \phi(z_2) = \left(\frac{\phi(z_1) - \phi(z_2)}{z_1 - z_2}\right)(z - z_2)$$

If we select z_3 so that $\phi(z_3) = 0$, we obtain the formula

$$z_3 = z_2 - \left(\frac{z_2 - z_1}{\phi(z_2) - \phi(z_1)}\right)\phi(z_2) \tag{3}$$

This procedure can be repeated to obtain a sequence of values $z_1, z_2, \dots, z_n$ by

$$z_n = z_{n-1} - \left(\frac{z_{n-1} - z_{n-2}}{\phi(z_{n-1}) - \phi(z_{n-2})}\right)\phi(z_{n-1})$$

This equation defines the secant method. (See Section 3.3, p. 94.) When several values of z have been obtained for which $\phi(z)$ is nearly 0, we can stop this procedure and use polynomial interpolation to estimate a better value. Here is how this should be done: Suppose that $\phi(z_1), \phi(z_2), \dots, \phi(z_n)$ are small. Find a polynomial p that interpolates the table

$\phi(z_1)$	$\phi(z_2)$	$\cdots$	$\phi(z_n)$
z_1	z_2	$\cdots$	z_n

Thus, the polynomial p has the property $p(\phi(z_i)) = z_i$ for $1 \le i \le n$. The next estimate, z_{n+1}, is determined by the equation $p(0) = z_{n+1}$. This procedure amounts to an approximation of the **inverse function** of ϕ by the polynomial p. Its success depends on ϕ having a differentiable inverse in a neighborhood of the root. This in turn requires the assumption that the root in question be *simple*.

Linear Function

The shooting method as outlined above can be quite costly in computational effort, and it is important to consider techniques for economizing. Obviously any partial information about the correct value of $x'(a)$ should be exploited. It is also possible to solve the initial-value problems with a large step size, because high precision is essentially wasted in the first stages of the shooting method. Only when $\phi(z)$ is nearly 0 should a small step size be used.

There is one class of problems in which the secant method yields the exact solution in one step. This occurs when ϕ is in fact a *linear* function, and that will happen when the differential equation is linear. In the linear case, the two-point boundary-value problem will have the form

$$\begin{cases} x'' = u(t) + v(t)x + w(t)x' \\ x(a) = \alpha \quad x(b) = \beta \end{cases} \tag{4}$$

We shall assume in what follows that the functions u, v, and w are continuous on the interval $[a, b]$. Suppose that we have solved the differential equation in (4) twice, with two different initial conditions, obtaining solutions x_1 and x_2, say,

$$\begin{cases} x_1(a) = \alpha \qquad x_1'(a) = z_1 \\ x_2(a) = \alpha \qquad x_2'(a) = z_2 \end{cases} \tag{5}$$

Now we form a linear combination of x_1 and x_2:

$$y(t) = \lambda x_1(t) + (1 - \lambda)x_2(t) \tag{6}$$

where λ is a parameter. We can easily verify that y solves the differential equation and meets the first of the two boundary conditions, that is, $y(a) = \alpha$. We select λ so that $y(b) = \beta$. Thus,

$$\beta = y(b) = \lambda x_1(b) + (1 - \lambda)x_2(b)$$

and

$$\lambda = \frac{\beta - x_2(b)}{x_1(b) - x_2(b)} \tag{7}$$

In a computer realization of these ideas for the linear problem (4), we can obtain x_1 and x_2 at the same time. Thus, the two initial-value problems to be solved could be specified as

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x'(a) = 0 \end{cases} \qquad \begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x'(a) = 1 \end{cases}$$

where $f(t, x, x') = u(t) + v(t)x + w(t)x'$. The solution of the first is x_1 and the solution of the second is x_2. To create a system of first-order equations without t appearing explicitly, we also define $x_0 = t$, $x_3 = x_1'$, and $x_4 = x_2'$. Then the system of differential equations with initial values is

$$\begin{cases} x_0 = 1 & x_0(a) = a \\ x_1' = x_3 & x_1(a) = \alpha \\ x_2' = x_4 & x_2(a) = \alpha \\ x_3' = f(x_0, x_1, x_3) & x_4(a) = 0 \\ x_4' = f(x_0, x_2, x_4) & x_5(a) = 1 \end{cases} \tag{8}$$

This system should be given as input to a computer program for solving the initial-value problem. The approximate values of the discrete functions $x_1(t_i)$ and $x_2(t_i)$ for $a = t_0 \le t_i \le t_m = b$ should be stored in the computer memory as one-dimensional arrays. Next, the value of λ should be computed by Equation (7). Finally, the solution y should be computed at each desired value of t from Equation (6).

THEOREM 1 First Theorem on Linear Two-Point Boundary-Value Problem

If the linear two-point boundary-value problem (4) *has a solution, then either* x_1 *itself is a solution or* $x_1(b) - x_2(b) \neq 0$ *(and* y *is a solution).*

Proof Let y_0, y_1, and y_2 be solutions of these initial-value problems:

$$\begin{array}{lll} y_0'' = u + vy_0 + wy_0' & y_0(a) = \alpha & y_0'(a) = 0 \\ y_1'' = vy_1 + wy_1' & y_1(a) = 1 & y_1'(a) = 0 \\ y_2'' = vy_2 + wy_2' & y_2(a) = 0 & y_2'(a) = 1 \end{array}$$

By the theory of second-order linear differential equations (in particular, by Theorem 3 below), the general solution of the differential equation in (4) is

$$y_0 + c_1 y_1 + c_2 y_2$$

in which c_1 and c_2 are arbitrary constants. We see immediately that the functions x_1 and x_2 in (5) are special cases of the general solution, given by

$$x_1 = y_0 + z_1 y_2 \qquad x_2 = y_0 + z_2 y_2 \tag{9}$$

Because it has been assumed that the problem in (4) has a solution, c_1 and c_2 exist such that

$$\alpha = y_0(a) + c_1 y_1(a) + c_2 y_2(a)$$

$$\beta = y_0(b) + c_1 y_1(b) + c_2 y_2(b)$$

The first of these equations reduces to $c_1 = 0$, and thus we infer the existence of c_2 such that

$$\beta = y_0(b) + c_2 y_2(b) \tag{10}$$

If $x_1(b) - x_2(b) \neq 0$, then the function y defined by Equations (6) and (7) solves (4). If $x_1(b) - x_2(b) = 0$, then from Equations (9) we have $y_2(b) = 0$. Equation (10) tells us that $y_0(b) = \beta$, and Equations (9) tell us that x_1 is a solution. ■

Newton's Method

Let us return to the more general (nonlinear) two-point boundary-value problem of Equation (1) and consider how to apply Newton's method. Recall that x_z is defined as the solution of the problem

$$\begin{cases} x_z'' = f(t, x_z, x_z') \\ x_z(a) = \alpha \qquad x_z'(a) = z \end{cases} \tag{11}$$

We want to select z so that

$$\phi(z) \equiv x_z(b) - \beta = 0$$

Newton's formula for the function ϕ is

$$z_{n+1} = z_n - \frac{\phi(z_n)}{\phi'(z_n)} \tag{12}$$

To determine ϕ', we differentiate partially with respect to z all the equations in (11):

$$\begin{cases} \dfrac{\partial x_z''}{\partial z} = \dfrac{\partial f}{\partial t}\dfrac{\partial t}{\partial z} + \dfrac{\partial f}{\partial x_z}\dfrac{\partial x_z}{\partial z} + \dfrac{\partial f}{\partial x_z'}\dfrac{\partial x_z'}{\partial z} \\ \dfrac{\partial}{\partial z}x_z(a) = 0 \qquad \dfrac{\partial}{\partial z}x_z'(a) = 1 \end{cases} \tag{13}$$

With some simplification and the introduction of the new variable $v = \partial x_z/\partial z$, this becomes

$$\begin{cases} v'' = f_{x_z}(t, x_z, x_z')v + f_{x_z'}(t, x_z, x_z')v' \\ v(a) = 0 \qquad v'(a) = 1 \end{cases} \tag{14}$$

We recognize this set of equations as an initial-value problem. The differential equation in (14) is called the **first variational equation**. It can be solved step-by-step along with (11). At the end, $v(b)$ will be available, and we have

$$v(b) = \frac{\partial x_z(b)}{\partial z} = \phi'(z)$$

This enables us to use Newton's method, Equation (12), to find a root of ϕ.

Multiple Shooting

An elaboration of the shooting method is called **multiple shooting**. The basic strategy here is to divide the given interval $[a, b]$ into subintervals and attempt to solve the global problem in pieces. Let us describe what would be done if the interval were subdivided into just two parts, $[a, c]$ and $[c, b]$.

The original problem is as before; namely,

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x(b) = \beta \end{cases}$$

On the two subintervals, we solve initial-value problems to get two functions x_1 and x_2:

$$\begin{cases} x_1'' = f(t, x_1, x_1') & x_1(a) = \alpha & x_1'(a) = z_1 & (a \le t \le c) \\ x_2'' = f(t, x_2, x_2') & x_2(b) = \beta & x_2'(b) = z_2 & (c \le t \le b) \end{cases}$$

Notice that z_1 and z_2 are parameters at our disposal. The function x_1 is required only on the interval $[a, c]$, and x_2 is required only on $[c, b]$. The numerical solution of x_2 will proceed in the direction of *decreasing* t.

The parameters z_1 and z_2 are now to be adjusted until the piecewise function

$$x(t) = \begin{cases} x_1(t) & a \le t \le c \\ x_2(t) & c \le t \le b \end{cases}$$

becomes a solution to the problem. Thus, we require continuity in x and x' at the point c: $x_1(c) - x_2(c) = 0$ and $x_1'(c) - x_2'(c) = 0$. These two conditions are to be fulfilled by an adroit choice of z_1 and z_2. Typically this would be done by Newton's method in dimension 2, as in Section 3.2 (p. 88).

Multiple shooting with k subintervals will involve k **subfunctions**, each obtained by numerically solving an initial-value problem. The initial values of these k subfunctions will form a set of $2k$ parameters. At each of the $k-1$ interior division points of the interval, the continuity of the global function and its first derivative must be imposed. This provides $2k - 2$ conditions. Two endpoint conditions exist, and so the number of conditions matches the number of parameters. The resulting system of nonlinear equations is solved iteratively—for example, by the higher-dimensional Newton method.

Much of the software that is available for two-point boundary-value problems is written for a system of first-order ordinary differential equations

$$X' = F(t, X)$$

where $X = (x_1, x_2, \ldots, x_n)^T$ and $F = (f_1, f_2, \ldots, f_n)^T$. The boundary conditions are often permitted to be quite general. For example, some codes allow the boundary conditions to be in the form

$$G\big(X(a),\ X(b)\big) = 0$$

where $G = (g_1, g_2, \ldots, g_n)^T$. Some software requires the user to furnish the Jacobian matrix J of F, which is the $n \times n$ matrix with entries $J_{ij} = \partial f_i / \partial x_j$.

EXAMPLE 1 Suppose that software of the type just described is to be used on the problem

$$\begin{cases} x'' = tx + \cos x' \\ x(3) + x'(5) = 7 \qquad x'(3)^2 x(5) = 10 \end{cases}$$

What are F, G, and J?

Solution Set $x_1 = x$ and $x_2 = x'$. The problem can now be expressed as

$$\begin{cases} x_1' = x_2 & x_1(3) + x_2(5) - 7 = 0 \\ x_2' = tx_1 + \cos x_2 & x_2(3)^2 x_1(5) - 10 = 0 \end{cases}$$

The functions F and G can be read off from these equations. The Jacobian function is given by

$$J(t, X) = \begin{bmatrix} 0 & 1 \\ t & -\sin x_2 \end{bmatrix}$$ ■

Second-Order Linear Equations

Two important theorems from the general theory of second-order linear differential equations are cited here. The standard reference for such matters is Coddington and Levinson [1955].

THEOREM 2 Second Theorem on Second-Order Linear Differential Equation

If u, v, and w are continuous functions on the closed interval $[a, b]$, then for any pair of real numbers α and α' the initial-value problem

$$\begin{cases} x'' = u + vx + wx' \\ x(a) = \alpha \qquad x'(a) = \alpha' \end{cases}$$

has a unique solution on $[a, b]$.

THEOREM 3 Third Theorem on Second-Order Linear Differential Equation

Every solution of the (nonhomogeneous) equation

$$x'' - vx - wx' = u \tag{15}$$

can be expressed in the form $x_0 + c_1x_1 + c_2x_2$, where x_0 is any particular solution of Equation (15), and x_1 and x_2 form a linearly independent set of solutions to the (homogeneous) equation

$$x'' - vx - wx' = 0$$

PROBLEMS 8.8

1. Using the true solution, find the function ϕ explicitly in this case:

$$\begin{cases} x'' = -x \\ x(0) = 1 \qquad x\left(\frac{\pi}{2}\right) = 3 \end{cases}$$

2. Determine the true solution and find the function ϕ explicitly in this case:

$$\begin{cases} x'' = -(x')^2 x^{-1} \\ x(1) = 3 \qquad x(2) = 5 \end{cases}$$

Solve the boundary-value problem using ϕ.

3. Solve analytically the three-point boundary-value problem:

$$\begin{cases} x''' = -e^t + 4(t+1)^{-3} \\ x(0) = -1 \qquad x(1) = 3 - e + 2\ln 2 \qquad x(2) = 6 - e^2 + 2\ln 3 \end{cases}$$

4. Show how the shooting method can be used to solve a two-point boundary-value problem of the following type, in which the constants α, β, and c_{ij} are all given:

$$\begin{cases} x'' = u(t) + v(t)x + w(t)x' \\ c_{11}x(a) + c_{12}x'(a) = \alpha \\ c_{21}x(b) + c_{22}x'(b) = \beta \end{cases}$$

Hint: Let x_1 solve the differential equation with initial conditions specified in such a way that $c_{11}x_x(a) + c_{12}x_1(a) = \alpha$. Let x_2 solve the differential equation with initial conditions $x_2(a) = -c_{12}$ and $x_2'(a) = c_{11}$. Consider $x_1 + \lambda x_2$.

5. What is the first variational equation corresponding to this linear differential equation?

$$x'' = a(t) + b(t)x + c(t)x'$$

6. What is the first variational equation for this differential equation?

$$x'' = \cos(tx) + \sin(t^2x')$$

Can the first variational equation be solved numerically by itself, or must it be solved simultaneously with the equation for x''?

7. Show that if x_1 and x_2 solve $x'' = u(t) + v(t)x + w(t)x'$ with the initial conditions $x_1(a) = x_2(a) = \alpha$, $x_1'(a) = 0$, and $x_2'(a) = 1$, then $x_2 - x_1$ is the solution of the first variational problem in Equation (12).

8. Show that if we solve a linear two-point boundary-value problem by using Newton's method on ϕ, and if we compute ϕ' by using the first variational equation, then the result is the same as the one given by Equations (6) and (7).

9. Prove that the function ϕ is linear in the case of a linear differential equation.

10. Solve the problem

$$\begin{cases} x'' = -9x \\ x(0) = 1 \qquad x(\frac{\pi}{6}) = 5 \end{cases}$$

by first finding the solution x_z to the problem

$$\begin{cases} x'' = -9x \\ x(0) = 1 \qquad x'(0) = z \end{cases}$$

Then adjust z so that $x_z(\frac{\pi}{6}) = 5$. Describe how the result would be altered if $x(\frac{\pi}{3}) = 5$.

11. Find the function ϕ explicitly in this case:

$$\begin{cases} x'' = -2t(x')^2 \\ x(0) = 1 \qquad x(1) = 1 + \frac{\pi}{4} \end{cases}$$

Using ϕ, solve the boundary-value problem.

12. If x_z is the solution of the initial-value problem $x'' = x$, $x(0) = 0$, $x'(0) = z$, what is $x_z(1)$? *Hint:* Multiply the differential equation by x and integrate.

13. For the two-point boundary-value problem $x'' = x$, with $x(0) = 0$ and $x(1) = 17$, what is the function $\phi(z)$?

14. If x_z is the solution of the initial-value problem $x'' = -x$, $x(0) = 5$, $x'(0) = z$, and if $\phi(z) = x_z(\pi/2) - 3$, then what is $\phi'(z)$?

15. In solving the two-point boundary-value problem

$$\begin{cases} x'' - 37t^2x' = 95 \\ x(6) = 1 \qquad x(12) = 2 \end{cases}$$

using the shooting method based on the secant method, we obtain two pairs of numbers $(z_i, x_{z_i}(b))$ for $i = 1, 2$, say, $(4, 5)$ and $(2, 9)$. What initial-value problem is solved for the next iteration in this procedure?

16. Refer to the discussion of the linear two-point boundary-value problem and show that $c_1 = 0$ and $c_2 = [\beta - y_0(b)]/y_2(b)$.

17. (Continuation) Prove that if $y_2(b) = 0$, then the boundary-value problem will not be solvable for all (α, β).

18. Prove that the following procedure will solve the two-point boundary-value problem (4), provided that a solution exists. Solve two initial-value problems

$$\begin{cases} x_1'' = u + vx_1 + wx_1' & x_1(a) = \alpha & x_1'(a) = 0 \\ x_2'' = vx_2 + wx_2' & x_2(a) = 0 & x_2' = 1 \end{cases}$$

Then either x_1 itself is the solution or the solution is $x_1 + cx_2$, where c is the constant $[\beta - x_1(b)]/x_2(b)$.

19. Prove that the problem $x'' = x$, $x(a) = \alpha$, $x(b) = \beta$ always has a unique solution. (Assume that $a \neq b$.)

COMPUTER PROBLEMS 8.8

1. Write out the algorithm for the shooting method. Solve the two-point boundary-value problem

$$\begin{cases} x'' = e^t + x\cos t - (t+1)x' \\ x(0) = 1 \qquad x(1) = 3 \end{cases}$$

by the shooting method. Notice that the problem is linear. Use the Runge-Kutta method of order 4 with step size $h = 0.01$.

2. Write an algorithm for solving the linear two-point problem as outlined in the text. Refer to Equations (4)–(8). Write a general-purpose code based on this algorithm. The functions u, v, and w should be furnished by the user of the code in the form of subroutines.

3. (Continuation) Test the code in the preceding computer problem on this example:

$$\begin{cases} u(t) = e^{t-3},\ v(t) = t^2 + 2,\ w(t) = \sin t \\ a = 2.6,\ b = 5.1,\ \alpha = 7.0,\ \beta = -3 \end{cases}$$

8.9 Boundary-Value Problems: Finite-Differences

Another approach to the two-point boundary-value problem consists of an initial discretization of the t-interval followed by the use of approximate formulas for the derivatives. These two formulas are especially useful:

$$x'(t) = (2h)^{-1}\big[x(t+h) - x(t-h)\big] - \tfrac{1}{6}h^2 x'''(\xi) \tag{1}$$

$$x''(t) = h^{-2}\big[x(t+h) - 2x(t) + x(t-h)\big] - \tfrac{1}{12}h^2 x^{(4)}(\tau) \tag{2}$$

Second-Order Differential Equations

Suppose that the problem to be solved is

$$\begin{cases} x'' = f(t, x, x') \\ x(a) = \alpha \qquad x(b) = \beta \end{cases} \tag{3}$$

Let the interval $[a, b]$ be partitioned by points $a = t_0, t_1, t_2, \ldots, t_{n+1} = b$. These need not be equally spaced, but in practice they usually are. Indeed, if the points are *not* uniformly distributed, then more complicated versions of (1) and (2) will have to be introduced. So, for simplicity, we assume

$$t_i = a + ih \qquad 0 \le i \le n+1 \qquad h = (b-a)/(n+1) \tag{4}$$

The approximate value of $x(t_i)$ is denoted by y_i. The discrete version of (3) is then

$$\begin{cases} y_0 = \alpha \\ h^{-2}\big(y_{i-1} - 2y_i + y_{i+1}\big) = f\big(t_i, y_i, (2h)^{-1}(y_{i+1} - y_{i-1})\big) \qquad (1 \le i \le n) \\ y_{n+1} = \beta \end{cases} \tag{5}$$

The Linear Case

In Equation (5), the unknowns are $y_1, y_2, \ldots, y_n$, and n equations are to be solved. If f involves y_i in a nonlinear way, these equations will be nonlinear and, in general, difficult to solve. However, let us assume that f is linear in x and x'. Then it has the form

$$f(t, x, x') = u(t) + v(t)x + w(t)x' \tag{6}$$

System (5) is now a linear system of equations, which can be written in the form

$$\begin{cases} y_0 = \alpha \\ \left(-1 - \tfrac{1}{2}hw_i\right) y_{i-1} + (2 + h^2 v_i) y_i + \left(-1 + \tfrac{1}{2}hw_i\right) y_{i+1} = -h^2 u_i \quad (1 \le i \le n) \\ y_{n+1} = \beta \end{cases} \tag{7}$$

We have written $u_i = u(t_i)$, $v_i = v(t_i)$, and so on.

Next, we introduce the abbreviations

$$\begin{aligned} a_i &= -1 - \tfrac{1}{2}hw_{i+1} \\ d_i &= 2 + h^2 v_i \\ c_i &= -1 + \tfrac{1}{2}hw_i \\ b_i &= -h^2 u_i \end{aligned}$$

so that the system of equations looks like this:

$$\begin{bmatrix} d_1 & c_1 & & & & \\ a_1 & d_2 & c_2 & & & \\ & a_2 & d_3 & c_3 & & \\ & & \ddots & \ddots & \ddots & \\ & & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & & a_{n-1} & d_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} b_1 - a_0\alpha \\ b_2 \\ b_3 \\ \vdots \\ b_{n-1} \\ b_n - c_n\beta \end{bmatrix} \tag{8}$$

Since the elements not shown are 0, this system is tridiagonal and can be solved by a special Gaussian algorithm. Note that if h is small and $v_i > 0$, the matrix of the system is diagonally dominant because

$$|2 + h^2 v_i| > \left|1 + \tfrac{1}{2}hw_i\right| + \left|1 - \tfrac{1}{2}hw_i\right| = 2$$

Here we must assume that $|\frac{1}{2}hw_i| \le 1$ because then the terms $1 \pm \frac{1}{2}hw_i$ are both nonnegative. Henceforth, we assume that $v_i > 0$ and that h is small enough to validate the inequality $|\frac{1}{2}hw_i| < 1$.

The following equality will be needed later:

$$\begin{aligned} |d_i| - |c_i| - |a_{i-1}| &= 2 + h^2 v_i - \left(1 - \tfrac{1}{2}hw_i\right) - \left(1 + \tfrac{1}{2}hw_i\right) \\ &= h^2 v_i \end{aligned} \tag{9}$$

Convergence

We shall undertake to show that when h converges to 0, the discrete solution converges to the solution of the boundary-value problem. To know that the boundary-value problem

$$\begin{cases} x'' = u + vx + wx' \\ x(a) = \alpha \qquad x(b) = \beta \end{cases} \tag{10}$$

has a unique solution, we invoke a theorem from Keller [1968, p. 9] that yields this conclusion under the hypotheses that u, v, and w belong to $C[a, b]$ and $v > 0$. These assumptions are therefore adopted. The theorem of Keller follows.

THEOREM 1 Theorem on Existence and Uniqueness of Solution to Boundary-Value Problem

The boundary-value problem

$$\begin{cases} x'' = f(t, x, x') \\ c_{11}x(a) + c_{12}x'(a) = c_{13} \\ c_{21}x(b) + c_{22}x'(b) = c_{23} \end{cases}$$

has a unique solution on the interval $[a, b]$ provided that:

1. *f and its first partial derivatives f_t, f_x, and $f_{x'}$ are continuous on the domain* $D = [a, b] \times \mathbb{R} \times \mathbb{R}$
2. $f_x > 0$, $|f_x| \le M$, *and* $|f_{x'}| \le M$ *on* D
3. $|c_{11}| + |c_{12}| > 0$, $|c_{21}| + |c_{22}| > 0$, $|c_{11}| + |c_{21}| > 0$, *and* $c_{11}c_{12} \le 0 \le c_{21}c_{22}$

We denote by $x(t)$ the true solution of the problem and by y_i the solution of the discrete problem. Notice that y_i depends on h. We shall estimate $|x_i - y_i|$ and show that it converges to 0 when $h \to 0$. Here, of course, x_i denotes $x(t_i)$.

With the aid of Formulas (1) and (2), we see that $x(t)$ satisfies the following system of equations, with $1 \le i \le n$:

$$h^{-2}(x_{i-1} - 2x_i + x_{i+1}) - \tfrac{1}{12}h^2 x^{(4)}(\tau_i) \tag{11}$$

$$= u_i + v_i x_i + w_i\left[(2h)^{-1}(x_{i+1} - x_{i-1}) - \tfrac{1}{6}h^2 x'''(\xi_i)\right] \tag{12}$$

On the other hand, the discrete solution satisfies the equation

$$h^{-2}(y_{i-1} - 2y_i + y_{i+1}) = u_i + v_i y_i + w_i(2h)^{-1}(y_{i+1} - y_{i-1})$$

If we subtract Equation (12) from Equation (11) and write $e_i \equiv x_i - y_i$, the result is

$$h^{-2}(e_{i-1} - 2e_i + e_{i+1}) = v_i e_i + w_i(2h)^{-1}(e_{i+1} - e_{i-1}) + h^2 g_i \tag{13}$$

where

$$g_i = \tfrac{1}{12}x^{(4)}(\tau_i) - \tfrac{1}{6}x'''(\xi_i)$$

After collecting terms and multiplying by $-h^2$, we have an equation similar to Equation (7):

$$\left(-1 - \tfrac{1}{2}hw_i\right)e_{i-1} + (2 + h^2 v_i)e_i + \left(-1 + \tfrac{1}{2}hw_i\right)e_{i+1} = -h^4 g_i \tag{14}$$

Using the coefficients introduced earlier, we write this in the form

$$a_{i-1}e_{i-1} + d_i e_i + c_i e_{i+1} = -h^4 g_i$$

Let $\lambda = \|e\|_\infty$ and select an index i for which

$$|e_i| = \|e\|_\infty = \lambda$$

Here e is the vector $e = (e_1, e_2, \dots, e_n)$. Then from Equation (14) we get

$$|d_i|\,|e_i| \le h^4|g_i| + |c_i|\|e_{i+1}| + |a_{i-1}|\|e_{i-1}|$$

and using Equation (9), we obtain

$$\begin{aligned} |d_i|\lambda &\le h^4\|g\|_\infty + |c_i|\lambda + |a_{i-1}|\lambda \\ \lambda\left(|d_i| - |c_i| - |a_{i-1}|\right) &\le h^4\|g\|_\infty \\ h^2 v_i \lambda &\le h^4\|g\|_\infty \\ \|e\|_\infty &\le h^2\left[\|g\|_\infty / \inf v(t)\right] \end{aligned}$$

By Equation (13), $\|g\|_\infty \le \|x^{(4)}\|_\infty/12 + \|x'''\|_\infty/6$. The expression $\|g\|_\infty/\inf v(t)$ is a bound independent of h. Thus, we see that $\|e\|_\infty$ is $\mathcal{O}(h^2)$ as $h \to 0$.

PROBLEMS 8.9

1. Solve the two-point boundary-value problem

$$\begin{cases} x'' + 2x' + 10t = 0 \\ x(0) = 1 \qquad x(1) = 2 \end{cases}$$

for $x(\frac{1}{2})$ using the finite-difference method with $h = \frac{1}{2}$.

2. Show what modifications must be made in the linear system of equations (7) to cope with boundary conditions of the form

$$c_{11}x(a) + c_{12}x'(a) + c_{13} = c_{21}x(b) + c_{22}x'(b) + c_{23} = 0$$

3. Use Keller's theorem (Theorem 1, p. 591) to prove that the boundary-value problem (10) has a unique solution provided that the coefficient functions are continuous and $v > 0$.

COMPUTER PROBLEMS 8.9

1. Write a general-purpose computer program to solve linear two-point boundary-value problems by the finite-difference method as described in the text. Allow the user to furnish a, α, b, β, and n and the functions u, v, and w, as in Equations (3), (4), and (6).

2. (Continuation) Test the program written in the preceding computer problem on the following examples:

a. $$\begin{cases} x'' = -x \\ x(0) = 3 \qquad x(\frac{\pi}{2}) = 7 \end{cases}$$

b. $$\begin{cases} x'' = 2e^t - x \\ x(0) = 2 \qquad x(1) = e + \cos 1 \end{cases}$$

Also, compute the errors in the numerical solutions of these two test cases. The solutions are: **a.** $x(t) = 7\sin t + 3\cos t$ **b.** $x(t) = e^t + \cos t$

8.10 Boundary-Value Problems: Collocation

The method of collocation provides a strategy by which we can attack many problems in applied mathematics. First, we give a general description. Suppose that we have a linear operator L (such as an integral operator or differential operator) and we wish to solve the equation

$$Lu = w \tag{1}$$

In this equation, w is given and u is sought. Several approximate methods for solving Equation (1) begin by selecting some set of basic vectors $\{v_1, v_2, \dots, v_n\}$ and then trying to solve Equation (1) with a vector u of the form

$$u = c_1v_1 + c_2v_2 + \cdots + c_nv_n \tag{2}$$

Because L is a linear operator, we have

$$Lu = \sum_{j=1}^{n} c_jLv_j$$

and thus Equation (1) leads to

$$\sum_{j=1}^{n} c_j L v_j = w \tag{3}$$

In general, we shall not be able to solve for the coefficients $c_1, c_2, \dots, c_n$ in System (3). But perhaps we can make Equation (3) *almost true*. In the method of **collocation**, the vectors u, w, and v_j are all functions on a common domain. We then require that the functions w and $\sum_{j=1}^{n} c_j L v_j$ agree in value at n prescribed points:

$$\sum_{j=1}^{n} c_j (L v_j)(t_i) = w(t_i) \qquad (1 \le i \le n) \tag{4}$$

This is a system of n linear equations from which we can compute values for the n unknown coefficients, c_j. Of course, the functions v_j and the points t_i should be chosen so that the matrix with entries $(L v_j)(t_i)$ is nonsingular.

Sturm-Liouville Boundary-Value Problems

Now let us consider how this method would work on a Sturm-Liouville two-point boundary-value problem:

$$\begin{cases} u'' + pu' + qu = w \\ u(0) = 0 \qquad u(1) = 0 \end{cases} \tag{5}$$

Here the functions p, q, and w are all given and are assumed to be continuous on the interval $[0, 1]$. The function u is unknown; it, too, is defined on $[0, 1]$, but we expect it to be twice continuously differentiable. This problem becomes an example of the scheme outlined previously if we define the operator L by setting

$$Lu \equiv u'' + pu' + qu \tag{6}$$

We look for a solution in the vector space

$$V = \left\{u \in C^2[0, 1] : u(0) = u(1) = 0\right\} \tag{7}$$

Accordingly, if we select our set of basic functions $\{v_1, v_2, \dots v_n\}$ from V, then the homogeneous boundary conditions will be satisfied automatically. One set of functions that suggests itself is the (doubly indexed) set

$$v_{jk}(t) = t^j (1 - t)^k \qquad (j \ge 1,\ k \ge 1) \tag{8}$$

We readily verify that

$$v'_{jk} = j v_{j-1,k} - k v_{j,k-1} \tag{9}$$

$$v''_{jk} = j(j-1) v_{j-2,k} - 2jk v_{j-1,k-1} + k(k-1) v_{j,k-2} \tag{10}$$

From Equations (9) and (10), it is a simple matter to write down the function $L v_{jk}$. If n functions are chosen from the set described in Equation (8), and if n points t_i are chosen in $[0, 1]$, we can attempt to solve the collocation Equations (4) and obtain an approximate solution of the problem (5).

Cubic B-Splines

Perhaps a better choice of basic functions for such problems would be a set of B-splines. In describing how the B-splines can be used to advantage, let us take as our *model problem* a slightly more general one:

$$\begin{cases} u'' + pu' + qu = w \\ u(a) = \alpha \qquad u(b) = \beta \end{cases} \tag{11}$$

In order that our basic functions possess two continuous derivatives, we consider only the B-splines B_i^k with $k \geq 3$. For simplicity, we take $k = 3$. Also, we take the knots to be equally spaced: $t_{i+1} - t_i = h$. Finally, we shall use the knots as the collocation points. Let n be the number of basic functions to be used (and the number of coefficients to be determined). A total of n conditions should exist for a determination of the n coefficients. Since there will be two end conditions, namely,

$$\sum_{j=1}^{n} c_j v_j(a) = \alpha \quad \text{and} \quad \sum_{j=1}^{n} c_j v_j(b) = \beta \tag{12}$$

we see that there should be $n - 2$ collocation conditions:

$$\sum_{j=1}^{n} c_j (Lv_j)(t_i) = w(t_i) \qquad (1 \leq i \leq n-2) \tag{13}$$

These considerations lead us to define

$$h = (b-a)/(n-3) \tag{14}$$

$$t_i = a + (i-1)h \qquad (i = 0, \pm 1, \pm 2, \ldots) \tag{15}$$

The knots t_i that lie in $[a, b]$ are $a = t_1, t_2, \ldots, t_{n-2} = b$ (as is easily verified). These knots are the collocation points. Some knots outside the interval $[a, b]$ will be needed for defining the B-splines B_j^3. The arrangement of knots is shown in Figure 8.4.

The cubic B-splines of interest are the ones that are not identically 0 on $[a, b]$. These are $B_{-2}^3, B_{-1}^3, B_0^3, B_1^3, \ldots, B_{n-3}^3$. Accordingly, we put $v_j = B_{j-3}^3$ for $1 \leq j \leq n$.

Because the knots are equally spaced, the functions v_j can be obtained from a single B-spline denoted by B and defined by

$$B(t) = \begin{cases} (t+2)^3/6 & \text{on } [-2,-1] \\ \left[1 + 3(t+1) + 3(t+1)^2 - 3(t+1)^3\right]/6 & \text{on } [-1, 0] \\ B(-t) & \text{on } [0, 2] \\ 0 & \text{elsewhere} \end{cases} \tag{16}$$

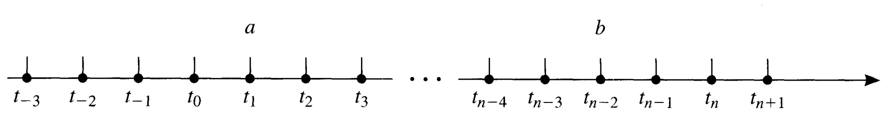

FIGURE 8.4
The configuration of knots

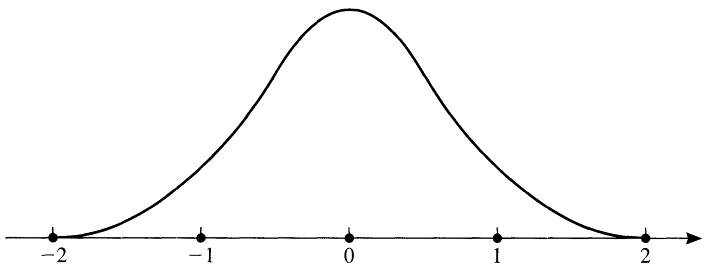

FIGURE 8.5
Cubic B-spline

Then, as we easily verify,

$$v_j(t) = B\left(\frac{t-a}{h} - j + 2\right) \tag{17}$$

The graph of the function B is shown in Figure 8.5, and the reader should verify that B is indeed a cubic spline.

The first and second derivatives of v_j are needed to compute $(Lv_j)(t_i)$. These derivatives are readily obtained from Equations (16) and (17). In a computer program to carry out the collocation method using the cubic B-splines, we would have to write subprograms to compute $B(t)$, $B'(t)$, and $B''(t)$. Then the matrix elements $(Lv_j)(t_i)$ could be computed efficiently. This will be a **banded matrix** because each function v_j has a small support. A general-purpose routine intended for *production* use should exploit the sparse nature of the coefficient matrix.

COMPUTER PROBLEM 8.10

1. a. Program a general-purpose code for solving a two-point boundary-value problem of the form

$$\begin{cases} u''(t) + p(t)u' + q(t)u(t) = w(t) \\ u(a) = \alpha \qquad u(b) = \beta \end{cases}$$

by the method of collocation. The user will specify: (1) functions p, q, and w; (2) real numbers a, α, b, and β, with $a < b$; (3) n, the number of *basic* functions to be used; and (4) the *basic* functions v_i and their derivatives v_i' and v_i''. The computer program will then generate $n-2$ collocation points t_i, for $1 \le i \le n-2$, equally spaced in the interval $[a, b]$, including the endpoints. The program next sets up a linear system of n equations with n unknowns $c_1, c_2, \ldots, c_n$, as follows:

$$\begin{cases} \displaystyle\sum_{j=1}^{n} c_j (Lv_j)(t_i) = w(t_i) \qquad (1 \le i \le n-2) \\ \displaystyle\sum_{j=1}^{n} c_j v_j(a) = \alpha \qquad \sum_{j=1}^{n} c_j v_j(b) = \beta \end{cases}$$

The program should then call a *linear equation solver* to compute the coefficients c_j. Finally, the approximate solution $u = \sum_{j=1}^{n} c_j v_j$ is tested by evaluating u and the *residual* $Lu - w$ at $2n - 5$ points equally spaced in $[a, b]$. These points will include the collocation points (where the residual should be 0) as well as the midpoints between collocation points.

b. Test the program written in part (a) by solving

$$\begin{cases} u'' + (\sin t)u' + (t^2 + 2)u = e^{t-3} \\ u(2.6) = 7 \qquad u(5.1) = -3 \end{cases}$$

Use cubic B-splines with equally spaced knots. The collocation points should be the knots on $[a, b]$.

8.11 Linear Differential Equations

Here we consider systems of n linear differential equations with constant coefficients. The systems are assumed to be **autonomous**, which means that the independent variable t is not present explicitly. Such a system has the appearance

$$\begin{cases} x_1' = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ x_2' = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \quad\vdots \\ x_n' = a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{cases} \tag{1}$$

In vector-matrix notation, this is simply

$$X' = AX \tag{2}$$

where $X = (x_1, x_2, \dots, x_n)^T$ and $X' = (x_1', x_2', \dots, x_n')^T$.

Eigenvalues and Eigenvectors

Let us attempt to solve this with a vector of the form $X(t) = e^{\lambda t}V$, where V is a constant vector. On substituting this *trial* solution in Equation (2), we obtain

$$\lambda e^{\lambda t} V = e^{\lambda t} AV \tag{3}$$

Thus, if

$$AV = \lambda V$$

then the vector function $e^{\lambda t}V$ is indeed a solution of Equation (2). We have proved the following theorem.

THEOREM 1 Eigenvalues and Eigenvectors, Linear Differential Equation

If λ is an eigenvalue of the matrix A, and if V is a corresponding eigenvector, then $X(t) = e^{\lambda t}V$ is a solution of the equation $X' = AX$.

This theorem indicates that solving the differential equation $X' = AX$ will involve some knowledge of the eigenvalues and eigenvectors of A. Furthermore, the theory of similar matrices will show us how to simplify a system of linear differential equations by changing variables. These matters will be taken up presently.

The most pleasant state of affairs for the equation $X' = AX$ is described in the next theorem.

THEOREM 2 Theorem on Basis of Solution Space

If the $n \times n$ matrix A has a linearly independent set of eigenvectors $V_1, V_2, \ldots, V_n$ with $AV_i = \lambda_i V_i$, then the solution space of the equation $X' = AX$ has a basis $X_i = e^{\lambda_i t} V_i$ with $1 \le i \le n$.

Proof The set $\{X_1, X_2, \ldots, X_n\}$ is linearly independent because $\sum_{i=1}^{n} c_i X_i = 0$ leads to $\sum_{i=1}^{n} c_i e^{\lambda_i t} V_i = 0$, $c_i e^{\lambda_i t} = 0$, and $c_i = 0$, for $1 \le i \le n$.

To prove that the set in question spans the solution space of $X' = AX$, let X be any solution. As an element of $\mathbb{R}^n$ or $\mathbb{C}^n$, the initial-value vector $X(0)$ is a linear combination of $V_1, V_2, \ldots, V_n$, say

$$X(0) = \sum_{i=1}^{n} c_i V_i$$

Define $Y = \sum_{i=1}^{n} c_i X_i$. Then

$$Y' = \sum_{i=1}^{n} c_i X_i' = \sum_{i=1}^{n} c_i \lambda_i e^{\lambda_i t} V_i = \sum_{i=1}^{n} c_i e^{\lambda_i t} A V_i = A\left(\sum_{i=1}^{n} c_i X_i \right) = AY$$

Thus Y and X are solutions of the differential equation, and they have the same initial value: $Y(0) = X(0)$. (Why?) By the uniqueness theorem for the initial-value problem, we conclude that $Y = X$, or in other words, $X = \sum_{i=1}^{n} c_i X_i$. ■

If A has the property mentioned in Theorem 2, then there is a nonsingular matrix P whose columns are the vectors $V_1, V_2, \ldots, V_n$. The equation $AV_i = \lambda_i V_i$ translates into matrix notation as

$$AP = P\Lambda \tag{4}$$

where Λ is the diagonal matrix having $\lambda_1, \lambda_2, \ldots, \lambda_n$ on its diagonal. Consider the change of (dependent) variables described by $X = PY$. Since P is nonsingular, we can recover Y from X. Now Y has this property:

$$Y' = P^{-1}X' = P^{-1}AX = P^{-1}APY = \Lambda Y \tag{5}$$

Thus, the differential equation for Y is much simpler than the one for X, because Λ is a diagonal matrix. The individual equations in the system $Y' = \Lambda Y$ are **uncoupled** and can be solved separately.

EXAMPLE 1 Solve the initial-value problem $X' = AX$ when

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \qquad X(0) = \begin{bmatrix} 5 \\ 7 \\ 6 \end{bmatrix}$$

Solution The matrix $A - \lambda I$ is

$$\begin{bmatrix} 1-\lambda & 0 & 1 \\ 0 & -\lambda & 0 \\ 0 & 0 & -1-\lambda \end{bmatrix}$$

and its determinant is the **characteristic polynomial of** A:

$$(1-\lambda)(-\lambda)(-1-\lambda)$$

The roots of this polynomial are the *eigenvalues of* A and are $\lambda_1 = 1$, $\lambda_2 = 0$, and $\lambda_3 = -1$. For each of these, we find an eigenvector by solving $AV_i = \lambda_i V_i$. Placing these as columns in a matrix P, we obtain

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Next, we find that

$$P^{-1} = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

If $Y = (y_1, y_2, y_3)^T$, then the initial-value problem for Y is $Y' = \Lambda Y$, where

$$\Lambda = P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Hence, we have

$$\begin{cases} y_1' = y_1 \\ y_2' = 0 \\ y_3' = -y_3 \end{cases} \qquad Y(0) = P^{-1}X(0) = \begin{bmatrix} 8 \\ 7 \\ -3 \end{bmatrix}$$

and its solution is

$$y_1 = 8e^t \qquad y_2 = 7 \qquad y_3 = -3e^{-t}$$

Because $X = PY$, the corresponding solution for $X = (x_1, x_2, x_3)^T$ is

$$x_1 = 8e^t - 3e^{-t} \qquad x_2 = 7 \qquad x_3 = 6e^{-t}$$ ■

Matrix Exponential

There is an elegant formal method of solving a system $X' = AX$. This method makes it unnecessary to refer to the eigenvalues of A until we wish to compute solutions numerically. We begin by defining the matrix exponential.

DEFINITION 1 Definition of Matrix Exponential

If A is a square matrix, we put

$$e^A = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots \tag{6}$$

This definition is derived from the standard series

$$e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \cdots \tag{7}$$

by substituting a matrix for the complex variable z. To see that the series for e^A converges, we take any norm on $\mathbb{C}^n$ and use the corresponding subordinate matrix norm for $n \times n$ matrices. The *tail* of our series can be estimated as follows:

$$\left\| \sum_{k=m}^{\infty} \frac{1}{k!}A^k \right\| \le \sum_{k=m}^{\infty} \frac{1}{k!}\|A^k\| \le \sum_{k=m}^{\infty} \frac{1}{k!}\|A\|^k \tag{8}$$

This last expression is the tail of the ordinary exponential series when $z = \|A\|$. Thus, the tail of the series for e^A converges to 0 as $m \to \infty$. (This argument assumes as known the completeness of the space of $n \times n$ matrices with the given norm.)

If t is a real variable, then $tA = At$, and our definition yields

$$e^{At} = \sum_{k=0}^{\infty} \frac{t^k}{k!}A^k \tag{9}$$

Differentiation with respect to t in the series and subsequent simplification give us

$$\frac{d}{dt}e^{At} = Ae^{At} \tag{10}$$

THEOREM 3 Theorem on Solution of Initial-Value Problem

The solution of the initial-value problem

$$\begin{cases} X' = AX \\ X(0) \textit{ prescribed} \end{cases}$$

is $X(t) = e^{At}X(0)$.

Proof From the formula $X = e^{At}W$, with $W = X(0)$, we have at once

$$\begin{cases} X' = Ae^{At}W = AX \\ X(0) = e^{A0}W = W \end{cases}$$

■

Diagonal and Diagonalizable Matrices

To use the preceding result in practice, it will be necessary to compute the matrix exponential in an efficient manner. Let us start with the case in which A is a *diagonal* matrix. If $A = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, then $A^k = \text{diag}(\lambda_1^k, \lambda_2^k, \ldots, \lambda_n^k)$, as is easily verified. Hence, for such an A,

$$\begin{aligned} e^{At} &= \sum_{k=0}^{\infty} \frac{t^k}{k!} \operatorname{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k) \\ &= \operatorname{diag}(e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}) \end{aligned} \tag{11}$$

In this special case, the solution of the differential equation $X' = AX$ has the components

$$x_i(t) = e^{\lambda_i t} x_i(0) \qquad (1 \le i \le n)$$

The analysis just presented carries over at once to the case in which A is not diagonal but is **diagonalizable**. This term means that A is similar to a diagonal matrix, or in other words, that $P^{-1}AP = \Lambda$ for some diagonal matrix Λ and some nonsingular matrix P. If this is true, the change of variable $X = PY$ changes the differential equation $X' = AX$ into $Y' = \Lambda Y$, as shown in Equation (5). The initial condition $X(0)$ becomes $Y(0) = P^{-1}X(0)$, and the solution is

$$X = PY = P\big(e^{\Lambda t} P^{-1} X(0)\big) = P \operatorname{diag}(e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}) P^{-1} X(0) \tag{12}$$

Jordan Blocks

We have postponed until now the discussion of the case when A is not diagonalizable. This means that $\mathbb{C}^n$ does not have a basis consisting of eigenvectors of A. Two simple examples of such behavior are given by

$$J(\lambda, 2) = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \qquad J(\lambda, 3) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

Let us consider $J(\lambda, 3)$ in detail. Since this matrix is upper triangular, its diagonal elements are its eigenvalues. Thus, $J(\lambda, 3)$ has the single eigenvalue λ, and if we write out the equation $J(\lambda, 3)X = \lambda X$, we have

$$\begin{cases} \lambda x_1 + x_2 = \lambda x_1 \\ \lambda x_2 + x_3 = \lambda x_2 \\ \qquad\ \ \lambda x_3 = \lambda x_3 \end{cases}$$

This obviously implies that $x_2 = x_3 = 0$, and therefore the only solutions have the form $X = (\beta, 0, 0)^T$. The solutions form a one-dimensional space. In other words, the eigenvectors of $J(\lambda, 3)$ span only a one-dimensional subspace in $\mathbb{R}^3$ or $\mathbb{C}^3$. Matrices of the form $J(\lambda, k)$ exist for each $k \ge 2$, and they all have the same property that we just observed for $J(\lambda, 3)$. These matrices are called **Jordan blocks**. The principal theorem in the subject follows.

THEOREM 4 Theorem on Jordan Blocks

Every square matrix is similar to a block diagonal matrix having Jordan blocks on its diagonal.

The special form referred to in the theorem is called the **Jordan canonical form** of the given matrix. Here are some examples of 3×3 matrices in Jordan canonical form:

$$\begin{bmatrix} 7 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

The first of these contains three Jordan blocks, whereas the second, third, and fourth contain two. The fifth matrix is itself a Jordan block.

A Jordan block can be written in the form

$$J(\lambda, k) = \lambda I_k + H_k \tag{13}$$

where I_k denotes the $k \times k$ identity matrix, and H_k denotes a $k \times k$ matrix of the form

$$H_k = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix} \tag{14}$$

The effect of multiplying a vector by H_k is easily seen. We have

$$\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \vdots \\ \xi_{k-1} \\ \xi_k \end{bmatrix} = \begin{bmatrix} \xi_2 \\ \xi_3 \\ \xi_4 \\ \vdots \\ \xi_k \end{bmatrix} \tag{15}$$

It now is evident that $H_k^k V = 0$ because each application of H_k to V removes one component. Thus, H_k is **nilpotent**; indeed, $H_k^k = 0$. This fact will be useful in computing e^{At} when A is a Jordan block. We have

$$\begin{aligned} e^{(\lambda I_k + H_k)t} &= e^{t\lambda I_k} e^{H_k t} \\ &= \sum_{j=0}^{\infty} \frac{(\lambda t I_k)^j}{j!} \sum_{j=0}^{\infty} \frac{(t H_k)^j}{j!} \\ &= e^{\lambda t} \left[I_k + t H_k + \frac{t^2}{2!} H_k^2 + \cdots + \frac{t^{k-1}}{(k-1)!} H_k^{k-1} \right] \end{aligned} \tag{16}$$

The series terminates as shown because the kth power (and all subsequent powers) of H_k is 0. Notice that in the preceding calculation we used a valid case of the formula $e^{A+B} = e^A e^B$. (See Problem **8.11.13**, p. 607.)

Now let us solve the differential equation $X' = AX$ when A is a Jordan block, say $A = \lambda I_k + H_k$. The solution is obtained by applying Theorem 3 and Equation (16); it is

$$X(t) = e^{At}X(0) = e^{\lambda t}\left[I_k + tH_k + \frac{t^2}{2!}H_k + \cdots + \frac{t^{k-1}}{(k-1)!}H_k^{k-1}\right]X(0) \tag{17}$$

EXAMPLE 2 Solve the initial-value problem $X' = AX$ when

$$A = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \qquad X(0) = \begin{bmatrix} 7 \\ 5 \\ 3 \\ 9 \end{bmatrix}$$

Solution The matrix A is of the form $3I_4 + H_4$. The solution, given by Equation (17), is

$$\begin{aligned} X(t) &= e^{3t}\left(I + tH_4 + \tfrac{1}{2}t^2H_4^2 + \tfrac{1}{6}t^3H_4^3\right)X(0) \\ &= e^{3t}\begin{bmatrix} 7 \\ 5 \\ 3 \\ 9 \end{bmatrix} + te^{3t}\begin{bmatrix} 5 \\ 3 \\ 9 \\ 0 \end{bmatrix} + \frac{1}{2}t^2e^{3t}\begin{bmatrix} 3 \\ 9 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{6}t^3e^{3t}\begin{bmatrix} 9 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 7 + 5t + 1.5t^2 + 1.5t^3 \\ 5 + 3t + 4.5t^2 \\ 3 + 9t \\ 9 \end{bmatrix} e^{3t} \end{aligned}$$

■

Solution in Complete Generality

It is now possible to describe the solution of the system of differential equations $X' = AX$ in complete generality, starting with the Jordan canonical form of A and the similarity transformation that brings it about. Suppose that

$$P^{-1}AP = C$$

where C is the Jordan canonical form of A. We know that the change of variables $X = PY$ will lead to the differential equation $Y' = CY$. (In this connection see Equation (5).) This differential equation, $Y' = CY$, can be divided into *uncoupled* blocks. To see this, consider an example such as

$$C = \begin{bmatrix} 5 & 1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 7 & 1 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}$$

The differential equations are

$$\begin{cases} y_1' = 5y_1 + y_2 \\ y_2' = 5y_2 + y_3 \\ y_3' = 5y_3 \\ y_4' = 7y_4 + y_5 \\ y_5' = 7y_5 \end{cases}$$

It is clear that the first group of three equations can be solved by itself, and the last pair can be solved by itself. The general case is quite analogous, and we see that each Jordan block in C gives rise to a set of differential equations that is uncoupled from the remaining equations. With each Jordan block, a piece of the solution is obtained as in Equation (17).

EXAMPLE 3 Using the matrix C above, solve the initial-value problem $Y' = CY$, with initial conditions $Y(0) = (3, 2, 8, 4, 1)^T$.

Solution The two uncoupled systems are

$$\begin{bmatrix} y_1' y_2' y_3' \end{bmatrix} = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \qquad \text{initial value} \quad \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} y_4' \\ y_5' \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} y_4 \\ y_5 \end{bmatrix} \qquad \text{initial value} \quad \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

By using the method illustrated in Example 2, we obtain these solutions:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = e^{5t}\left(I + tH_3 + \frac{1}{2}t^2 H_3^2\right) \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix} = e^{5t} \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix} + te^{5t} \begin{bmatrix} 2 \\ 8 \\ 0 \end{bmatrix} + \frac{1}{2}t^2 e^{5t} \begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 + 2t + 4t^2 \\ 2 + 8t \\ 8 \end{bmatrix} e^{5t} \begin{bmatrix} y_4 \\ y_5 \end{bmatrix}$$

$$= e^{7t}(I + tH_2) \begin{bmatrix} 4 \\ 1 \end{bmatrix} = e^{7t} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + te^{7t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 + t1 \end{bmatrix} e^{7t}$$ ■

In the theory of the linear differential equation $X' = AX$, the exponential e^{At} is called the **fundamental matrix** of the equation. We have seen that it is the key to solving the initial-value problem associated with the differential equation. If we possess the Jordan canonical form C of A and know the similarity transformation

$$P^{-1}AP = C$$

then we can change variables by $X = PY$, solve the equation $Y' = CY$, and return to X, obtaining ultimately

$$X = PY = Pe^{Ct}Y(0) = Pe^{Ct}P^{-1}X(0) \tag{18}$$

On the other hand, we know another form for the solution, namely, $X = e^{At}X(0)$. Comparing this to Equation (18), we conclude that

$$e^{At} = Pe^{Ct}P^{-1}$$

This is one way of computing the fundamental matrix.

Nonhomogeneous Problem

The principles that have been established can now be applied to the **nonhomogeneous problem**

$$X' = AX + W \tag{19}$$

where W can be a vector function of t. Let us consider first the case when A is diagonalizable. Then a similarity transform $P^{-1}AP = \Lambda$ produces a diagonal matrix Λ. The change of variable $X = PY$ transforms Equation (19) to

$$Y' = \Lambda Y + P^{-1}W \tag{20}$$

This is a completely decoupled set of n equations, of which a typical one is of the form

$$\eta'(t) = \lambda\eta(t) + g(t) \tag{21}$$

The solution of this is

$$\eta(t) = e^{\lambda t}\left[\eta(0) + \int_0^t e^{-\lambda s}g(s)\,ds\right] \tag{22}$$

EXAMPLE 4 Solve the equation $X' = AX + W$ when

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \qquad W = \begin{bmatrix} t^2 \\ t \\ \sin t \end{bmatrix} \qquad X(0) = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Solution The system comprises the following individual uncoupled equations:

$$\begin{cases} x_1' = t^2 \\ x_2' = x_2 + t \\ x_3' = 2x_3 + \sin t \end{cases}$$

The solutions, obtained by using Equation (22), are

$$x_1(t) = 5 + \int_0^t s^2\,ds = 5 + \frac{1}{3}t^3$$

$$x_2(t) = e^t\left[7 + \int_0^t e^{-s}s\,ds\right] = 8e^t - t - 1$$

$$x_3(t) = e^{2t}\left[9 + \int_0^t e^{-2s} \sin s \, ds\right] = \frac{46}{5}e^{2t} - \frac{2}{5}\sin t - \frac{1}{5}\cos t \qquad \blacksquare$$

If the matrix A in the equation $X' = AX + W$ is not diagonalizable, we can use the Jordan canonical form and a change of variables to split the problem into uncoupled subsystems. Thus, it suffices to illustrate the procedure with a single Jordan block.

EXAMPLE 5 Solve the equation $X' = AX + W$ when

$$A = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \qquad W = \begin{bmatrix} t^2 \\ t \\ \sin t \end{bmatrix} \qquad X(0) = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Solution The individual equations are

$$\begin{cases} x_1' = 5x_1 + x_2 + t^2 \\ x_2' = 5x_2 + x_3 + t \\ x_3' = 5x_3 + \sin t \end{cases}$$

The recommended procedure here is to begin at the bottom and solve the equations in reverse order. This leads to

$$\begin{aligned}
x_3(t) &= e^{5t}\left\{9 + \int_0^t e^{-5s} \sin s \, ds\right\} \\
&= \left(9 + \frac{1}{26}\right)e^{5t} - \frac{5}{26}\sin t - \frac{1}{26}\cos t \\
x_2(t) &= e^{5t}\left\{7 + \int_0^t e^{-5s}\left[x_3(s) + s\right] ds\right\} \\
&= \left(7 + \frac{1}{25} - \frac{10}{26^2}\right)e^{5t} + \frac{1}{2}\left(9 + \frac{1}{26}\right)te^{5t} \\
&\quad + \frac{2}{26^2}(12\sin t + t\cos t) - \frac{1}{5}t - \frac{1}{25}x_1(t) \\
&= e^{5t}\left\{5 + \int_0^t e^{-5s}\left[x_2(s) + s^2\right] ds\right\} \\
&= \left(5 + \frac{74}{26^3}\right)e^{5t} + \left(7 + \frac{1}{25} - \frac{10}{26^2}\right)te^{5t} + \frac{1}{2}\left(9 + \frac{1}{26}\right)t^2e^{5t} \\
&\quad - \frac{2}{26^3}(55\sin t + 37\cos t) - \frac{1}{5}t^2 - \frac{1}{25}t \qquad \blacksquare
\end{aligned}$$

The computation of a matrix exponential e^A is a task that should be undertaken with full cognizance of the possible pitfalls. We recommend to the reader the article of Moler and Van Loan [1978]. Their investigation indicates that the following four-step procedure should work well in most cases:

1. Let j be the first positive integer such that $\|A\|/2^j \le 1/2$.
2. If a relative error tolerance ε is given, select p to be the first positive integer for which $2^{p-3}(p+1) \ge 1/\varepsilon$.
3. Compute $e^{A/2^j}$ from the truncated Taylor series $e^z = \sum_{k=0}^{p} z^k/k!$
4. Square the computed $e^{A/2^j}$ from Step **3** j times to obtain $e^A = (e^{A/2^j})^{2^j}$.

The computed value of e^A from Step **4** is e^{A+E}, where E is a matrix satisfying $\|E\|/\|A\| \le \varepsilon$. This conclusion is Corollary 1 in the cited reference.

PROBLEMS 8.11

1. Find the general solution of the system $X' = AX$ when

$$A = \begin{bmatrix} 1 & 0 & 3 \\ -1 & 1 & -1 \\ 3 & 0 & 1 \end{bmatrix}$$

2. (Continuation) Find the solution of the equation in the preceding problem subject to the initial condition $X_0 = (-1, 4, 7)^T$.
3. Find the general solution of the system

$$\begin{cases} x_1' = 3x_1 - 5x_2 \\ x_2' = 2x_1 + x_2 \end{cases}$$

4. Prove that for any $n \times n$ matrix A, e^A is nonsingular. *Caution:* Do not suppose that $e^A e^B = e^{A+B}$ for all $n \times n$ matrices A and B. (See the next problem.)
5. (Continuation) Show by example that the equation $e^A e^B = e^{A+B}$ is not always true. *Hint:* Consider e^{At}, e^{Bt}, and $e^{(A+B)t}$.
6. Prove that if $A = P^{-1}BP$, then $e^A = P^{-1}e^B P$.
7. Show in detail how the estimate in Equation (8) and a completeness argument are used to establish convergence in Equation (6).
8. Use induction on k to prove that if $V_1, V_2, \dots, V_k$ are eigenvectors of a matrix corresponding to k distinct eigenvalues, then $\{V_1, V_2, \dots, V_k\}$ is linearly independent.
9. Prove Equation (10) by differentiation in the series for e^{At}.
10. Use the results in Problem **8.11.6** (above) to establish the result in Equation (12) directly—that is, without the change of variable.
11. Let B be an $n \times n$ matrix, and let $V_1, V_2, \dots, V_k$ be vectors in $\mathbb{R}^n$ such that $V_1 \ne 0$, $BV_1 = 0$, $BV_2 = V_1, \dots, BV_k = V_{k-1}$. Use induction on k to prove that $\{V_1, V_2, \dots, V_k\}$ is linearly independent. How large can k be?
12. Find the exact conditions on a matrix of order 2 so that its Jordan canonical form will contain only one Jordan block.
13. Prove that $e^{A+B} = e^A e^B$ if and only if $AB = BA$. (See Problem **8.11.5**, above.)
14. Prove that if A and B are diagonalized by the same similarity transformation (i.e., PAP^{-1} and PBP^{-1} are diagonal), then $AB = BA$.
15. Prove that the solution of the linear system

$$X' = AX + V(t) \qquad X(0) = W$$

is given by

$$X(t) = e^{At}W + e^{At}\int_0^t e^{-As}V(s)\,ds$$

Explain what is meant by the indefinite integration appearing here.

16. What is the solution of the initial-value problem $X' = AX$ when X is specified at a point t_0 other than 0?

17. Show that the fundamental matrix for the system

$$X' = AX \quad \text{with} \quad A = \begin{bmatrix} -1 & 6 \\ 1 & -2 \end{bmatrix}$$

is

$$\frac{1}{5}\begin{bmatrix} 2e^{-4t} + 3e^t & -6e^{-4t} + 6e^t \\ -e^{-4t} + e^t & 3e^{-4t} + 2e^t \end{bmatrix}$$

18. Prove that the jth column in the fundamental matrix is a solution of the initial-value problem $X' = AX$, $X(0) = U_j$, where U_j is the jth standard unit vector.

19. Make a guess as to the matrix inverse of e^A and then prove that you are right. *Caution:* In general, $e^{A+B} \neq e^A e^B$.

20. Solve $X' = AX$ when

$$A = \begin{bmatrix} 5 & 4 & 3 \\ -1 & 0 & -3 \\ 1 & -2 & 1 \end{bmatrix} \quad \text{and} \quad X(0) = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$$

21. Complete Example 4.

22. Prove that a matrix that is not diagonalizable is the limit of a sequence of diagonalizable matrices.

23. (Continuation) Consider the linear system $X' = AX$, $X(0) = V$, in which A is not diagonalizable. If B is a diagonalizable matrix close to A, what can be said about the solution of $Y' = BY$, $Y(0) = V$?

24. Prove that $\det e^A = e^{\mathrm{tr}(A)}$, where A is any $n \times n$ matrix and $\mathrm{tr}(A)$ is the **trace** of A—that is, the sum of the diagonal elements in A.

25. For an $n \times n$ matrix A, let $|A| = \sum_{i=1}^n \sum_{j=1}^n |a_{ij}|$. Prove that

$$|e^A| \leq n - 1 + e^{|A|}$$

26. Although in general $e^A e^B \neq e^{A+B}$, prove that $e^{A+B} = \lim_{k\to\infty}[e^{A/k}e^{B/k}]^k$.

8.12 Stiff Equations

Stiffness, in a system of ordinary differential equations, refers to a wide disparity in the time scales of the components in the vector solution. Some numerical procedures that are quite satisfactory in general will perform poorly on stiff equations. This happens when stability in the numerical solution can be achieved only with very small step size.

Stiff differential equations arise in a number of applications. For example, in the control of a space vehicle the path is expected to be quite smooth, but very rapid corrections in the course can be made if any deviation from the programmed flight path is detected. Another source of such problems is in the monitoring of chemical processes, because a great diversity of time scales can exist for the physical and chemical changes occurring. In electrical circuit theory, stiff problems arise because transients with time scales on the order of a microsecond can be imposed on the generally smooth behavior of the circuit.

Euler's Method

The numerical difficulties can be illustrated by using Euler's method on a simple model problem. Euler's method is the Taylor method of order 1, and for the initial-value problem

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \tag{1}$$

it proceeds by using the equation

$$x_{n+1} = x_n + hf(t_n, x_n) \qquad (n \geq 0) \tag{2}$$

Consider now the result of using Euler's method on this simple test problem:

$$\begin{cases} x' = \lambda x \\ x(0) = 1 \end{cases} \tag{3}$$

Euler's method produces the numerical solution

$$\begin{cases} x_0 = 1 \\ x_{n+1} = x_n + h\lambda x_n = (1 + h\lambda)x_n \end{cases} \tag{4}$$

Thus, at the nth step, the approximate solution is

$$x_n = (1 + h\lambda)^n \tag{5}$$

On the other hand, the actual solution of Equation (3) is

$$x(t) = e^{\lambda t} \tag{6}$$

If $\lambda < 0$, the solution in Equation (6) is exponentially decaying. It is tending to the *steady state* of 0 as t goes to infinity. The numerical solution in Equation (5) will tend to 0 if and only if $|1 + h\lambda| < 1$. This forces us to choose $h > 0$ so that $1 + h\lambda > -1$, and because $\lambda < 0$, we must impose the condition $h < -2/\lambda$.

For example, if $\lambda = -20$, we have to take $h < 0.1$, although the solution that we are trying to track is extremely flat (and practically 0) shortly after the initial instant, $t = 0$, where $x = 1$. We note that $x(t) = e^{-20t} \leq 2.1 \times 10^{-9}$ for $t \geq 1$. Thus, the numerical solution must proceed with *small* steps in a region where the nature of the solution indicates that *large* steps may be taken. This phenomenon is one aspect of stiffness. A function such as e^{-20t} that decays to 0 almost immediately (i.e., with very large negative slope) is said to be a **transient** because its physical effect is of brief duration. We expect that the numerical tracking of a transient function will require a small step size until the transient effect has become negligible; after that, a good numerical method should permit a large step size. Euler's method does not fulfill this desideratum.

Modified Euler's Method

By contrast, the implicit Euler's method *will* meet the criterion just mentioned. The implicit Euler's method is defined by the equation

$$x_{n+1} = x_n + hf(t_{n+1}, x_{n+1}) \qquad (n \geq 0) \tag{7}$$

When this method is applied to the test problem in Equation (3), we obtain

$$\begin{cases} x_0 = 1 \\ x_{n+1} = x_n + h\lambda x_{n+1} \end{cases} \tag{8}$$

This gives

$$x_{n+1} = (1 - h\lambda)^{-1} x_n \tag{9}$$

Hence, at the nth step, we have

$$x_n = (1 - h\lambda)^{-n} \tag{10}$$

For negative λ, our requirement that the numerical solution should mimic the actual solution becomes

$$|1 - h\lambda|^{-1} < 1 \tag{11}$$

This is obviously true for *all* (positive) step sizes h.

Systems of Differential Equations

Similar considerations apply to systems of differential equations. Again we appeal to the principle that a good numerical method should perform well on simple linear test cases. (Recall that this principle was used in Section 8.5, p. 558, to deduce that *stability* and *consistency* are essential attributes of acceptable linear multistep methods.) Here is a simple test case involving a system of two differential equations:

$$\begin{cases} x' = \alpha x + \beta y & x(0) = 2 \\ y' = \beta x + \alpha y & y(0) = 0 \end{cases} \tag{12}$$

The solution of this system is

$$\begin{cases} x(t) = e^{(\alpha+\beta)t} + e^{(\alpha-\beta)t} \\ y(t) = e^{(\alpha+\beta)t} - e^{(\alpha-\beta)t} \end{cases} \tag{13}$$

If Euler's method is used to compute a numerical solution to Equations (12), the formulas for advancing the solution are

$$\begin{cases} x_{n+1} = x_n + h(\alpha x_n + \beta y_n) & x_0 = 2 \\ y_{n+1} = y_n + h(\beta x_n + \alpha y_n) & y_0 = 0 \end{cases} \tag{14}$$

The solutions of these difference equations are

$$\begin{cases} x_n = (1 + \alpha h + \beta h)^n + (1 + \alpha h - \beta h)^n \\ y_n = (1 + \alpha h + \beta h)^n - (1 + \alpha h - \beta h)^n \end{cases} \tag{15}$$

The case in which we are interested is defined by $\alpha < \beta < 0$. With this supposition, the solutions in Equations (13) are decaying exponentially to 0. In order that the numerical solutions in Equations (15) will mimic this behavior, we want

$$|1 + \alpha h + \beta h| < 1 \qquad |1 + \alpha h - \beta h| < 1 \tag{16}$$

The inequalities in (16) are equivalent to the single condition

$$0 < h < -2/(\alpha + \beta)$$

because $\alpha < \beta < 0$. To see what can happen, suppose that $\alpha = -20$ and $\beta = -19$. Then our solutions are combinations of e^{-39t} and e^{-t}. The first of these is a *transient* function, which after a short time interval is negligible compared to e^{-t}. Yet the transient will control the allowable step size throughout the entire numerical solution.

General Linear Multistep Methods

Let us examine the general linear multistep method of Section 8.5 (p. 557) to see what additional properties it should have in order to perform satisfactorily on our single test equation (3). The general method has the form

$$a_k x_n + a_{k-1} x_{n-1} + \cdots + a_0 x_{n-k} = h[b_k f_n + b_{k-1} f_{n-1} + \cdots + b_0 f_{n-k}] \tag{17}$$

When this is applied to the test problem, we obtain

$$a_k x_n + a_{k-1} x_{n-1} + \cdots + a_0 x_{n-k} = h\lambda(b_k x_n + b_{k-1} x_{n-1} + \cdots + b_0 x_{n-k}) \tag{18}$$

Thus, our numerical solution will solve the homogeneous linear difference equation

$$(a_k - h\lambda b_k)x_n + (a_{k-1} - h\lambda b_{k-1})x_{n-1} + \cdots + (a_0 - h\lambda b_0)x_{n-k} = 0 \tag{19}$$

The solutions of this equation will be combinations of certain *basic* solutions of the type $x_n = r^n$, where r is a root of the polynomial

$$\phi(z) = (a_k - h\lambda b_k)z^k + (a_{k-1} - h\lambda b_{k-1})z^{k-1} + \cdots + (a_0 - h\lambda b_0) \tag{20}$$

Notice that the polynomial defined in Equation (20) is of the form $\phi = p - \lambda hq$, where p and q are the polynomials that were useful in Section 8.5 (p. 558), namely,

$$p(z) = a_k z^k + a_{k-1} z^{k-1} + \cdots + a_1 z + a_0 \tag{21}$$

$$q(z) = b_k z^k + b_{k-1} z^{k-1} + \cdots + b_1 z + b_0 \tag{22}$$

A-Stability

If $\lambda < 0$ in the test problem (3), then the solution is exponentially decaying. For the numerical solution obtained from the multistep method in Equation (17) to reflect this behavior, it is necessary that all roots of the polynomial ϕ in Equation (20) lie in the disk $|z| < 1$. For complex values of λ, say $\lambda = \mu + i\nu$, the solution of the test problem is

$$x(t) = e^{\lambda t} = e^{\mu t} e^{i\nu t} = e^{\mu t}(\cos \nu t + i \sin \nu t)$$

Exponential decay in this case corresponds to $\mu < 0$. In order that the multistep method perform well on such problems, we would like the roots of ϕ to be interior to the unit disk whenever $h > 0$ and $\mathrm{Re}(\lambda) < 0$. This property is called **A-stability**.

The implicit Euler's method in Equation (7) is A-stable, as shown in Equation (11). The implicit trapezoid method, defined by

$$x_n - x_{n-1} = \tfrac{1}{2} h[f_n + f_{n-1}] \tag{23}$$

is A-stable because the polynomial ϕ is

$$\phi(z) = z - 1 - \lambda h\left(\tfrac{1}{2} z + \tfrac{1}{2}\right)$$

and its root is $z = (2 + \lambda h)/(2 - \lambda h)$; we easily see that this root is interior to the unit disk when $h > 0$ and $\mathrm{Re}(\lambda) < 0$.

An important theorem, Dahlquist [1963], states that an A-stable linear multistep method must be an implicit method, and its order cannot exceed 2. This result places a severe restriction on A-stable methods. The implicit trapezoid rule is often used on stiff equations because it has the least truncation error among all A-stable linear multistep methods.

Region of Absolute Stability

Every multistep method possesses a **region of absolute stability**. This is the set of complex numbers ω such that the roots of $p - \omega q$ lie in the interior of the unit disk. From the preceding discussion it follows that a method will work well on the test problem $x' = \lambda x$ if λh lies in the region of absolute stability. We also notice that a method is A-stable if and only if its region of absolute stability contains the left half-plane.

EXAMPLE 1 What is the region of absolute stability for the Euler method?

Solution The Euler method is defined by the equation

$$x_n - x_{n-1} = hf_{n-1}$$

Hence, $p(z) = z - 1$, $q(z) = 1$, and $\phi(z) = z - 1 - \omega$, where $\omega = \lambda h$. Now the root of ϕ is $z = 1 + \omega$. For *all* roots of ϕ to be interior to the unit disk, we require $|1 + \omega| < 1$. This is a disk of radius 1, in the complex plane, with its center at -1. ■

Because A-stability is such a severe restriction, methods lacking the property are nevertheless used on stiff problems. When this is done, we hope that $\omega = \lambda h$ will lie in the region of absolute stability for the method. The quantity λ pertains to the differential equation being solved. For a single linear equation, λ is the coefficient of x in the equation. For a single nonlinear equation, λ can be a constant that arises by locally approximating the equation by a linear equation. For a system of linear equations, say $X' = AX$, λ can be any one of the eigenvalues of A. Thus, h times each eigenvalue of A should lie in the region of absolute stability for the method used. For a system of nonlinear equations, we can contemplate local approximation of the given system by linear systems and application of the preceding criterion. This idea is usually impossible to implement, and accordingly, the only safe strategy in the case of difficult stiff problems may be to revert to the trapezoid rule. Some of the best current codes for the initial-value problem make an effort to detect stiffness in the course of the step-by-step solution and to take appropriate defensive action when stiffness occurs. The reader is referred to Gear [1971], Shampine and Allen [1973], and Shampine and Gordon [1975] for details. In general, the higher-order methods have smaller regions of absolute stability, and in the presence of stiffness these adaptive codes shift to methods of lower order having more favorable regions of absolute stability. (See Byrne and Hindmarsh [1987] or Shampine and Gear [1979].)

Nonlinear Equations

To see the relevance of the foregoing considerations to systems of *nonlinear* equations, let us turn to a typical system, assumed to be autonomous:

$$X' = F(X) \tag{24}$$

Here X is a vector function of t having n component functions $x_1, x_2, \dots, x_n$. On the right side there is a function $F : \mathbb{R}^n \to \mathbb{R}^n$. This function has component functions f_i. The **Jacobian matrix** of F is the matrix $J = (J_{ij})$, defined by $J_{ij} = \partial f_i / \partial x_j$. At a point X_0, the linearized form of Equation (24) is

$$X' = F(X_0) + J(X_0)(X - X_0) \tag{25}$$

where $J(X_0)$ indicates the evaluation of J at X_0. Equation (25) is a linear differential equation, and the eigenvalues of the matrix $J(X_0)$ figure in the theory. If Equation (24) is stiff in the vicinity of X_0, these eigenvalues satisfy the inequalities

$$\text{Re}(\lambda_1) \le \text{Re}(\lambda_2) \le \cdots \le \text{Re}(\lambda_n) < 0$$

and $\text{Re}(\lambda_1)$ is much smaller than $\text{Re}(\lambda_n)$.

If a multistep method is to perform well on such a problem, then $h\lambda_i$ should lie in the region of absolute stability for that method. If the λ_i were known, this information could be used to select h appropriately.

PROBLEMS 8.12

1. Find the region of absolute stability for the implicit Euler method defined by Equation (7).
2. Determine whether the problem $x'' = (57 + \sin t)x$ is stiff.
3. Determine whether this problem is stiff:

$$\begin{cases} x'' = -20x' - 19x \\ x(0) = 2 \qquad x'(0) = -20 \end{cases}$$

4. Consider the multistep method

$$x_n + \alpha x_{n-1} - (1+\alpha)x_{n-2} = \tfrac{1}{2}h\big[-\alpha f_n + (4+3\alpha)f_{n-1}\big]$$

Determine α so that the method is stable, consistent, convergent, A-stable, and of second order. (See Sections 8.4 and 8.5, p. 549 and p. 557, respectively.)

5. Verify that Equations (15) provide the solution to Equations (14).
6. Find the region of absolute stability of the implicit trapezoid rule, Equation (23).
7. Show that the two inequality relations (16) are equivalent to $0 < h < -2/(\alpha+\beta)$ when $\alpha < \beta < 0$.

COMPUTER PROBLEMS 8.12

1. Solve the differential equation in Problem **8.12.2** (above) with initial conditions $x(0) = 1$, $x'(0) = -\sqrt{57}$. (The **Riccati transformation** is $y = x'/x$. It can be used to obtain an equivalent pair of equations, $y' = 57 + \sin(t) - y^2, x' = xy$, with $y(0) = -\sqrt{57}$ and $x(0) = 1$, so that available software can be applied.)
2. Test the **implicit midpoint method**,

$$x_n - x_{n-1} = hf\big(t_{n-1} + \tfrac{1}{2}h,\ \tfrac{1}{2}(x_n + x_{n-1})\big)$$

on the equation $x' = \lambda x$, with $\lambda < 0$, to determine whether its performance will be good on stiff problems.

3. A certain chemical reaction is described by a stiff system

$$\begin{cases} x_1' = -1000x_1 + x_2 & x_1(0) = 1 \\ x_2' = 999x_1 - 2x_2 & x_2(0) = 0 \end{cases}$$

Show that x_1 decays rapidly, whereas x_2 decays slowly. Such a system is difficult to solve because a very small step size will be necessary.